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STUDY INTO THE POTENTIAL AND FEASIBILITY OF A STANDALONE SOLAR- WIND HYBRID ELECTRIC ENERGY SUPPLY SYSTEM

For Application in Ethiopia

Doctoral Thesis

By

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Stockholm, December 2009

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TRITA REFR Report No 09/64
ISSN 1102-0245
ISRN KTH/REFR/09/64-SE
ISBN 978-91-7415-329-3

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Printed by Universitetservice US AB
Stockholm, 2009

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Abstract

The tendency to use renewable energy resources has grown continuously over the past few decades, be it due to fear over warnings of global warming or because of the depletion and short life of fossil fuels or even as a result of the interest which has developed among researchers doing scientific research into it. This work can be considered as joining any of these groups with an objective of giving electric light to the poor population living in one of the poorest nations in the world.

The aim of the work is to investigate supplying electric energy from solar-wind hybrid resources to remotely located communities detached from the main grid line in Ethiopia. The communities in mind are one of two types; the first is the majority of the poor population residing in the countryside; and the other is people relocated by the Government from the over used and dry regions to relatively productive and fertile ones in line with the long-term poverty reduction plan.

The work was begun by investigating wind energy and solar energy potentials at four geographically different locations in Ethiopia by compiling data from different sources and analyzing it using a software tool. The locations are Addis Ababa (09:02N, 038:42E), Mekele (13:33N, 39:30E), Nazret (08:32N, 039:22E), and Debrezeit (8:44N, 39:02E).

The results related to wind energy potential are given in terms of the monthly Average wind speed, the wind speed probability density function (PDF), the wind speed cumulative density function (CDF), the wind speed duration curve (DC), and power density plots for all four selected sites. According to the results obtained through the analysis, the wind energy potential, even if it is not exceptional, is irrefutably high enough to be exploited for generating electric energy.

The solar energy potential, based on sunshine duration data collected over a period of 7 - 11 years and radiation data obtained from different sources, has been calculated using regression coefficients specific to the sites in question. Based on the sunshine duration data, the monthly average daily sunshine amount for each of the places has also been computed and given in a form of plot. Through additional work on the

results of the calculations, the solar energy potential has been given in the form of solar radiation plots for each of the selected sites. As expected, the results indicated an abundance of solar energy potential.

It is based on the promising findings of these two energy resource potentials, wind and solar, that the feasibility study for a standalone solar-wind hybrid energy supply system has proceeded, targeting the community mentioned earlier. The hybrid system consisted of Wind turbine, Photovoltaic panel, diesel generator and a bank of batteries, with a power conditioning converter included in the system.

The hybrid standalone supply system is intended to provide electricity to a model community of 200 families with five to six family members in each. The community is equipped with a primary load, a deferrable load, a community school and a health post. An electric load which includes lighting, water pumping, a radio receiver, and some clinical equipment has been suggested. **Hybrid Optimization Model for Electric Renewables, HOMER**, software has been used for the analysis. The average wind speed and average solar radiation calculated from the data for all of the selected sites has been used to input into the software.

The hybrid system design is approached in three different ways. The first approach is to include within the hybrid system those components which are locally available, without giving special attention to their efficiencies and proceed with the design work. The second approach is to thoroughly search the market for the best and most efficient technological products and to select the best components for the analysis. A third approach considered in an attempt of cost minimization is to see if a self-contained type of design can be a better solution. What this means is every household will have its own supply system that may consist of any combination of PV and wind turbine including converter, battery and charge controller.

After running the simulations, lists of power supply systems have been generated, sorted according to their net present cost. Sensitivity variables, such as range of wind speeds, range of radiation levels and diesel price have been defined as inputs into the software and the optimization process has been carried out repeatedly for the sensitivity variables and the results have been refined accordingly.

Keywords: Wind Speed; Sunshine Duration; Solar Radiation; Feasibility Study; Standalone System; Solar-Wind Hybrid.

Acknowledgements

I sometimes wonder why I even wanted to continue my studies after my second degree. Was it my interest in further education or was it just because the chance was available or was it even because someone pushed me to continue? I don't really have a precise answer for these questions. I can just say it was a mix of all of them. Yes, I had the interest but not the courage to go through the application processes. Yes, I can also in a way say that the chance was available, and again yes, someone pushed (encouraged) me to continue my studies further. That someone was *Dr. Frehinot Woldehanna* who interfered in the struggles of my mind and put my name in the application he was writing to SIDA for project funding, which was later accepted, thus implicitly realizing my further studies. *Dr. Frehinot Woldehanna* not only did that but also encouraged further study and that is why my heartfelt thanks go to him. "What if the application failed?" could be a question which comes to mind and that is why I would like to thank the Swedish Government and its tax-paying public for providing me the funding, through SIDA, which has completely covered all my expenses for the whole duration of my stay here in Sweden.

I would like to take this opportunity to wholeheartedly thank my supervisor, *Prof. Björn Palm*. I thank him not only for his valuable advice on my work but also, I can comfortably say, for his help in personal matters. He was much more than an academic supervisor to me. I have always felt comfortable having him behind me and life for me in Sweden has been enjoyable because of this. I also wish to express my thanks to *Dr. Tesfaye Bayou* and *Prof. Woldegiorgis Woldemariam* for their helpful ideas during the course of my work.

My special thanks go to *Inga Du Rietz*, the "iron" secretary, whom I have always run to with my endless questions; and also to the people at the Division of Applied Thermodynamics and Refrigeration, *Anders Johansson (Dr.)*, *Benny Andersson*, *Benny Sjöberg*, *Birger Söderström*, *Cecilia Hägg*, *Claudi Martin*, *Eric Granryd (Prof.)*, *Erik Björk*, *Hans Jonsson (Dr.)*, *Hatef Madani*, *Jaime Arias (Dr.)*, *Jan-Erik Nowacki (Dr.)*, *Joachim Claesson (Dr.)*, *Jose Acuna*, *Marino Grozdek*, *Monika Ignatowicz*, *Muhammad Mamayun Maqbool*, *Nabil Kassem (Dr.)*, *Oxana Samoteeva*, *Per Lundqvist (Prof.)*, *Peter Hill*, *Primal*

Fernando (Dr.), Rahmatollah Khodabandeh(Dr.), Rashid Ali, Raul Anton(Dr.), Richard Furberg, Samer Sawalha(Dr.), Shota Nozadze, Simanic Branko, Stina Gustafsson, Susy Mathew, Teclemariam Nemariam, Tony Chapman, Wabib Omhai(Dr.), Wimolsiri Pridasawas (Dr.), Yang Chen, and Ake Melinder(Dr.).

I am also very much indebted to my family; my wife Genet Wube, who has been handling all family matters back at home throughout my period of absence; my daughter Yohanna Getachew who was born in my absence and missed all the fatherly care from me in her childhood.

Most of all I would like to thank the almighty God who I strongly believe gave me the power and the strength to accomplish this task. Praised be the lord! Amen.



Stockholm, December 2009

Publications

Journal Papers:

Published:

Bekele G, Palm B. Wind energy potential assessment at four typical locations in Ethiopia, *Applied Energy* 2009; 86: 388–396.

This paper has been selected by the Scientific Secretariat of Eni (an Italian multinational oil and gas company) for the 2010 edition of the Eni award and is currently a candidate.

Bekele G, Palm B. Feasibility Study for a Standalone Solar-Wind Based Hybrid Energy System for Application in Ethiopia, *Applied Energy* 2010; 86: 487–495.

In Review:

Bekele G, Palm B. Assessment of Solar Energy Potential at Four Typical Locations in Ethiopia, submitted to the journal *Energy for Sustainable Development*

Bekele G, Palm B. Solar-Wind-Based Village Electrification in Ethiopia: A Comparison of Technologies. Submitted to the journal *Renewable Energy*

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Introduction

Background

Ethiopia is known as being unique among other African countries for its historical background. The country has a total area of 1,127,127 sq km of which land makes up 1,119,683 sq km and water coverage 7,444 sq km. The terrain is mainly high plateau with mountain ranges divided by The Great Rift Valley. The elevation generally ranges between 1,500 and 3,000 meters above sea level with extremities of 125 m below sea level in The Denakil Depression and 4,620 m above sea level at Ras Dashen.





Figure 1 Map of Ethiopia and the locations [Genesis, 2009] and [Googlemap, 2009]

It is well known that the country has been suffering from cyclical drought since the early 1970s and is listed as one of the poorest nations in the world. The cause of the problem is not difficult to see if one takes a closer look at what has been going on in the country for a time long. Indeed, it is for the most part a man-made problem, to which natural disasters have also contributed to some extent. A modern energy supply system, such as electricity, is lacking and therefore most of the people depend on fuel-wood for their daily energy needs, which has caused unimaginable deforestation and desertification of the land to almost irreversible levels. The lack of re-plantation and rehabilitation schemes for the vegetation consumed and for the degraded soil has worsened the problem further. Years and years of erosion have washed fertile top soil out into the neighboring countries and have changed the land into one of rock-strewn, pebbly fields and dry soil. A typical example is The River

Nile with its tributaries, which have been carrying away the top soil out into the Sudan and Egypt. Today, the percentage of arable land remaining is only 10% of the total area in the country. Problems such as this, along with many other irregularities have caused the country to become so dry and unproductive that its people have lost their pride and feel ashamed of being associated to famine and drought. The dilemma is that even today the situation is not showing any signs of ending due to the fact that the country still suffers shortages of modern electrical and petroleum gas energy fuels and the vast majority of the population is still heavily dependent on biomass-based resources for their daily energy needs. With this situation prevalent, it is clear to see that there is no hope of coming out of the cycle in the foreseeable future. The vegetation is being depleted at an alarming rate and whatever biomass stock is available will not last for more than a couple of decades, which clearly indicates that the country is changing into another of the continent's deserts.

As can be imagined, in this poor country, where more than half of the population lives below the poverty line, agriculture is the main source of livelihood for more than 80 % the population. It is well known that agricultural production, unless of a modern and mechanized type, is extremely vulnerable to factors such as climatic conditions, the impacts of war and civil conflict, disease, etc. and unfortunately all of these are typical problems in the case of Ethiopia. The current climatic conditions and other factors, as mentioned earlier, have reduced the total fertile land to just 10 %, the war and the civil conflicts have been rocking the country for several years and the country is also home to many diseases, such as malaria, HIV and the like, which have been killing the people non-stop, putting the country high up on the list of mortality rates. The recurring droughts that have been observed in the country for so many years have left the poor without food crops, causing periodic famines. The persistent lack of rainfall, which is probably exacerbated by the removal of vegetation, is also one of the major factors behind the poor harvest. Other factors to mention include the wide fluctuations in agricultural production as a result of drought, an ineffective and inefficient agricultural marketing system, underdeveloped transport and communication networks, underdeveloped production technologies, and most importantly environmental degradation.

As part of the solution to the recurring drought and poverty of recent years the government has instigated plans to resettle about two million people from the over-used and infertile regions to better, unused ones all across the country where they can farm and bring themselves out of the poverty they are bogged down in. The resettlement program has

been partially implemented and since 2003 about 1.3 million people have been resettled in four of the regions the country is divided into [Bekele and Palm, 2009]. Ethiopia is currently divided into 11 regions of which two are Addis Ababa (The capital) and Dire Dawa (a city in the Eastern part of the country). The four regions where the resettlement is implemented are Tigray, Oromia, Amhara, and Southern Nations, Nationalities and Peoples Regions (SNNPR) The resettlers, as can be imagined once again do not have access to any modern energy supply systems, such as electricity, and ironically, what they have to rely on for their daily energy needs is the diminishing biomass stock from their surroundings. The first time they arrive to the new locations they have to use the wood in the surroundings to build their homes and then they continue to use the same wood for their daily energy needs, such as for cooking and for lighting while sitting and chatting in the evenings. Once again there do not seem to be any re-plantation schemes included in the resettlement program, to regenerate the biomass that is being used. It is therefore clear to see that the situation further diminishes the hope of rehabilitating the vegetation. What then is the solution to the entire crisis that is threatening the country? This is an important question that needs to be raised and addressed.

Currently, the Ethiopian Electric Power Corporation, EEPCo, which is the sole electric power producer in the country, generates considerably less than 1000 MW [EEPCo, 2007] of electric power. In total amount of electricity consumed in the year 2005/06 was less than 3 TWh/year [EEPCo, 2007] in a country of, according to the recent census, approximately 77 million people [CIA, (July 2007 est.)]. From this it is not difficult to see how many of the people are short of electric power. The power generation resource is almost exclusively hydro, while there are also quite a few others, such as diesel and geothermal generated power which make up just a small part, 5% at most [EEPCo, 2007]. The reason for using hydro is because the country is rich in large rivers such as the Nile, Gibe, Tekeze and many others. But hydro power has got its own drawbacks as it requires huge dams for the water storage, which occupies a large area. Besides the ecological disturbance that may be caused, by submerging the already threatened vegetation, it also has adverse effects on the already meagre land per capita of the country. This is further worsened by rapid population growth and, if the present trend continues unabated, the population in 10 years time will be over 100 million.

Other sources used for generating electricity are geothermal (steam), mini hydro - power plants and a number of isolated diesel generators scattered across the country as Self-Contained Systems. The use of fossil

fuel resources is impractical as it is becoming increasingly evident that it entails more and more problems, both in terms of fluctuating prices and its environmentally-unfriendly nature because of CO₂ and other polluting emissions, which are believed by many to be the main cause of global warming. Added to this are its rate of depletion and the political turmoil that fossil fuels are causing across the globe. On the other hand, up to now the country has not had its own sources of this fuel type and has been importing it which will have to continue for an unknown length of time into the future. Despite government subsidies, the price of gas has more than doubled in a period of less than five years. With the escalating price of oil and the country's shaky economy it does not seem the country will emerge from its economic problems for the foreseeable future.

The other sources mentioned, such as geothermal and mini – hydro power plants are not yet in extensive use and are still to be developed.

All in all, as stated previously, the total sum of electricity produced in the country, by the sole electric power authority, EEPCo, from all the aforementioned sources, currently amounts to well under 1000 MW (2004: 670 MW installed capacity [EEPCo, 2009]) of power and this is for a nation of almost 80 million inhabitants. Moreover, as would be expected, the distribution of power across the country is restricted to just the urban populace, a fraction which makes up just 20% of the total population. This is the on the ground reality regarding electric power in the whole country and it is this situation that calls for a radical change, a rethinking of the energy path that the country is perusing.

Which other sources are possible is a straightforward question that should be asked. With the present capacity of the country, nuclear energy and the like is not likely to be practical in the foreseeable future. Therefore it is fair to think of other resources such as wind and solar energy. The annual solar radiation reaching the ground is well over 2000 kWh/m² as is the case in most tropical regions. Wind Speed Data, collected as early as the 1930s, during the late 1960s and early 1970s and also as recently as the early 2000s, at different sites in the country, showed that the wind energy potential is also something which should not be underestimated. Given the current situation, the author of this work believes that these two resources are immediate candidates for investigation and the most feasible resources to work on. It is therefore these two resources which are focused on in this study. Investigations into these resources are also a contemporary phenomenon, which researchers and scientists all across the globe are continually working on. Furthermore, these resources are clean and environmentally friendly

while at the same time being free and/or inexpensive once they have been made available.

Structure of the Thesis

To put it simply, the thesis is written by first discussing the potential of renewable resources, i.e. wind and solar energy potentials at four typically habitable locations in Ethiopia. Following that the thesis discusses how the feasibility study into the standalone solar/wind hybrid electric energy supply system has been conducted and gives the results obtained.

Part I begins with the basic background theory for the determination of the wind energy potentials being discussed. This includes sources of wind energy, the energy in the wind, the energy output of a wind turbine, how to measure wind speed, where to place the wind turbine, a brief note on turbine technology, an introductory note about the turbine generator, etc. Following on from the background theory, a short note is provided regarding the assessment of wind energy potential at the four typically selected locations and the associated findings. This is a short note as full detail is given in the published paper, entitled Wind Energy Potential Assessment at Four Typical Locations in Ethiopia [Bekele and Palm, 2009b].

Part II continues in the same way as Part I, with the basic theoretical background of research into solar energy potential outlined first. In this part, discussions are made regarding the sun as a source of energy, the determination of solar radiation from sunshine duration data, with the utilization of empirical formulas derived from different authors involving regression coefficients, error determination techniques using statistical formulas. Following that, the assessment of solar energy potentials at the aforementioned locations and the associated findings are given in detail.

In Part III, a brief note on the basic working principles of a solar/wind hybrid system, its constituent components and advantages is given first. Then the basic operating principles of the system components are discussed; the PV, the diesel generator, the inverter, and the battery. Following that, details of the feasibility study into the standalone solar-wind hybrid electric energy supply system are discussed. This includes the model and schematic diagram of the hybrid setup, the step-by-step details of the electric load, the required information input into the software, etc. Regarding the results obtained, each location is treated separately.

First Mekele is considered and for this location the renewable resource data was input to the software and the resulting alternatives for implementable hybrid setups obtained. From the list of alternative setups, the best was selected, the basis for the selection discussed, and so on. The cost break down for the implementable setups is also given.

The same procedure is followed to analyze the other two locations and it is Nazret and Debrezeit which are subsequently discussed. The analysis and results for the fourth location have been submitted for publication as a journal paper and the paper is attached at the end.

Short review of publications by the author

Paper 1:

Bekele G, Palm B. Wind energy potential assessment at four typical locations in Ethiopia, *Applied Energy* 2009; 86: 388–396

This paper has been selected by the Scientific Secretariat of Eni (an Italian multinational oil and gas company) for the 2010 edition of the Eni award and is currently a candidate. The paper discusses wind energy potential at four different sites in Ethiopia; Addis Ababa (09:02N, 38:42E), Mekele (13:33N, 39:30E), Nazret (08:32N, 39:22E), and Debrezeit (8:44N, 39:02E). Data from different sources have been compiled and used for the analysis. As none of the data obtained is complete, efforts have been made with the analysis in order to come up with a reasonably complete set of data. The analysis is supported by a piece of software known as **Hybrid Optimization Model for Electric Renewables**, HOMER, [HOMER, ver. 2.19]. The results regarding wind energy potential are given in terms of the monthly Average wind speed, wind speed probability density function (PDF), wind speed cumulative density function (CDF), wind speed duration curve (DC), and power density for all the selected sites. The results confirmed that the wind energy potential can be exploited for generating electric energy, at least for standalone systems. The paper is published in the journal *Applied Energy*.

Paper 2:

Bekele G, Palm B. Solar Energy Potential Assessment at four typical locations in Ethiopia, submitted to the journal *Energy for Sustainable Development*.

As a continuation to the study of wind energy potential in the first paper, solar energy potential at the same four locations is investigated. Here, data is again obtained from different sources. The available data is sunshine duration data. There is no radiation data available except for one of the locations, Addis Ababa. Hence the work in this part is to find out the radiation data for the other locations based on the available sunshine hour data and therefore this is the theme of this part of the work. The results obtained are given in the form of solar radiation plots for all the selected locations.

Paper 3:

Bekele G, Palm B. Feasibility Study for a Standalone Solar-Wind Hybrid Energy System for Application in Ethiopia, *Applied Energy* 2010; 86: 487–495.

This paper discusses the supply of electric energy from a solar-wind hybrid source to a remotely located model community. The community may be classified as one of two types; native people, and those relocated by the Government in line with the poverty reduction program and in each case the community is detached from the main grid line. Based on the findings of the wind and solar energy potentials in the earlier studies a feasibility study into the supply of electricity to a model community of 200 families with five family members in each is scrutinized. Primary and deferrable loads are provided to the community in addition to a community school and a health post. The electric load includes lighting, water pumping, a radio receiver, and some clinical equipment. Here HOMER software is again used for the analysis and it is only one of the sites, Addis Ababa, that is analyzed in this paper. The results for wind and solar potential obtained in the earlier findings are used as an input to the software. After running the simulation for optimum results, a list of power supply systems are obtained sorted according to their net present cost. Sensitivity variables, such as range of wind speed, radiation levels, diesel price and price of PV cells are defined as inputs and the simulation is rerun in search of optimum results. The results obtained include alternative realizable setups along with their net present cost.

Paper 4:

Bekele G, Palm B. Solar-Wind-Based Village Electrification in Ethiopia: A Comparison of Technologies, Submitted to the journal *Renewable Energy*

The target of the paper is to study the feasibility of supplying electric light and potable water to the community in the resettlement villages, which the government has established in the remote areas within the different regions of the country where the main grid line does not reach. The study is conducted by a comparison of the most efficient and up-to-date technologies of the components used in the hybrid system both at the load and the supply side. Individual solution, i.e., on per household basis is also checked in an attempt of cost reduction. The results showed significant change in the cost compared to what was done in the previous study in which standard available system components were used.

1 PART I:

Basic Theory and Wind Energy Potential

1.1 Basic Related Wind Theory

1.1.1 What is the source of wind energy?

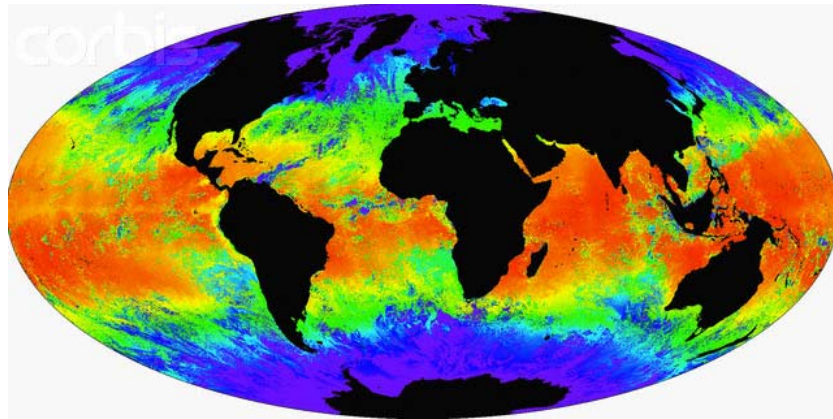


Figure 1-1 NASA satellite sea surface temperature image of the globe

The regions around equator are heated more by the sun than the rest of the globe. The warm colors, red, orange and yellow indicate the hot areas in the infra-red image of sea surface temperatures (taken from a NASA satellite, NOAA-7 in July 1984).

Most renewable energy ultimately comes from the sun and 1-2 % of the sun's energy reaching the earth is converted into wind [Danish wind, 2008]. Differences in air pressure caused by the uneven heating of the Earth's surface by the sun forces air circulation; and air flows from areas of high pressure to areas of low pressure .

As a result of temperature and pressure differences, and also the Coriolis Effect, there are different global wind patterns at different latitudes. Trade winds, prevailing westerlies, and polar easterlies are some of the types that can be mentioned in this regard.

The Coriolis force is the apparent deflection of air from its path as it moves from high to low pressure areas because of the rotation of the earth.

Other wind resources such as Geostrophic Winds, Surface Winds, Local Winds (as in Sea Breezes), Mountain winds, etc. should also be noted [Danish wind, 2008].

1.1.2 Energy in the Wind

The calculation procedures for determining the power available in the wind can be found in many standard text books on wind power. The following basic relationships can be found, for example, in (Gasch R, Tvele J, 2002, , Manwell J.F, 2002,, Gipe P,1999)

The energy the wind transfers to the rotor of a wind turbine is proportional to the density of the air, the rotor area, and the cube of the wind speed.

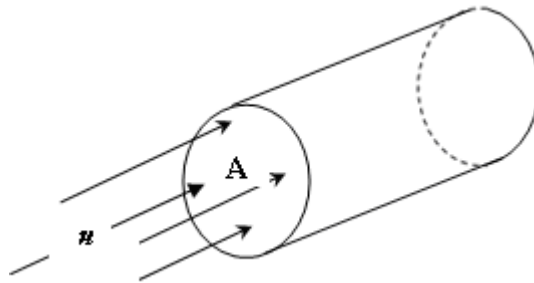


Figure 1-2 Air flow through a rotor area, A , at speed u m/s

$$P = \frac{1}{2} \rho A u^3 \quad \text{Eq. 1-1}$$

where:

- P - Power in the wind (W)
- ρ - Density of the air (at normal atmospheric pressure and at 15° Celsius air weighs some 1.225 kilograms per cubic meter)
- A - Rotor Area (A typical 1,000 kW wind turbine has a rotor diameter of 54 meters, i.e. a rotor area of some 2,300 square meters.)[Danish wind, 2008]
- u - The wind speed (m/s)

It is to be noted that the mean wind speed should not be simply inserted into Eq.1-1, as this will give an erroneous result because of the fact that the mean of the cubes of wind velocities will almost always be greater than the cube of the mean wind speed.

The most accurate estimate for wind power density is that given by Eq.1-2.

$$P/A = \frac{1}{2} \cdot \frac{1}{n} \cdot \sum_{j=1}^n (\rho_j \cdot u_j^3) \quad \text{Eq. 1-2}$$

Where n is the number of wind speed readings and ρ_j and u_j are the j^{th} readings of the air density and wind speed.

For a known pressure and temperature:

$$\rho = \frac{P_r}{RT} \quad \text{Eq. 1-3}$$

Where p_r is air pressure (Pa) and R is the specific gas constant (287 Jkg⁻¹ K⁻¹) and T is air temperature in °K.

For the available temperature data:

$$\rho = \frac{P_0}{RT} \exp\left(-\frac{gz}{RT}\right) \quad \text{Eq. 1-4}$$

where P_o is standard sea level atmospheric pressure (101,325 Pa), g is the gravitational constant (9.8 m/s²); and z is the region's elevation (m) [Oklahoma Wind power, 2008].

If pressure and temperature data is not available, the following correlation may be used for estimating the density [Oklahoma Wind power, 2008].:

$$\rho = 1.225 - (1.194 \times 10^{-4}) * z \quad \text{Eq. 1-5}$$

1.1.3 Energy Output

The power available from a wind turbine is usually shown by the machine's power curves $P(u)$ and a typical curve is shown in figure 1-3.

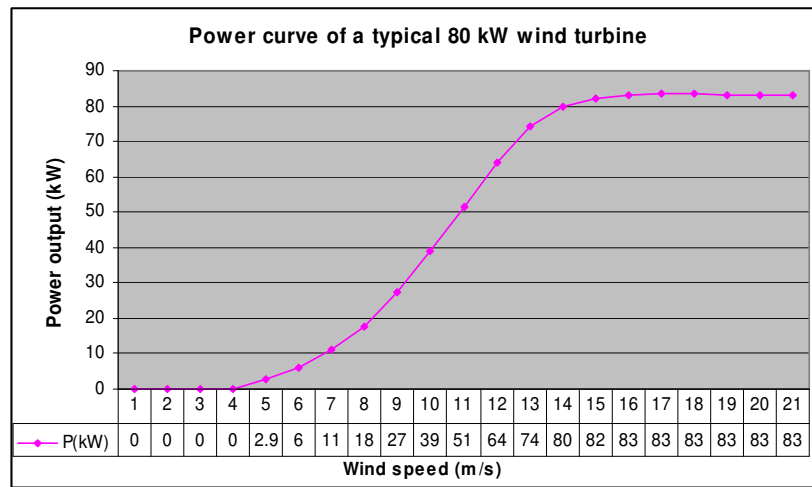


Figure 1 -3 Typical power curve for an 80 kW wind turbine (WES18, 18m rotor diameter) [HOMER, Ver. 2.19]

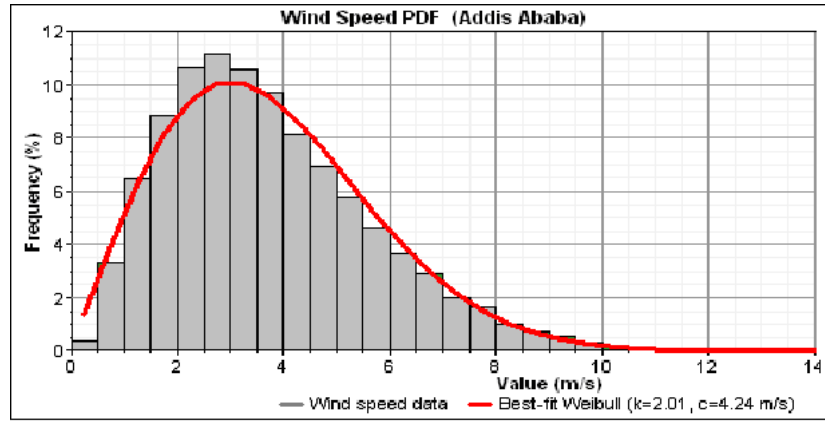


Figure 1-4 Wind speed probability density function for Addis Ababa

A Weibull distribution graph is usually used to describe wind variation over a certain period of time at a particular site. Figure 1-4 shows a typical distribution plot for wind speed data based on wind speed measured five times every day for three years, 2000–2003, in Addis Ababa. As can be seen in the graph, the mean wind speed is about 4 m/s. The mean wind speed can be obtained by summing up the products of each wind speed interval and the probability of getting that wind speed.

The Weibull probability density function (PDF) is given by equation Eq.1-6 [Manwell, 2002].

$$f(u) = \frac{k}{c'} \left(\frac{u}{c'} \right)^{k-1} \cdot \exp \left[- \left(\frac{u}{c'} \right)^k \right] \quad \text{Eq. 1-6}$$

where:

u = the wind speed,

k = a constant known as shape factor, as the value of k increases the curve will have a sharper peak

c' = a scale parameter in m/s; the larger the scale parameter, the more spread out the distribution.

The area under the curve is always unity.

The power density can in this case be expressed by *Eq.1-7*. This is the same equation as *Eq. 1-1* but for a median (average of a series of recorded wind speed) wind speed in which case the frequency of the recording is considered.

$$P/A = \frac{1}{2} \sum_{j=1}^n \rho \cdot V_j^3 \cdot f_j \quad \text{Eq. 1-7}$$

where V_j is the median velocity in class j and f_j is the frequency of occurrence in the same class.

For $k = 2$ the Weibull PDF is commonly known as the Rayleigh density function in which case *Eq. 1-6* may be rewritten as in *Eq.1-8*.

$$f(u) = \frac{2u}{c'^2} \cdot \exp \left[- \left(\frac{u}{c'} \right)^2 \right] \quad \text{Eq. 1-8}$$

1.1.4 Wind Speed Measurement

Among the various types of anemometer, such as the ultrasonic or laser, the most common type is the cup anemometer, which is used for measuring wind speeds. The wind direction is detected with a wind vane, which is normally fitted together with the anemometer. A data logger collects wind speed and wind direction data from the anemometer and wind vane respectively. Wind speeds are usually recorded as a 10 minute average.

1.1.5 Turbine Siting

Finding a place for a wind turbine is one of the most challenging aspects of using wind energy. If located too close to homes, in addition to the uncomfortable noise it creates for surrounding families, the turbine suffers building interference. If it is too far away, then the cost of cabling and the burial of cables should not be overlooked. Rarely is there an ideal site [Gipe P, 1999].

With regard to the wind, nature itself is usually an excellent guide for finding a suitable wind turbine site. The inclination of trees and bushes

reveals information about the prevailing wind of the region. However, the best guide is Meteorology data collected for more than 30 years and compiled in the form of wind rose diagrams [Danish wind, 2008]. Nonetheless, such data are rarely available, especially in a country such as Ethiopia. It is under such circumstances that observing the surroundings gives significant clues about the wind regime of the area [Danish wind, 2008]. Furthermore, the site to be selected should be free of nearby obstacles (such as trees, small houses or other buildings). It has to be wide and open and of as low roughness as possible in the prevailing wind direction. Such sites are quite common in the country. The surface roughness causes wind shear close to the ground and suppresses the wind's speed within a certain distance.

Wind speed increases with height and therefore a higher tower captures more wind energy. Wind speed at any height, before tapering off, can be estimated using equation 1-9, if the wind speed ($u(z_r)$) is known at a certain reference height (z_r) above a surface with a known roughness length (z_0) [Danish Wind, 2008]. Figure 1-5 illustrates a typical wind speed profile for a surface roughness length of 0.1.

$$u(z) \cdot \ln\left(\frac{z_r}{z_0}\right) = u(z_r) \cdot \ln\left(\frac{z}{z_0}\right) \quad \text{Eq. 1-9}$$

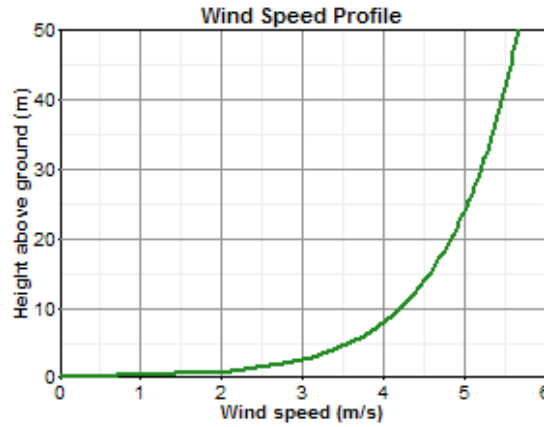


Figure 1-5 A typical wind speed profile for a surface roughness length of 0.1 [HOMER, Ver. 2.19]

When selecting sites, infrastructural facilities, such as roads, should also be considered.

Based on the literature survey and theoretical notes, such as those given thus far, the wind energy potential of the four selected locations, assumed to be models for most habitable parts of the country, is investigated.

1.1.6 Brief Note on Wind Turbine Technology

When designing a wind turbine, there are several factors that must be taken into consideration: the dynamic behavior, the strength, the fatigue properties of the materials and the entire assembly. Hence, manufacturers have developed a variety of turbines with this in mind, or with other advantages and disadvantages. In general, the wind turbine must be designed to:

- Withstand high wind loads; optimum robustness and solidity
- Compliant to accommodate shade loads
- Manage loads mechanically and/or electrically

The most important design variables are:

- Number of blades
- Power control system and
- generator types

Regarding the number of blades, three-bladed horizontal-axis wind turbines are currently the most commonly used types for grid-connected wind turbines. Stability is the most important reason for this. Turbines with even number of blades give stability problems. The reason is because of the fact that in the instant of time when the tip of one of the blades passes the top most side and is forced to bend because of the force of the wind the tip of the opposite side blade passes into the wind shade in front of the tower. [Danish wind, 2008].

With regard to the power control system, the power regulation mechanisms must be implemented in such a way that power output is limited close to the rated value, as wind turbines have their highest efficiency at the wind speed they are designed for. There are three commonly used types of power control in the industry.

- Stall Control
- Pitch Control
- Active stall regulation

Using stalling regulation, the aerodynamic design principle is to increase the angle at which the relative wind strikes the blades (angle of attack) and to reduce the induced lifting force at the moment the wind speed becomes too high. This happens because of turbulence created on the side of the rotor blade which is not facing the wind. Stall controlled wind turbines have their rotor blades bolted onto the hub at a fixed angle.

The Pitch control mechanism is usually hydraulically operated. An electronic controller, which depends on the output power, sends a signal to the blade pitch mechanism so as to turn the rotor blades out of the wind to the exact degree required and to keep the rotor blades at the optimum angle for maximized output at all wind speeds. In pitch control mechanism the rotor blades are rotated around their longitudinal axis.

With an active stall regulation mechanism the machine is usually programmed to pitch the blades much like a pitch-controlled machine at low wind speeds, so as to get a reasonably large torque at low wind speeds. If the generator is about to be loaded, then the machine also pitches its blades to increase the angle of attack of the rotor blades forcing the blades to go into a deeper stall thus wasting the excess energy in the wind [Danish wind, 2009]. In this control mechanism the machine can be run almost exactly at rated power at all high wind speeds.

There are also other control mechanisms such as the use of ailerons (flaps) to alter the geometry of the wings or yawing to turn the rotor partly out of the wind to decrease power.

1.1.7 Wind Turbine Generators

Wind turbine generators are a bit different from other generating units in that the input power to the generator shaft is taken from the wind turbine rotor which fluctuates greatly in terms of mechanical power (torque). The transmission system consists of the rotor shaft with bearings, brake(s), an optional gearbox, as well as a generator and optional clutches. There are two types of generator, synchronous and asynchronous. Synchronous generators are more expensive compared to asynchronous (induction) generators. Six-pole asynchronous generators are the most commonly used types.

The speed of the asynchronous generator varies with the turning force (torque) applied to it. It has a slightly softer connection to the network frequency than the synchronous generator, as it allows a limited amount of slip, or variation, in generator RPM.

In the case of the synchronous generator, the speed is set by the grid frequency and the number of pairs of poles of generators. The generator runs at a fixed frequency (line frequency), and hence at a fixed speed. Equation 1-10 gives the relationship between the frequency and the synchronous speed.

$$n_s = \frac{60f}{P_l} \quad \text{Eq. 1-10}$$

where n_s is the synchronous speed, f the line frequency and P_l the number of pole pairs.

That what has been discussed thus far regarding wind turbine technology mainly applies to larger size wind turbines. The design principles of smaller wind turbines are somewhat different to the larger ones in that the distinctive purpose of small wind turbines is to produce power frequently over short periods, e.g. for battery charging. It is important that small turbines generate in weak winds and respond quickly when harnessable winds occur. The rapid starting of the rotor before the generator cuts in is a further requirement [Joliet; 2008]. Small wind turbines often have direct drive generators (without a gearbox) and give out direct current. Their blades could be aeroelastic types and usually use a vane to point into the wind. Figure 1-6 shows the power curve of a typical 20 kW wind turbine.

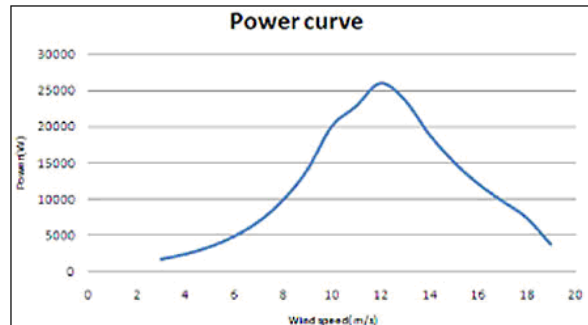


Figure 1-6 typical 20 kW wind turbine power curve [Joliet, 2008]

The wind turbine may have the following technical specification [Joliet, 2008]:

Rotor Diameter (m):	10
Start up wind speed (m/s):	2.5
Rated wind speed (m/s):	10
Cut out wind speed (m/s):	15
Max. output power (W):	25000
Output voltage (VDC):	360
Furling:	3 stage motorized yaw control
Noise level:	38.2db

1.2 Assessment of Wind Energy Potential

1.2.1 Previous Studies

Two previous studies have given substantial results regarding the wind energy potential in the country [Wolde-Ghiorgis W, 1988] [Drake and Mulugetta,1996] by identifying the wind regimes in several areas. However, the data used in these studies is relatively old; the most recent data used in the first study is from 1968-1973 and was recorded only three times a day, at 6:00, 12:00, and 18:00. The remaining data used was also recorded three times a day at 8:00, 14:00 and 19:00 during the period 1937 – 1940.

Data used in the second study was collected during the period 1979-1990 at 60 different locations across the country and recordings were made, according to the author, 4 to 7 times per day at a height of 2 m. However, the data couldn't be found in the archives of the source, the National Meteorological Service Agency (NMSA), which the author provided.

Unlike the previous studies, this study focuses on four specific locations, carefully selected in such a way that they represent a significant portion of the habitable parts of the country.

1.2.2 The Wind Energy Potential

Compared to the previous studies, the data used within this work is relatively recent, from 2000 – 2003, and it is data which has been

recorded five times daily, at 6:00, 9:00, 12:00, 15:00, and 18:00, at a height of 10 meters, for three consecutive years. According to the NMSA, wind speed and direction data have been collected using various types of Lambrecht cup anemometer. The wind vane, which is used to measure wind direction, is integrated with the instrument. The cup and wind vane sensor are mounted on the same shaft. The problem with cup anemometers is their inertia, i.e. their starting threshold wind speed value is approximately 1 m/s [Miodrag, 2009], and once they gain momentum in gusty conditions they over-speed. This may result in a lack of accurate measurements when dealing with low wind speeds and an over-estimation of the mean wind speed under high wind speed conditions. The accuracy of such an instrument is $\pm 2\%$ FS [Lambrecht, 2009].

The four locations investigated are Addis Ababa, 09° 02'N, 38° 42'E, 2408 m (AMSL); Mekele, 13° 33'N, 39° 30'E, 2130 m; Nazret 08°32'N, 39°22'E, 1690 m; and Debrezeit, 08°44'N, 39° 02'E 1850 m. The data is from the same source, NMSA. The data can be claimed to be fairly complete for the given period of time, with only a few recordings missing here and there. The missing data has been replaced by the averages of the preceding and following readings. For verification purposes, data from other sources has been investigated for those sites that data is available for [NASA, 2008].

Hybrid Optimization Model for Electric Renewables (HOMER) software is used to analyze the data. The software is a micro-power optimization model for both off-grid and grid-connected power systems in a variety of applications. More detail about HOMER is given in section 3.2.2. It should be noted that the data used in the previous studies, and also as a basis for this one, is recorded between 6:00 and 18:00, which means that there is no recorded data for the period between 18:00 in the evening and 6:00 in the morning, except the unpublished data recently collected by GTZ [GTZ, 2005] at a location close to one of the sites, Mekele. Hence, the data that could be obtained from NMSA can be said to be somewhat incomplete.

The major task of this part of the study has therefore been to convert this incomplete data set into a relatively complete one and this has been achieved. Different methods have been followed for this purpose. One of the methods used, which enabled to determine the wind energy potential, is to use HOMER to manipulate the data. What was done in this regard was to first assume the available wind data as a complete data recorded over 24 hours daily and then calculate the monthly average. This average is fed into the software like any standard monthly average wind data, which HOMER therefore recognized it as any standard

monthly average wind speed data. The data then is processed and hourly data of a year (8760 hours) is synthesized. What is done next was to handpick those particular data generated at the particular times during which the actual (measured) data was recorded and then their monthly average was calculated. This means from the synthesized hourly data those at the hours 6:00, 9:00, 12:00, 15:00, and 18:00 are handpicked and their monthly average is calculated. The monthly average is fed again to the software so that it synthesizes another set of hourly data and once again HOMER synthesized hourly data of a year. This time the level of the curve of the wind speed is raised by about 8 % more than the curve of the earlier synthesized data. This is because the daytime wind speed is higher than the night time.

By using appropriate scaling factor the second synthesized data is scaled down so that it is equal to the data measured and this was achieved to accuracy of 2 % error. Hence, the simultaneously generated hourly data during the night time is what is considered to have filled the gap of the missing night time data. This is illustrated in figure [1-7]. In the figure monthly averages of: the measured (curve A), the synthesized hourly data from the measured (curve B), the synthesized hourly data from the monthly average of the handpicked daytime data (curve C), and the scaled down of curve C (curve D).

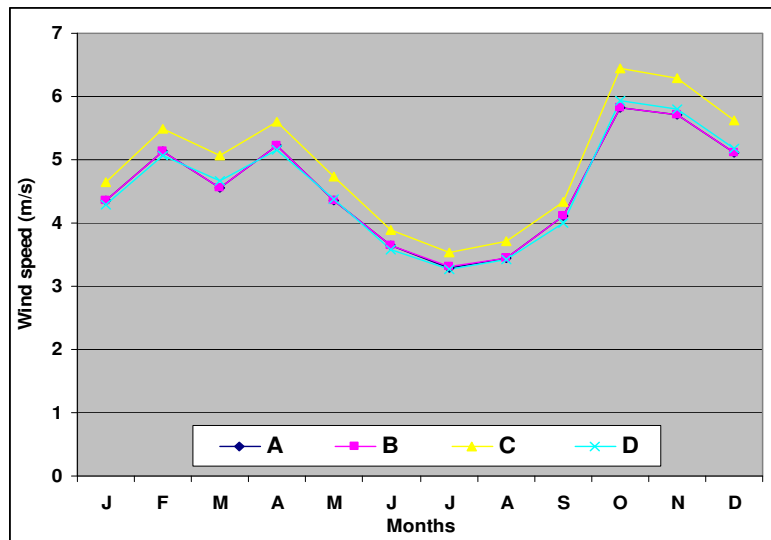


Figure 1-7 Monthly average wind speed of: the measured (A), of the synthesized hourly data from the measured (B), the synthesized data from the filtered out daytime data (C), and of the scaled down synthesized data (D)

It should be noted that the principal parameter used by the software, to synthesize the potential wind speed is the monthly average wind speed data measured only during the daytime over the years. It is from this data the hourly data of a year (8760 hours) is synthesized. The detail of the work is well explained in the published paper [Bekele and Palm, 2009b].

Other data input into the software are given in section 3.3. in detail. These include the shape parameter k , the anemometer height at which data is collected, the diurnal pattern strength, the autocorrelation factor, etc.

The final result, i.e. the most probable wind speed distribution or wind energy potential for each of the locations selected is given in Figure 1-8.

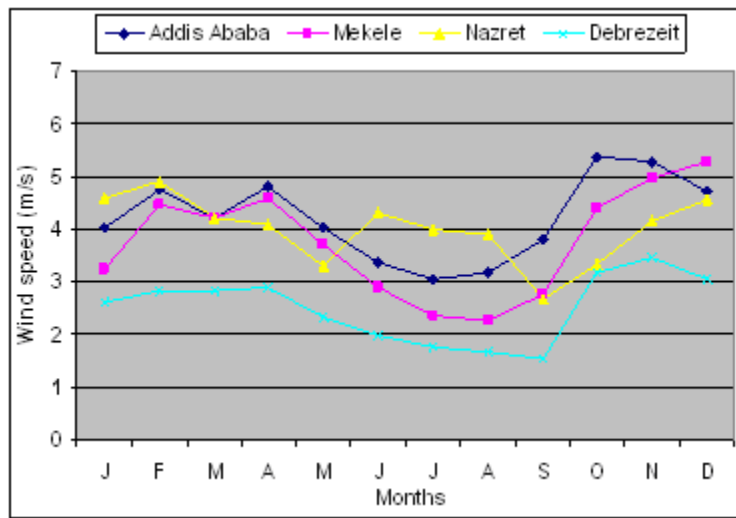


Figure 1-8 Software generated monthly average wind speeds given the measured data as input

The potential of each location has been evaluated against the wind power classification of the US Department of Energy (DOE). Accordingly, Addis Ababa and Nazret are found to be of class 2 type while Mekele and Debrezeit are of class 1. While class 2 potential is considered marginally good for wind energy development, class 1 potential is, in general, considered unsuitable [Bekele and Palm, 2009b]. However, average annual wind speeds of 3 to 4 m/s may be adequate for non-grid-connected electrical and mechanical applications, such as battery charging and water pumping, which is indeed the goal of this research. In

general, with the help of a software tool and based on the somewhat incomplete data collected in recent years (2000 – 2003), the most probable wind energy potentials of the four selected locations have been determined. Although the wind potentials may not be sufficient for independent wind energy conversion systems, it is believed that they are usable if integrated with other energy conversion systems such as PV, diesel generator and battery.

2 PART II:

Basic Theory and Solar Energy Potential

2.1 Basic Related Theory

2.1.1 Solar Energy

General information about solar power is found in the following references [Duffie and Beckman, 1991] [Markvart, 2000]. The sun radiates energy radially, from an effective surface temperature of about 5760 K, as electromagnetic radiation known as 'solar energy' or sunshine.

The earth is situated at about 150 million km from the sun with a total surface area of about 510 million km², of which only about 21% is land. A substantial portion of the solar radiation, on its way to reaching the earth's surface, is attenuated due to atmospheric interventions.

Additionally, because of the sun-earth angle concept, the solar radiation received at the earth's surface varies on hourly, daily, or monthly basis. Hourly variation is due to the motion of the sun from east to west, and also due to the presence of clouds, whereas daily variation and monthly (seasonal) variation is due to the position of the sun.

Longitude and latitude give the location of a place on the earth's surface. The Sun comes overhead twice a year in the tropical belt. Ethiopia is in the equatorial region which is probably the most favorable region for solar energy. According to the findings of this work, disregarding the rainy season, July and August, the average daily duration of sunshine is approximately 8-10 hours [Bekele and Palm, 2009a].

It is well known that most developing countries do not have properly recorded radiation data. What usually available is sunshine duration data. Solar radiation data is the best source of information for estimating the

solar energy potential of a certain location, which is necessary for the proper design of a solar energy conversion system.

Ethiopia is one of the developing countries without properly recorded solar radiation data and, like many other countries, what is available is sunshine duration data. However, given a knowledge of the number of sunshine hours and local atmospheric conditions, sunshine duration data can be used to estimate monthly average solar radiation, with the help of empirical equation 2-1 [Duffie and Beckman, 1991].

$$\overline{H} = \overline{H}_0 \left(a + b \frac{\overline{n}}{N} \right) \quad \text{Eq. 2-1}$$

where:

- $\overline{H}$ = the monthly average daily radiation on a horizontal surface (MJ/m²)
- $\overline{H}_0$ = the monthly average daily extraterrestrial radiation on a horizontal surface (MJ/m²)
- $\overline{n}$ = the monthly average daily number of hours of bright sunshine
- N = the monthly average of the maximum possible daily hours of bright sunshine (i.e. the day length of the average day of the month)
- a** and **b** are regression coefficients

Solar radiation, known as extraterrestrial radiation, H_0 , on a horizontal plane outside the atmosphere, is given by equation 2-2.

$$H_0 = \frac{24 * 3600 * G_{sc}}{\pi} \left(1 + 0.033 * \cos \left(\frac{360 n_d}{365} \right) \right) * \left(\cos \phi \cos \delta \sin \omega_s + \frac{\pi \omega_s}{180} \sin \phi \sin \delta \right) \quad \text{Eq. 2-2}$$

where

- n_d = the day number,
- $G_{sc} = 1367 \text{ W/m}^2$, the solar constant,
- ϕ = the latitude of the location,

δ = the declination angle given as

$$\delta = 23.45 \sin\left(360 \frac{248 + n_d}{365}\right) \quad \text{Eq. 2-3}$$

ω_s = is the sunset hour angle given as

$$\omega_s = \cos^{-1}(-\tan\phi \tan\delta) \quad \text{Eq. 2-4}$$

The maximum possible sunshine duration N is given by

$$N = \frac{2}{15} \omega_s \quad \text{Eq. 2-5}$$

Equations (2-2) and (2-5) are used to calculate the extraterrestrial radiation and the maximum possible daily hours of bright sunshine respectively at the specified locations.

The regression coefficients **a** and **b** for M number of data points can be calculated from the following equations (2-6) and (2-7).[Nguyen and Pryor, 1997]

$$a = \frac{\sum \frac{\bar{H}}{\bar{H}_0} \sum \left(\frac{\bar{n}}{\bar{N}}\right)^2 - \sum \frac{\bar{n}}{\bar{N}} \sum \frac{\bar{n}}{\bar{N}} \frac{\bar{H}}{\bar{H}_0}}{M \sum \left(\frac{\bar{n}}{\bar{N}}\right)^2 - \left(\sum \frac{\bar{n}}{\bar{N}}\right)^2} \quad \text{Eq. 2-6}$$

$$b = \frac{M \sum \frac{\bar{n}}{\bar{N}} \frac{\bar{H}}{\bar{H}_0} - \sum \frac{\bar{n}}{\bar{N}} \sum \frac{\bar{H}}{\bar{H}_0}}{M \sum \left(\frac{\bar{n}}{\bar{N}}\right)^2 - \left(\sum \frac{\bar{n}}{\bar{N}}\right)^2} \quad \text{Eq. 2-7}$$

Results estimated in this way can be further improved by comparing them with data which can be obtained from sources such as NASA's surface solar energy data set or the Meteonorm global meteorological

database for applied climatology. The comparison can be made using the root mean square error (RMSE) formula given in equation 2-8.

$$\text{RMSE } (\%) = 100 \left(\frac{1}{\overline{H}_{ob}} \right) \left(\sum \frac{E_i^2}{M} \right)^{1/2} \quad \text{Eq. 2-8}$$

where:

$$\begin{aligned} E_i &= H_{estimated} - H_{obtained} \\ i &= 1, 2, \dots, M \end{aligned} \quad \text{Eq. 2-9}$$

M = the total number of observation points and
 $\overline{H}_{ob}$ = the arithmetic mean value of the obtained data

NB. The subscript “obtained” refers to the data obtained from Meteoronorm [Meteoronorm, Ver. 5.1x] and NASA [NASA, 2008] and also the measured data for Addis Ababa, obtained from NMSA.

The correlation of the radiation levels calculated based on the different models given by different authors is compared against the radiation level obtained in this work. The correlation coefficient, r , by which the depth of the correlation of the two radiation levels is compared is given by another statistical formula (Eq. 2-10) [Nguyen and Pryor, 1997].

$$r = \frac{\sum (H_{estimated} - \overline{H}_e)(H_{obtained} - \overline{H}_{ob})}{\sqrt{\sum (H_{estimated} - \overline{H}_e)^2 \sum (H_{obtained} - \overline{H}_{ob})^2}} \quad \text{Eq. 2-10}$$

where:

$\overline{H}_e$ = the arithmetic mean value of the M estimated values of the solar radiation and

$\overline{H}_{ob}$ = the arithmetic mean value of the M obtained values of the solar radiation

2.2 Solar Energy Potential

There are a couple of previous studies concerning solar energy distribution across the country [ENEC, 1986] [Drake and Mulugetta, 1996]. They both provided a considerable set of results on a countrywide basis. However, differences can be observed between these and the results achieved in this study [Bekele and Palm, 2009a]. This study, as mentioned earlier, focuses on finding the solar energy potential of the four selected locations.

Unfortunately, as in the case of the wind, there is no properly recorded radiation data available throughout the country, except for one particular location, Addis Ababa. It is clear that the best source of information for evaluating the solar energy potential of a given location is radiation data. However, where no such data is available, empirical relationships involving information on sunshine duration, temperature and cloudiness can be used to determine the potential.

The data used as a basis for this study is sunshine duration data recorded by NMSA for a period of more than 10 years at the selected locations. The data was recorded relatively recently, from the early 1990s to 2003. Radiation measurement is taken using an Eppley model pyranometer and this is done at one of the sites, Addis Ababa. A Campbell-Stokes sunshine recorder is used to measure the sunshine duration [Mulugeta, 1996]. This instrument is widely used in many countries and probably it is the most common type of sunshine recorder in use today. The unit is designed to record sunshine duration by burning a hole through a card. The advantages of using this instrument are its simplicity and ease of use. Furthermore, it requires less maintenance, as there are no moving parts within the instrument. The disadvantage of this instrument is its inability to burn a hole in the card when the sun is low in the sky. Thus, it can be said that it only measures the amount of bright sunshine, not visible sunshine. Reading the cards is another major problem, as the presence and absence of clouds affects the amount of burn on the card.

Regression coefficients, developed by other authors for locations of similar climatic conditions, together with equations related to solar energy given earlier, Eq.2-1 to Eq.2-10, have been used to determine the potential.

Furthermore, radiation data obtained from the Global Meteorological Database for Solar Energy and Applied Meteorology [Meteonorm, Ver. 5.1x] and the renewable energy resource web site, sponsored by NASA's Science Mission Directorate, Earth-Sun System Division, and Applied Sciences Program [NASA, 2008] have also been checked against the results obtained. The deviations of the comparisons have been evaluated using Root Mean Square Error (RMSE)(Eq. 2-8). The details of this work is explained thoroughly in the attached paper [Bekele and Palm, 2009a]. Figure 2-1 shows the results obtained. Regression coefficients relevant to each of the locations have also been calculated using equations 2-6 and equation 2-7. In general, the findings clearly indicate that the available solar energy potential is excellent.

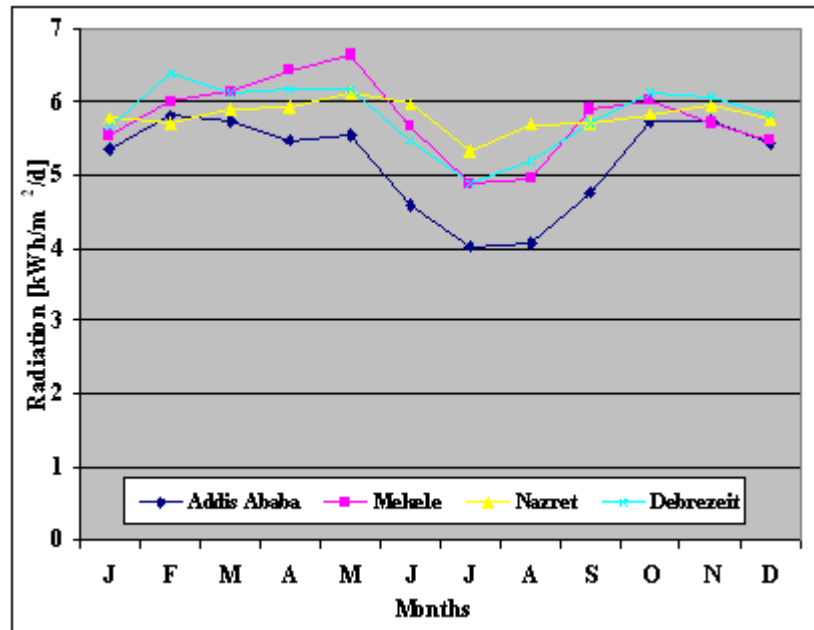


Figure 2-1 Global solar radiation of the locations on a horizontal surface

3 PART III:

Basic Theory and the Hybrid System

As previously introduced, the hybrid system studied is one combining solar and wind with diesel generator(s) and a bank of batteries, which is included for backup purposes. Power conditioning units, such as converters, are also a part of the supply system. It is conceivable that a solar/wind hybrid system has numerous advantages. One of the advantages is reliability; when solar and wind power production resources are used together, reliability is improved and the system's energy service is enhanced. What this means is that in the absence of one type of energy another would be available to carry out the service, and, as a result the size of the battery storage can be reduced. Illustrative schematic diagram of the set up is given in figure 3-7 of section 3.2.1

Other advantages are the stability and immobility of the system (fewer moving parts) and a lower maintenance requirement, thus reducing downtime during repairs or routine maintenance. In addition to this, as well as being indigenous and free, renewable energy resources also contribute to the reduction of emissions and pollution.

The operational concept of the hybrid system is that renewable resources are the first choice for supplying load and any excess energy produced is stored in the battery. The diesel generator is a secondary source of energy. Electronic controller circuitry is used to manage energy supply and load demand.

The main actors, or elements, of the hybrid system are the wind turbine and the PV generators. Diesel generator(s), a battery system, and an inverter module are additional parts of the system. In the following sections the basic principles of these components will be discussed. In addition to the theoretical notes considered as a background literature survey is also part and parcel of the foundation of this work. It is known that researchers have been working in the area of standalone hybrid system since long and numerous research results for a variety of

applications have been published. For this particular work attention is given towards those concerning African countries.

In one of African countries, Cameroon, Off-grid generation options for remote villages have been simulated for a load of 110 kWh/day and 12 kWp [Nfah EM, et al, 2008]. In the research HOMER is used to simulate the energy cost of the different design options. The study is based on solar, hydropower, diesel generator, and battery sources. Using load data to which an hourly and daily noise of 5 % is added and also based on hydro and solar resources data, the levelized costs of energy for different renewable energy options have been calculated and the levelized cost of energy found was 0.296 Euro/kWh. This cost is for a micro-hydro hybrid system comprising a 14 kW micro-hydro generator, a 15 kW LPG (liquefied petroleum gas) generator and 36kWh of battery storage.

In another simulation comprising of photovoltaic (PV) hybrid systems, an 18kWp PV generator, a 15 kW LPG generator and 72kWh of battery storage the levelized cost was also found as 0.576 Euro/kWh for remote petrol price of 1 Euro/l and LPG price of 0.70 Euro/m³. The authors concluded that both simulation options prove to be the cheapest depending on where the location is within the country.

In the same country, Cameroon, another research is carried out by modeling solar/diesel/battery hybrid power systems for typical rural households and schools [Nfah EM, et al, 2007]. Based on hourly solar radiation computed from the global horizontal solar radiation, the average daytime temperature, and parameters of selected solar modules the monthly energy production of the modules was computed. As a result, the selected solar modules rated power in the range 50–180 Wp produced energy in the range 78.5–315.2 kWh/yr. With the energy produced by the solar module a hybrid power system comprising of solar/diesel/battery to meet the energy demand of typical rural households in the range 70–300 kWh/yr is modeled. The supply to the secondary school has been found to be 2585 kWh/yr from 1440Wp solar array and a 5kW single-phase generator operating at a load factor of 70%. In the study cost analysis is not treated.

Another study conducted in another part of Africa, Algeria, presents techno-economic assessment for off-grid hybrid generation systems with an aim of achieving a share of 10–12 % renewable energy sources in primary energy supply by 2010 [Himri Y, et al, 2008]. The model used to evaluate the energy production, life-cycle costs and greenhouse gas emissions reductions is HOMER. The aim of the study is to perform an economical feasibility study of adding wind turbine to an existing grid-

connected diesel power plant supplying energy to a remotely located village in order to reduce the diesel consumption and environmental pollution.

The authors concluded that for wind speed less than 5.0 m/s the diesel power plant would be feasible solution over the range of fuel prices used in the simulation (0.05-0.179 \$/L) and the wind diesel hybrid system becomes feasible at a wind speed of 5.48 m/s or more and a fuel price of 0.162\$/L or more. This is for a case where the carbon tax is not taken into consideration and subsidy is not abolished otherwise the hybrid system will become feasible, according to the authors.

A study by Magda Moner-Girona proposes an alternative approach to the promotion and support schemes of renewable energy technologies in isolated areas based on the generation of renewable electricity [Magda Moner-Girona, 2009]. The study presents evaluation of the renewable energy premium tariff (RPT) scheme, a locally adapted Feed-in Tariff modified for decentralized grids of developing countries that motivates the operation of renewable energy technologies by paying for renewable electricity generated. In the study it is deduced that a good quality performance is attainable as the support is given based on the renewable electricity production and not on the initial capital investment.

The support scheme has been designed to provide a cost-effective method for the introduction of renewable energy technologies to remote villages, to provide sustainable and affordable electricity to local users, to make renewable energy projects attractive to policy-makers, and concurrently decrease financial risk to attract private sector investment.

Energy situation for Ruanda was published quite recently [Safari B, 2009]. Ruanda is a neighboring country with all the power shortage problems similar to Ethiopia and has been experiencing energy crisis. According to the article, the reason for that is lack of investment in the energy sector. It also mentions that the population growth and increasing industrialization in urban areas, existing hydro and thermal power plants energy supply is increasingly scarce with high energy costs, and energy instability.

Just similar to Ethiopia, wood fuel is being the most important source of energy in the country and the author predicts that the dependence on it will continue to impact on the process of environmental degradation.

The author states Rwanda as rich in renewable energy resources such as methane gas, solar, biomass, geothermal etc. which is more or less are the same as resources in Ethiopia and that the Government is working towards the development of the rural energy through alternative energy projects where access to national grid is still difficult. The paper in general presents a review of existing energy resources and energy applications together with the recent developments on renewable energy.

Wind-diesel-battery hybrid system modeled for north Cameroon is another African based study assessed in the literature survey [Nfah EM, Ngundam JM, 2008]. The objective of the study is to electrify households and schools in remote areas of Cameroon from Wind/Diesel/battery hybrid power system. The wind resource used is of the period 1991–1995 and the diurnal pattern is in the range 3–6 m/s from 9:00 to 15:00 for eight months. This is more or less the objective of this study and the data also looks to be the same for most regions of Ethiopia.

In the study it has been found that two wind turbines with power rating of 180 W and 290W were found to be enough for the hybrid system for typical rural households energy needs in the range 70–300 kWh per year. Another combination consisting of two wind turbines rated as 290W and a 5 kW single phase generator requiring only 106 generator hours/yr has been found to supply 2585 kWh/yr or 7 kWh/day load to a typical secondary school. Indeed the load is much smaller than what has been suggested for this study however the general idea of the design resembles a lot.

Several other articles have been studied and those mentioned here are results related to the different parts of Africa.

3.1 Basic Theory of the Hybrid System Components

3.1.1 Photovoltaics

The theoretical note under this subtopic is primarily based on the reference material [Duffie and Beckman, 1991] [Markvart, 2000]. Photovoltaic (photo = light; voltaic = produces voltage) or PV systems convert light energy directly into electricity using semiconductor technology (see figure 3-1).

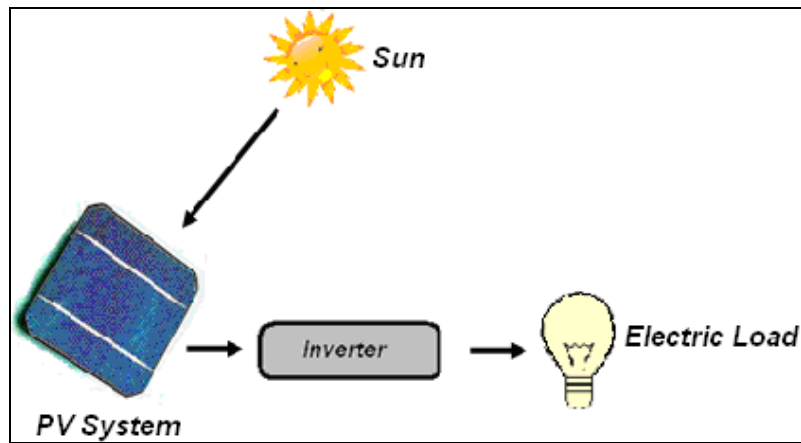


Figure 3-1 Light energy converted to electricity through PV system

The most basic power conversion unit of a photovoltaic (PV) system is the solar cell. As shown in figure 3-1, sunlight strikes a PV cell and a direct current (D.C.) is generated. An inverter inverts the D.C. to an Alternating Current (A.C.) and by connecting the electric load to the output terminals the current can be utilized.

Currently, there are many different types of solar cell available on the market with the proportion given in figure 3-2

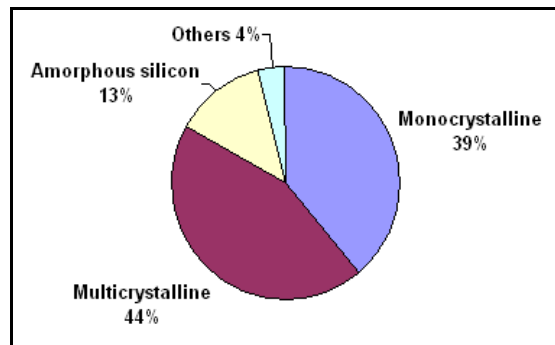


Figure 3-2 Proportion of PV technologies on the market [Markvart, 2000]

The intensity of light energy determines the amount of electricity generated. In other words, the conversion of energy relies on the quantum nature of light, whereby we perceive light as a flux of particles - photons - which carry the energy, E_{ph} , as given by equation 3-1.

$$E_{ph}(\lambda) = \frac{hc}{\lambda} \quad \text{Eq. 3-1}$$

where:

h = the Planck constant,
 c = the speed of light (m/s), and
 λ = the wavelength of light (m).

On a clear day, approximately 4.4×10^{17} photons strike a square centimeter of the earth's surface every second. Those photons with energy in excess of the band gap energy of the semiconductor material being used can be converted into electricity by the solar cell.

A rough estimate of the current that can be generated by a solar cell is given by equation 3-2. Ignoring losses in the cell, and assuming each photon produces one electron charge, for an electron charge of 1.6×10^{-19} coulomb, and 4.4×10^{17} photons striking a square centimeter of cell area, the current density is approximately 70 mA/cm².

$$I_L = qNA \quad \text{Eq. 3-2}$$

where N is the number of photons, A the area exposed to light, and q the charge in coulomb

The maximum voltage, V, that a solar cell can generate is equal to the band gap of the semiconductor in use and is expressed in electronvolts. This means that the separation of electrons and holes at the terminals of the solar cell can only continue until the electrostatic energy of the charges after separation, E_g , equals to the pair energy in the semiconductor. Hence, the maximum voltage is given by equation (3-3). In other words, the maximum voltage that can be generated by a solar cell is numerically equal to the band gap of the particular semiconductor in use expressed in electronvolts [Markvart, 2000].

$$V = E_g / q \quad \text{Eq. 3-3}$$

A diagram of a typical solar cell is given in figure 3-3. The current generated is extracted via contacts on the front and rear sides of the cell. A thin layer of dielectric material, known as an anti-reflection coating or ARC, covers the cell to minimize light reflection from the uppermost surface.

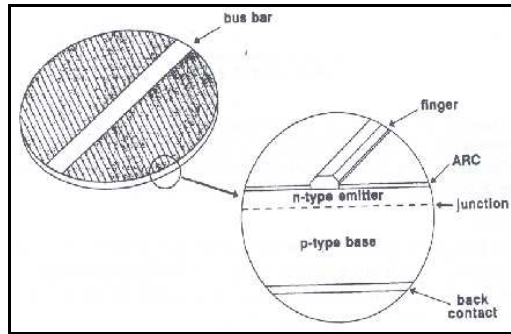


Figure 3-3 A typical silicon solar cell [Markvart, 2000]

The assessment of solar cell operation or the design of solar-cell-based power systems requires an understanding of the electrical characteristics or voltage-current relationship of the cell under various temperature and radiation levels. A typical model or equivalent circuit for a solar module is given in figure 3-4. For practical operation, solar cells are usually assembled into modules consisting of several cells or an array consisting of several modules.

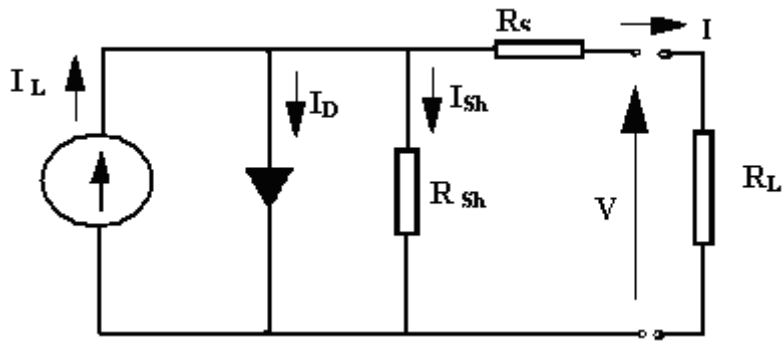


Figure 3-4 A solar cell equivalent circuit [Duffie and Beckman, 1991]

The equation governing the I-V characteristics of the cell in the figure is given by Eq.3-4.

$$I = I_L - I_0 \left[\exp \left(\frac{q(V + IR_s)}{mKT} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}} \quad \text{Eq. 3.4}$$

$$= I_L - I_D - I_{sh}$$

where:

- I_D = the diode current (A)
- I_{sh} = the shunt resistance current (A)
- I = the load current (A)
- I_L = current produced by the cell (A)
- I_0 = reverse saturation current of the diode (A)
- q = charge on an electron (C)
- V = output voltage (V)
- K = Boltzmann's constant
- T = working temperature of the cell in (K)
- m = the diode quality factor

The current-voltage (I-V) and power-voltage (P-V) characteristics of a typical PV module corresponding to equation 3.4 are shown in figure 3-5 with the short circuit current at radiation level G, $I_{sc}(G)$, the maximum power current (I_{mp}), maximum power point (P_{mp}), maximum power voltage (V_{mp}), and open circuit voltage (V_{oc}) labeled at their respective points.

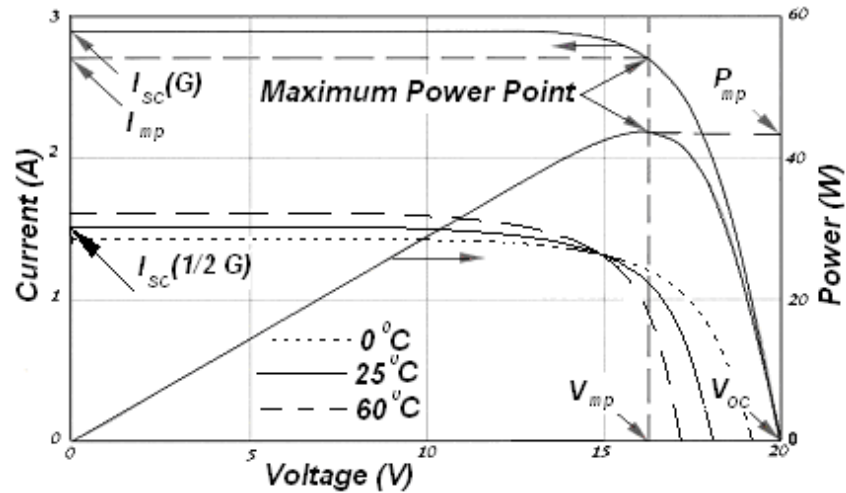


Figure 3-5 I-V and P-V sketches for a typical PV module

P_{mp} is the maximum power that can be obtained from the module and it corresponds to the maximum rectangular area under the I-V curve. V_{oc} increases logarithmically, whereas I_{sc} increases almost in proportion to the radiation as long as the current axis does not intersect the curved portion of the I-V characteristic [Duffie and Beckman, 1991]. In the figure it is worth taking a note of the effect of temperature upon the current generated by the solar cell (module).

PV cells are the fundamental building blocks of PV systems. The power generated by a PV cell is only enough to power small devices such as electronic calculators. Each silicon solar cell produces about 0.5 volts. To meet higher loads the PV cells must be connected in series and/or in parallel depending on the magnitude of the voltage and current required. Typically, 36 cells are connected in series to form a module which is capable of producing enough voltage to charge 12 volt batteries and run pumps and/or motors. It is important to note that losses of voltage occur due to the temperature rise of the cells in the heat of the sun and also that a 12 V battery typically needs about 14 V in order to be charged.

Modules are the basic building blocks of systems. For more voltage or current modules are connected in series or in parallel respectively to form a panel and then panels can be assembled into a group to form a complete PV array.

The power output of a PV system is given by equation (3.5) [Duffie and Beckman, 1991].

$$P = A_c \eta_{mp} \eta_e G_T \quad \text{Eq. 3-5}$$

where:

- A_c = the array area
- η_{mp} = the maximum power point efficiency of the array ($\approx 14\%$)
- η_e = the efficiency of power conditioning equipment ($\approx 90\%$)
- G_T = the incident solar radiation on the array

PV technology has numerous advantages:

- the energy resource is free, renewable, and inexhaustible everywhere

- it counterbalances greenhouse gas and pollutant emissions from fossil fuel-based generation, and toxic waste from nuclear generation
- it produces energy during the daytime when demand peaks, which is when power is typically most expensive to produce
- after the initial setup, the facilities can operate with little maintenance
- operating costs are extremely low compared to other power technologies
- it is highly reliable and generates electricity at the actual point of use
- PV systems are easily expandable, allowing for initial set-up even with a small budget and the addition of more modules in the future, when convenient.

With regard to the application of PV in a country such as Ethiopia and more generally in the developing world, where there is abundant sunlight, a large rural population and where there is no proper infrastructure nor resources to develop an electrical grid, it should be a very attractive option.

Considering the case of Ethiopia, solar energy is available almost everywhere across the country than other renewable resources, such as hydro and/or wind. Where wind and hydro are available, they too are good sources of energy, but only selected areas have good wind and hydro power potential is not evenly distributed either. The impact that water storage for hydropower may have on the already meager availability of arable land is also a concern that needs to be addressed before embarking on the process of harnessing this resource.

3.1.2 Diesel Generator

A diesel generator is simply a normal electric generator driven by a diesel engine (prime mover). An electrical generator is an electromechanical system that converts mechanical energy into electrical energy through the interaction of electromagnetic and electrostatic fields within the system.

Figure 3-6 shows the per-phase equivalent circuit of a synchronous generator driven by a prime mover, which in this case is the diesel engine. T is the mechanical torque of the prime mover, E_a is the internal voltage generated, I_a is the armature current, $R_a + jX_s$ is the synchronous impedance, and V_a is the load voltage.

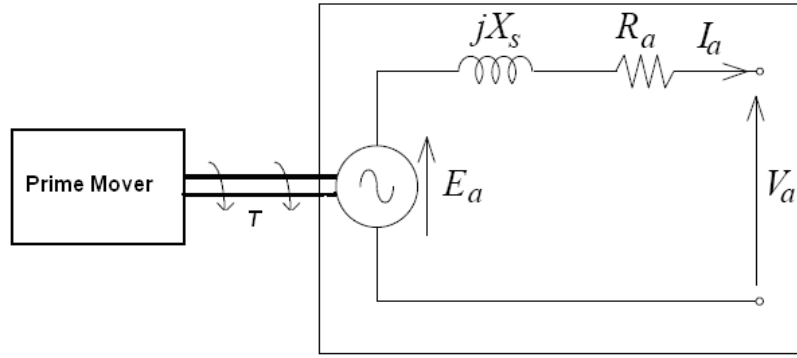


Figure 3-6 the per-phase equivalent circuit of a synchronous generator driven by a diesel generator (prime mover)

With the generator under load the voltage V_a is given by equation 3-6.

$$V_a = E_a - (R_a + jX_s)I_a \quad \text{Eq. 3-6}$$

The winding resistance, R_a , is generally much smaller than the synchronous reactance, jX_s . Hence, equation 3-6 can be rewritten as in equation 3-7.

$$V_a = E_a - jX_s I_a \quad \text{Eq. 3-7}$$

Equation 3-7 is the per-phase terminal voltage of the generator.

The electrical power output is given by equation 3-8. .

$$P_{\text{out}} = mV_a I_a \cos \phi \quad \text{Eq. 3-8}$$

where:

$m =$ the number of phases; $m=3$ for a 3-phase generator
 $\cos \phi =$ the cosine of the angle between the voltage V_a and the current I_a .

3.1.3 Inverter

An inverter is an electrical device that converts DC power to AC power at a desired output voltage or current. Its typical application is to convert DC power from a battery or PV array into AC power for use with conventional, utility-powered household appliances.

Basically, there are three kinds of DC-AC inverters; square wave, modified sine-wave, and pure sine wave inverters. Of the three, the square wave type is the simplest and least expensive, but with the poorest quality output signal. The modified sine wave type is suitable for many load types and is the most popular low-cost inverter. Pure sine-wave inverters produce the highest quality signal and are used for sensitive devices such as medical equipment, laser printers, stereos, etc.

The working principle of most inverters is to use a low DC voltage input and to first step-up the voltage to a level corresponding to that of the peak value of the desired AC voltage and then generate the desired AC voltage by using a full-bridge or half bridge electronic circuit configuration. The output voltage of the inverter is controlled by electronic circuitry.

3.1.4 Battery

Batteries are a key component in a stand-alone renewable energy system. Basically, a battery is a device that stores energy for later use. It is a combination of electrochemical cells that can store chemical energy that has the potential to be converted into electric voltage or, to put it simply, it is a device that converts chemical energy directly to electrical energy.

Lead-acid battery is the type of battery commonly used in stand-alone power systems. Batteries can be classified in two ways: by their application (the way they are used) and their construction (how they are built). The major construction types are flooded (wet), gelled, and AGM (Absorbed Glass Mat). The construction aspect is beyond the scope of this work. With regard to their applications, the major ones are automotive (starting), marine, and deep-cycle. Deep-cycle batteries are used in renewable energy applications.

In terms of the automotive (starting) type, it is designed to provide a large amount of current for a short period of time. To achieve a sufficiently large amount of current, car batteries use thin plates in order

to increase the surface area. Such batteries are not suitable for storing the energy that PV or hybrid systems produce.

On the other hand, Deep Cycle batteries are designed to be discharged as low as 80 % and recharged over and over again and therefore have much thicker plates. It is important to note that companies recommend that hybrid system batteries should not be discharged beyond 50% of their capacity. Deep Cycle batteries deliver a consistent voltage as the battery discharges.

The Marine type is usually hybrid and falls somewhere between the starting and deep-cycle battery types. The plates may be composed of lead sponge, but it is coarser and heavier than that used in starting batteries.

With regard to the effect of temperature on batteries, the battery capacity is reduced as the temperature goes down, and is increased as the temperature goes up. The standard rating for batteries is 25 °C. Battery Ah capacity drops to 50% at approximately -27 °C. At freezing ($\approx 0^{\circ}\text{C}$), capacity is reduced by 20%.

3.2 Feasibility Study of the Hybrid System

Based on the theoretical background discussed thus far, the feasibility study into the establishment of a standalone solar-wind hybrid electric energy supply system to a model community of about 200 families, with five to six family members in each, at four typically selected areas was carried out.

The locations are Addis Ababa, 09° 02'N, 38° 42'E, 2408 m (AMSL); Mekele, 13°33'N, 39° 30'E, 2130 m; Nazret 08°32'N, 39° 22'E, 1690 m; and Debrezeit, 08°44'N, 39° 02'E 1850 m. The results of the study for one of the four selected areas, Addis Ababa, which is published as a paper [Bekele and Palm, 2009c] is appended towards the end of the thesis. Similarly, another results table for another paper written based on the solar and wind energy potential of Mekele [Bekele and Palm, 2009d] is also appended at the end of the thesis. The purpose of the second paper [Bekele and Palm, 2009d], as it was highlighted in a previous section, is to supply electric lighting and potable water to community resettled by the government and the approach followed is to compare products in the market and use the most efficient and up-to-date

technologies for the components used in the hybrid system. Design on individual basis is also investigated in the same paper. Further details about the results of the two papers will be given in a later section. The outcomes of the study for the remaining three locations is discussed in the subsequent sections.

3.2.1 The Model and the Hybrid Setup

The Hybrid Optimization Model for Electric Renewables (HOMER) software is used as a tool to carry out the research. As mentioned earlier, the main objective of the research is to assess the feasibility and economic viability of utilizing hybrid PV–Wind–diesel–battery based standalone power supply systems to meet the load requirements of the hypothetical community specified earlier.

A schematic diagram of the standalone hybrid power supply system sought is shown in figure 3-7 and its representation by HOMER is shown in figure 3-8. The power conditioning units are DC-DC and AC-DC converters, with the sole purpose of matching the PV and wind turbine generated voltages to that of the bus voltage at the DC centre. The AC load is of both primary and deferrable types.

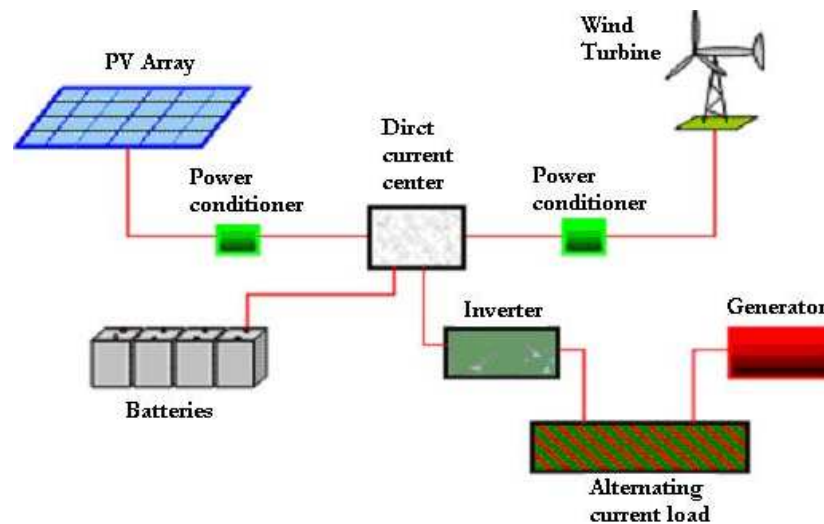


Figure 3-7 General schemes for the standalone hybrid power supply system

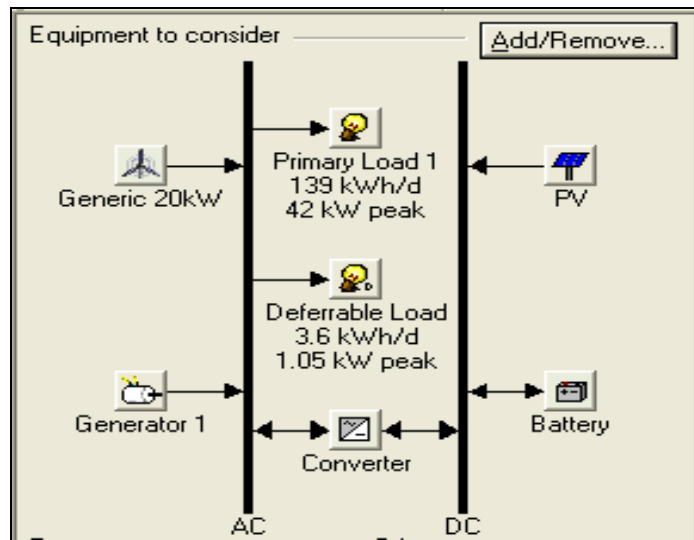


Figure 3-8 HOMER diagram for the hybrid PV-wind-gen-battery-converter set-up

HOMER requires input information in order to analyze the system and to give the feasible set-ups. One such piece of information is the electrical load condition, which the hybrid system is expected to supply.

3.2.2 Introducing HOMER

A brief note about the Hybrid Optimization Model for Electric Renewables (HOMER) software, what it can do and where it can be applied will be outlined in this section. According to its website, the software is copyrighted by the Midwest Research Institute (MRI) and provided by the National Renewable Energy Laboratory (NREL) operated by MRI for the U.S. Department of Energy (DOE).

The software, in its complete form, is provided for free. This includes information about websites providing data sources, such as for solar radiation and wind speed, and also information about sources of power components such as wind turbines, generators, batteries, etc.

In short, HOMER helps find the least cost combination of components that meet a required load, based on an hourly analysis of the input variables, such as wind and solar data. For systems that meet the yearly load, the life-cycle cost is also estimated by the software.

HOMER can be applied to a number of system designs: grid-connected or off-grid, stand-alone or distributed generation and conventional or renewable technologies. The renewable technologies can be classified into three groups:

- Power sources
- Storage and
- Loads

In the category of power sources, the most common types are the following: photovoltaic (PV); wind turbine; hydro power; generators with different prime movers such as diesel, bio-gas, or coal-fired; electric utility grid; micro-turbine or fuel cell. In the storage class, bank of batteries and hydrogen can be mentioned. With regard to loads, there are two types: primary and deferrable loads. The Primary load is the electrical load that must be met at a specified time (e.g. lighting) and the deferrable load is the load that need not be met within a specified time but should be met within a certain time period (e.g. water pumping).

The software is sufficiently intelligent to identify the proper timings of energy supplied to the components which are connected to the system. For systems which include batteries and a generator, the software can decide the times at which the batteries should be charged and when the generator should be operated. It also gives the deferrable load a lower priority than the primary load but a higher priority than charging the batteries. For this there are two dispatch strategies that HOMER follows. A dispatch strategy is a set of rules by which the operation of the generator and the batteries is controlled whenever there is a shortage of energy from the renewable resources. There are two types of dispatch strategy; load following and cycle charging. The load following strategy enables HOMER to serve the deferrable load under two conditions. These are a) when the storage tank is empty and b) when the system produces excess electricity. Under a cycle charging strategy, the generator serves the deferrable load when it is able to produce more electricity than that needed by the primary load. If the storage tank is empty, then the deferrable load is considered a primary load and all the power sources serve the deferrable load as much as possible.

The software also provides a feature for carrying out sensitivity analysis, which enables the evaluation of the economic and technical feasibility of several technological options. This feature can also be used when there is doubt over the exact value of a certain input, such as the annual average wind speed, annual average solar radiation, diesel price or the price of

PV cells. Furthermore, when the data represents a range of applications this feature can be used. The sensitivity analysis performs energy balance calculations on hourly basis for a whole year (8,760 hours). It compares the electric load for each hour to the energy that the system can supply during that hour.

While carrying out the sensitivity analysis, the optimization process is repeated for each input value in the range so that the effect that changes in the value have on the results can be examined. As many sensitivity variables as required can be specified, and the results are displayed in the form of a graph.

Nonetheless, this is not as simple as it may seem; as the number of sensitivity variables increases, the computational time of the software also increases, which could be considered a limitation or a challenge to using the software. When using the sensitivity analysis feature of the software, several sizes of each component must be considered in order to meet the load and the computation time is dependent on how many of them are used. To minimize the computation time an iterative process need to be followed. This is done by first considering just a small number of sizes and/or variables over a relatively large range to decrease the initial running time. After each successive run, a greater number of options and variables is added within the range in order to increase the resolution. Indeed, this takes quite a long time and can be considered as a limitation of the software.

Furthermore, similar to the several sensitivity variables that can be input into the sensitivity table, there is another search space table to which can be applied different sizes and quantities of the different system components, such as the size of PV array, generator, inverter and the quantities of batteries and wind turbines. Again, this further lengthens the computation time of the software, as the software tries each and every component size and quantity. As in the previous case, the simulation can first be run coarsely, by minimizing the number of variables within the range. The results are then refined by adding a greater number of variables within the range. It should also be noted that the greater the number of variables within a certain range supplied to HOMER, the better the result. However, attention needs to be paid to the computation time.

After running the simulation, the results are given as a list of feasible system configurations, sorted by life-cycle cost. From the results, the least-cost systems, which are displayed in the first few rows of the list, can be chosen for implementation. The designer can also scan for other

feasible systems in the list and decide to take any particular setup by evaluating the pros and cons of the setup against cost, renewable resource contribution, future price trend of the components, etc.

When using the software, the first thing to do is to formulate the problems to be addressed. What is studied in this work, as mentioned previously, concerns the supply of electricity to a model community located at a remote area, using mainly renewable resources. By considering the magnitude of the solar and wind resources, different combinations of set-ups have been evaluated. As all the locations are rich in solar radiation the first set-up scenario investigated was the cost-effectiveness of using only PV together with a diesel generator. Another was to see how the cost would change if a wind turbine was added to the PV-generator system, and so on. Once the problems were established then the components to be included in the system design, the quantities, and the sizes, could be identified (Table 3-3).

Following on from this, the technology options, component costs, and available resources were input to the software. HOMER used the inputs to provide different feasible system configurations, which were sorted according to their net present cost.

3.2.3 Electric Load

Deciding on the load is one of the most important steps in the design of a hybrid system. The community is equipped with a school and health post. Primary load, which must be met immediately, and deferrable load, which must be met within a certain time (the exact timing of which is not important), are both considered. The electrical loads are lighting, water pumping, a radio receiver, and the electric supply necessary for some clinical equipment.

The calculation of the load or, in general, the system design solution is approached in three different ways. The first is to include in the hybrid system those components which are locally available without considering their efficiencies and calculate the load. This is done for both the load and the supply side. The second approach is to thoroughly survey the market for the latest and most efficient types of system components and calculate the load [Bekele and Palm, 2009d]. The third approach in the effort of cost minimization is to see if a self-contained type of design can be effective i.e., on individual basis where every household will have its own supply system [Bekele and Palm, 2009d].

In the first approach, for the deferrable load, seven water pumps, one for the school and health-post and the remainder for household use, are assumed. Each water pump has a 150 W power rating, with a pumping capacity of 10 l/m. The pumps supply 20000 l/day for the 200 families (100 liters per family); and 2400 l/day for the school and clinic. The average deferrable load (total consumption of electricity by the pumps) is calculated to be 5.4 kWh/day for the households and 0.6 kWh/day for the school and health post, totaling 6 kWh/day. The peak deferrable load (rated power of the pumps) is 0.9 kW for the households and 0.15 kW for the school and health post. A sufficient water storage capacity for 4 days is assumed and the corresponding electricity storage capacity is 20 kWh for the households and 2.4 kWh for the school and health post.

The proposed primary load per household is a 5 W night light, a 3 W radio receiver and two 60 W light bulbs to be used between 18:00 and 23:00 in the evening and the daily consumption is calculated to be approximately 138 kWh.

Electric lighting for the school in the evenings (18:00-21:00) for those who wish to pursue basic education is suggested. For 4 classrooms with 4 lamps (energy saving type) of 40 watt capacity in each classroom and a lamp for a toilet is calculated to 2.04 kWh/day.

A typical two-room healthcare facility, equipped with vaccine refrigerator, light bulbs, stand-by communication VHF radio, microscope, and AM/FM radio receiver, is suggested. The assumption is that the “health post” will not provide health services constantly, with a permanent doctor or nurse present; instead a doctor or a nurse will periodically give treatment for minor illnesses and tend to any minor injuries. Patients with more serious problems will be referred to a nearby hospital. Hence, only the most basic health equipment is proposed and the daily consumption is calculated to be just 1 kWh .

The sum total of the daily energy consumption of the community is approximately 147 kWh.

Exceptions: July and August are the rainy months and the schools are closed. January is a semester break. During the rainy season, water consumption from the pumps is expected to be shared by river and rain water ponds. Assuming that this eases the load of the pumps by approximately 30 %, the daily load in these months would be 140 kWh and that in January 145 kWh. Thus the yearly load pattern is that given in Table 3-1.

Table 3-1 Monthly average daily electrical load (kW/h)

Months	Jan	Feb-May	June	July	Aug	Sep	Oct-Dec
Deferrable Load	6	6	5	4	4	5	6
primary load	139	141	141	139	139	141	141
Total Load	145	147	146	143	143	146	147

The 24 hour primary and deferrable load profiles are given in figures 3-9 and 3-10 respectively.

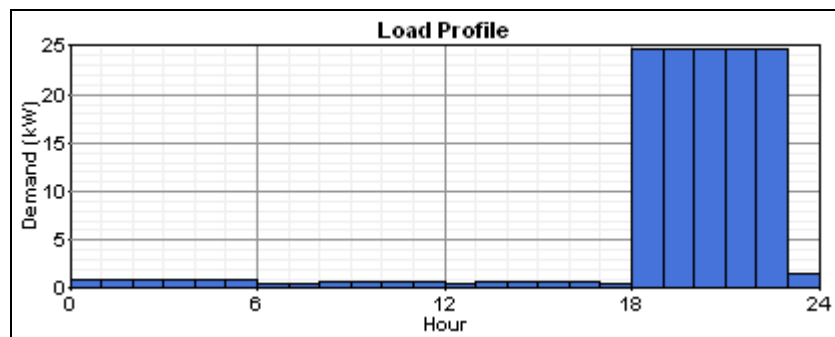


Figure 3-9 Primary load profile of the community

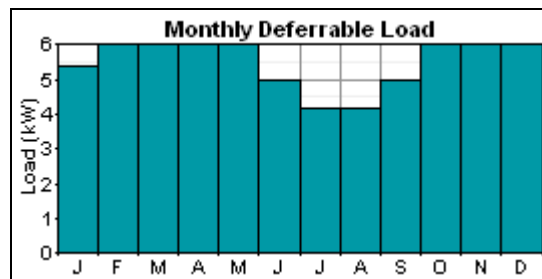


Figure 3-10 Monthly average deferrable load profile

3.3 Additional specifications input to the Software

Additional information input into HOMER is summarized in Table 3-3. The values given in this table are primarily chosen according to the size of the load for the assumed hypothetical community. The monthly average daily electrical load is given in Table 3-1. This load, in kWh/day, varies between 143 kWh/d and 147 kWh/d from month to month. In terms of variation of power on a daily basis, this is given in figure 3-9 for the primary load and figure 3-10 shows the community's deferrable load power demand on a monthly basis. These are the principal guidelines for selecting the size of the power components listed in Table 3-3. However, some of the components are also chosen by considering the local availability of the component. In this regard, the diesel engine driven generators selected can be mentioned as an example. The sizes of the available generators are 12, 17.6, 26.4 and 44 kW. The chosen generator size, 44 kW, was selected after repeatedly running the simulation, as it turned out to be the most cost effective in terms of minimizing the Net Present Cost. Smaller sized generators either couldn't supply the required load individually or in the case of more than one being required the net present cost was much higher.

The fuel for the generator engine is diesel oil and based on the minimal information obtained from the sole local vendor, and an additional web search, the generator is a Cummins brand and the model name is C55 D5. The capacity is 44 kW on standby, the engine type is 4BT.3G2, and the frequency of operation is 50 Hz. As the data for the fuel curve was not provided by the local vendors, the fuel curve calculation is based on fuel consumption data for a 50 kW generator obtained from another supplier [Generator, 2009] and from the software itself. Hence, the intercept coefficient (L/Hr/kW rated) used in the simulation is 0.06, while the value calculated for the 50 kW unit is 0.04; the corresponding slopes (L/Hr/kW output) are 0.25 and 0.24 respectively. It should be noted that the smaller the capacity of the generator, the higher the values of the intercept coefficient and the slope. Figure 3-11 shows the fuel efficiency curve of the selected generator.

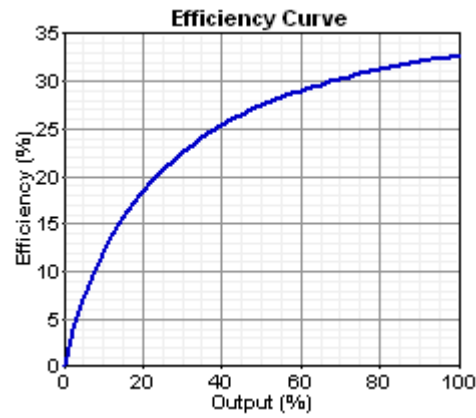


Figure 3-11 Fuel efficiency curve for the selected generator

Cost minimization is another criterion considered when selecting the components. One of the main criteria for the selection of the wind turbine is the cost. The wind turbines have been selected from different sources; the various wind turbine websites and those suggested by the HOMER program itself. Other selection criteria used are: the type of current they generate (AC or DC), how low the cut-in wind speed is, how expensive the wind turbine is and for what application the wind turbine be used for. The cut-in wind speed is one of the main criteria, as the wind resource at the sites is not very high. As the turbine price would also affect the total net present cost, this has been checked with the respective vendors. The type of current they generate, whether AC or DC, is also considered as this would have repercussions on the size of the inverter. A wind turbine that generates AC current has been chosen, as the load assumed for the households is of an AC type. As the aim of the research is to supply electric energy to remotely located communities, the wind turbines selected should be those which are applicable for home or off-grid use.

Based on the selection criteria mentioned above, different wind turbines have been tested by running the simulation several times. From those wind turbines which were candidates for this application, the Generic 20 type has been found to be the best in terms of the cut-in wind speed and also in respect to other criteria mentioned earlier. This wind turbine was selected from those suggested by HOMER. The turbine is a 20 kW turbine commonly available on the market such as the one for which the power curve is given in figure 1-6. The power curve for this turbine is given in figure 3-12. Table 3-2 shows the database for the power curve.

Table 3- 2 Power curve data for 20 kW wind turbine [HOMER, 2.19]

Wind speed (m/s)	0	3	4	5	6	7	8	9	10	11
Power output (kW)	0	0	0.4	0.7	1.9	4	6.7	10.4	14.1	17

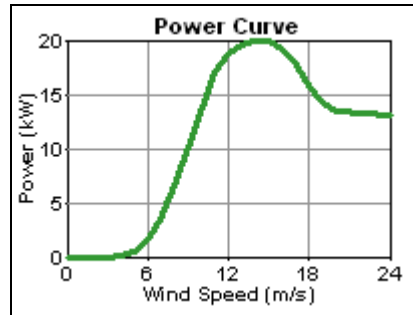


Figure 3-12 Power curve of the 20 kW generic 20 type wind turbine [HOMER, Ver. 2.19]

Table 3-3 Inputs to the software

	PV	Wind Turbine	Diesel Generator	Battery (Surrette 6CS25P)	Converter
Size (kW)	1	20	44	1156 Ah	1
Capital (\$)	1200-6000	45,000	11,000	833	700
Replacement cost (\$)	1200-6000	30,000	7,000	555	700
O & M cost (\$/yr)	0	900	0.4 (\$/hr)	15	0
Sizes considered (kW)	0, 5, 10, 15, 20, 30, 50, 70, 100		0, 44, 88		0, 20, 40, 60, 80, 100
quantities considered		0, 1, 2, 3		0, 40, 60, 80, 100, 200	
Life time	25 yrs	25 yrs	40,000 hrs	9,645 kWh	15 yrs

The costs are estimated according to the current local price of the components. Other inputs into the software, such as the range of sizes for the PV and the converter and the number of batteries, are given so as to give flexibility to the software and optimize the output results. More detail is given in the published paper [Bekele and Palm, 2009c].

Additional information required by HOMER software was calculated and/or estimated. Regarding the wind, the shape parameter k , which is an indication of the breadth of the distribution of wind speeds, is calculated by applying equation 3-9, which is appropriate where $1 \leq k < 10$ [Manwell, .2002], and also by repeatedly running the program, by way of trial and error, checking the results against the measured data. The value that fits best for k is found to be 2.

$$k = \left(\frac{\sigma_u}{\bar{u}} \right)^{-1.086} \quad \text{Eq. 3-9}$$

where: $\bar{u}$ is the mean wind speed and σ_u is the standard deviation.

The anemometer height at which data was collected is 10 m according to the data source, NMSA. Typical values for diurnal pattern strength range from 0 to 0.4 [HOMER, ver. 2.19]; by varying the values within the range, repeatedly running the software and checking the results against the measured data, a value of 0.25 has been selected. The autocorrelation function is a measure of the tendency of what a wind speed is likely to be, given what it was earlier [HOMER, ver. 2.19]. For complex topography the autocorrelation factor is (0.70 - 0.80) while for a uniform topography the range is higher, (0.90 - 0.97). A typical range for the autocorrelation factor is 0.8 – 0.95 [HOMER, ver. 2.19]. An average value of 0.85 is used here because the selected areas are of averagely uniform topography. The typical range for the time of peak wind speed, which is the time of day that tends, on average, to be the windiest throughout the year, is 14:00-16:00 [HOMER ver. 2.19]. This has also been observed in the available raw data for some of the months. In addition to this, the software has been run for different times between 14:00 and 18:00, the results have been checked against the measured data and the time of 15:00 has been chosen for the calculations.

The generator fuel is diesel and the prices considered are 0.4, 0.5, 0.7, 0.9, and 1.1 US dollars per liter. The current price considered is 0.5 USD. Inverter and converter efficiencies assumed are 90%. The project life

time is 25 years, and the interest rate is assumed to be the present rate, 6.67 %.

Except for the solar and wind resource data, the information input into HOMER is the same for all four locations under investigation. Specific information corresponding to each location is to be given under the respective subtopics of the region.

The results obtained for all four cases is a list of feasible power supply systems sorted according to their net present cost. Furthermore, sensitivity variables, such as the range of wind speed, radiation level, and diesel price are supplied, the software is tuned for optimum results and refined results are thus obtained.

In the second design approach, a comprehensive market survey is carried out in the Internet for the selection of light bulbs and the submersible water pumps. Obviously, the lightbulb industry is growing rapidly with many lighting products added into the market every year making the selection of the right bulb to the right application more and more difficult. An easier way to look into this problem is to group them into a certain major varieties; as Incandescent, Fluorescent (FL), Compact Fluorescent (CFL), Metal Halide, and LED (light emitting diode).

Incandescent lamps are relatively low in their efficiency; approximately 98% of the energy input is emitted as heat [Bekele and Palm, 2009d]. The halogen bulbs are quite similar to the incandescent bulbs, with only a few modifications. The FL and the CFL types have long average life and have good energy saving feature. For the same luminous intensity the CFL usually takes only 20-33% energy of what the incandescent bulbs take. The average rated life of a CFL is 8-15 times of that of the incandescent lamp and is between 6,000 and 15,000 hours, whereas that of incandescent lamps is usually between 750-1,000 hours.

The LED Light bulbs are in progress and are expected to be twice as energy efficient as fluorescent lamps and 10 times more efficient than incandescent lamps. In January 2009, researchers at Cambridge University had developed a LED bulb which is 12 times as energy efficient as a tungsten bulb and it can last for 100,000 hours [Bekele and Palm, 2009d]. Newer technologies are still coming into the market and these bulbs may soon supersede the other types of lightbulbs in the market.

Similar to the case of the light bulbs, finding the best brand of submersible pump in the market is not a simple task as the pump technology currently is well developed and all are too competitive.

Most of the available brands are of a higher capacity than those required for this study. It is clear that the efficiency of a pump is the energy delivered by the pump to the energy supplied to the pump shaft. This is best represented by the wire-to-water efficiency, which combines the overall efficiency of the pump and the motor [Bekele and Palm, 2009d]. The theoretical efficiency of a pump may reach over 90 %, however, the maximum practically attainable is somewhere between 80 to 90%. It is known that in general pump efficiencies are quite low; for smaller capacity pumps it is even much lower.

A number of brands have been assessed in the search for the best type of pumps that are available. The SQ and SQE pumps from Grundfos are reported to be of a high efficiency output within a wide load range as the motors are of a permanent-magnet type. The high and flat efficiency curve of the permanent-magnet motor covers a wider power range compared to conventional AC motors [Bekele and Palm, 2009d]. The pump technology, as was mentioned earlier, is improving and it is reported that the best new pump from Grundfos, Alpha2, needs less than half of the energy required by the old type from the same manufacturer [Bekele and Palm, 2009d]. Another brand reported is the SHURflo 9300 which is a lightweight submersible pump that is useful in low water applications such as remote homes, livestock and irrigation. Detailed report may be found in the attached paper [Bekele and Palm, 2009d].

In the third approach a self-contained system design is attempted for cost minimization. What this means is each household would have its own supply system, which may consist of any combination of PV and wind turbine including converter, battery and charge controller [Bekele and Palm, 2009d]. The price of these components capable enough to meet the peak load of the households calculated has been found to be much higher than the one found on an aggregate basis [Bekele and Palm, 2009d].

4 PART IV: Results and Conclusions

The wind energy [Bekele and Palm, 2009b] and solar energy potentials [Bekele and Palm, 2009a] of the locations were studied. Figures 1-8 and 2-1 respectively show the findings. It is based on the findings of these energy potentials that this feasibility study is carried out, by investigating the possibility of supplying electricity from a solar-wind based hybrid system to the model community of 200 families introduced earlier. As previously mentioned, HOMER is used for the analysis.

4.1 Brief note on the results of the feasibility study articles

First, before going into the discussion of the results of the other three locations, Mekele, Nazret, and Debrezeit, the results of the investigations for which articles are written [Bekele and Palm, 2009c,d] will be discussed in brief as was done for the cases of the articles for wind and solar energy potentials in section 1.2 and 2.2.

As was mentioned in the previous sections three different approaches were followed in this study. In the first approach, in which system components that are commonly available have been included in the design without much concern to the efficiency, a paper has been published based on the solar and wind energy potential of Addis Ababa [Bekele and Palm, 2009c]. In the second and the third approaches the solar and wind energy potential used is that of Mekele and the investigation has been prepared in the form of an article [Bekele and Palm, 2009d]. The results of these two investigations will be briefed in the following few paragraphs.

First considering the results for Addis Ababa, for finding optimum solutions the simulation model was run repeatedly by varying parameters that have a controlling effect over the output. These parameters are given in table 3-3. Also, variable prices of diesel oil and PV panels have been used for sensitivity analysis. The resulting output of the simulation

are in either of two forms; an overall form in which the top-ranked system configurations are listed according to their net present cost (NPC) and a categorized form, where only the least-cost system configuration is considered from each possible system type. Table 4-1 shows the resulting list of feasible combinations of system components in the overall form. As the table is too long to fit in this section, it has been truncated and only a selected part is kept here. The remaining part is given in the appendices Table 5-1.

Table 4-1 Overall optimization results according to the NPC for Addis

	pV (kW)	G20 wind turbine	Generator (kW)	Battery	Converter (kW)	Dispatch Strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (L/yr)	Generator (hrs/yr)
			44	40	20	CC	\$ 58,320	\$ 201,609	0.322	0	18,623	1,785
5			44	40	20	LF	\$ 76,320	\$ 220,728	0.353	0.16	18,115	2,391
			44	40	20	LF	\$ 58,320	\$ 222,616	0.356	0	21,056	2,725
10			44	40	20	LF	\$ 94,320	\$ 226,668	0.362	0.3	16,217	2,293
			44	60	20	CC	\$ 74,980	\$ 227,227	0.363	0	18,605	1,788
5			44	60	20	CC	\$ 92,980	\$ 230,456	0.369	0.15	16,320	1,638
15			44	40	20	LF	\$ 112,320	\$ 231,855	0.371	0.43	14,260	2,137
	1		44	40	20	LF	\$ 103,320	\$ 233,435	0.373	0.39	14,355	2,021
5			44	40	40	LF	\$ 90,320	\$ 238,842	0.382	0.16	18,083	2,379
5	1		44	40	20	LF	\$ 121,320	\$ 239,756	0.383	0.51	12,599	1,858
.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.
20	1		44	60	40	LF	\$ 205,980	\$ 289,942	0.464	0.81	5,662	817
10	2		44	40	20	LF	\$ 184,320	\$ 290,411	0.464	0.75	9,177	1,460
.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.
30			44	80	40	LF	\$ 213,640	\$ 300,698	0.481	0.77	6,379	914
10	1		44	80	40	LF	\$ 186,640	\$ 300,866	0.481	0.65	8,812	1,222
			44	80	40	CC	\$ 105,640	\$ 300,887	0.481	0	23,005	2,243
			88	40	40	LF	\$ 83,320	\$ 300,892	0.481	0	27,598	2,294
15			44	40	100	LF	\$ 168,320	\$ 300,932	0.481	0.43	13,767	1,919
15	1		44	60	60	LF	\$ 201,980	\$ 300,959	0.481	0.73	7,239	1,022
5			44	60	80	LF	\$ 134,980	\$ 301,094	0.481	0.16	18,051	2,374
5			44	40	60	CC	\$ 104,320	\$ 301,644	0.482	0.13	24,133	3,577

From the table the followings are remarkable results. The most cost effective system is that in the first row, the generator- battery- converter setup, with a total net present cost (NPC) of \$ 201 609 and cost of energy (COE) of 0.322 \$/kWh. The setup does not have any contribution from renewable resources and uses 18623 liters of diesel oil annually.

The PV-Gen-battery-Converter setup in the following row is the second most cost effective system with NPC of \$220,728 and COE of 0.353 \$/kWh. For this setup the part contributed by renewable resources is rather small, being only 16%. However, down in the list, there is another system comprising of PV-Wind-Gen-Battery-Converter having about 50 % contribution from the renewable resources for a total NPC of \$239,756 and COE of 0.383 \$/kWh. This is an increase of some 8 % in the cost over the earlier setup but the renewable fraction has increased from 16 % to 51 % making it a good candidate for implementation.

Another interesting setup further down in the list is that with 81 % utilization of renewable resources for a total NPC of \$289,942 and COE of 0.464 \$/kWh. Despite the higher cost, this is quite significant in view of the renewable fraction. In general we can see in the list numerous feasible setups with different levels of penetration into the renewable resources; the selection, however, depends on whether the initial cost is the principal concern or the benefits gained from utilizing the renewable resources. A detailed analysis is given in the published paper [Bekele and Palm, 2009c], which is attached at the end of the thesis.

The results obtained from the second and the third approaches are also prepared in the form of a paper [Bekele and Palm, 2009d]. In section 3.3 it is explained that a careful market-survey is conducted to compare and use the latest and most efficient types of system components both on the load and supply side of the hybrid system. Based on the selected components the electric load is calculated and used for running the simulation [Bekele and Palm, 2009d]. In the same paper, a self-contained system design is also examined as another alternative in the effort of cost minimization. The results of the investigation are briefed as follows.

As in the previous case for optimum solution the model was run repeatedly using different types, sizes, capacities and numbers of wind turbines, PV panels, diesel generators, batteries and converters; etc. The sensitivity analysis has been also done by varying important parameters

such as: diesel oil price and cost of PV panels. The simulation results obtained are in either of two forms; an overall form and a categorized form. Table 4-2 shows a list of feasible combinations of system components in the overall form. As in the previous case the table is truncated and it is only a selected part that is displayed. The complete table is given in the appendices (Table 5-5)

Table 4-2 Feasible setups selected from the results table

PV (kW)	WES 5 Tulipo	Gen1 (kW)	Battery	Converter (kW)	Display strategy	Initial capital	Operating cost (\$/Yr)	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (L)/yr	Gen (hrs)/yr
2		5	10	4	CC	\$ 21,330	3,798	\$ 66,940	0.327	0.21	4,639	3,090
.	.	.	.	.	.	.	.	.	.	.	.	.
2	1	5	10	4	CC	\$ 30,330	3,425	\$ 71,463	0.349	0.37	3,676	2,431
.	.	.	.	.	.	.	.	.	.	.	.	.
4	1	5	10	4	CC	\$ 37,530	3,273	\$ 76,833	0.375	0.51	3,435	2,727
.	.	.	.	.	.	.	.	.	.	.	.	.
4	1	5	15	4	CC	\$ 41,695	3,313	\$ 81,482	0.397	0.53	3,233	2,605
.	.	.	.	.	.	.	.	.	.	.	.	.
4	1	5	20	4	CC	\$ 45,860	3,452	\$ 87,313	0.426	0.54	3,158	2,656
.	.	.	.	.	.	.	.	.	.	.	.	.
10		5	25	10	CC	\$ 66,825	2,276	\$ 94,155	0.459	0.8	1,634	1,331
.	.	.	.	.	.	.	.	.	.	.	.	.
4	1	5	25	6	CC	\$ 51,425	3,616	\$ 94,852	0.463	0.53	3,111	2,168
4	1	5	20	10	CC	\$ 50,060	3,776	\$ 95,402	0.465	0.5	3,474	2,534
.	.	.	.	.	.	.	.	.	.	.	.	.
6		5	30	4	CC	\$ 52,390	3,624	\$ 95,912	0.468	0.55	3,315	3,080
2	1	5	20	15	CC	\$ 46,360	4,139	\$ 96,065	0.469	0.36	3,862	2,535
12		5	25	10	CC	\$ 74,025	1,837	\$ 96,090	0.469	0.88	1,028	854
12		5	30	10	CC	\$ 78,190	1,708	\$ 98,699	0.481	0.93	591	471
14		5	30	10	LF	\$ 85,390	1,393	\$ 102,119	0.498	0.99	153	192

From the table the following interesting setups can be considered for practical implementation. The most cost effective system is the 2 kW PV- 5 kW generator- 10 S6CS25P type batteries- 4 kW converter setup listed in the first row of the table. Operating cost for this setup is \$ 3,798 per year, the total net present cost (NPC) is \$ 66,940, the cost of energy (COE) is 0.327 \$/kWh, the contribution from the renewable resources is 21 %, and the annual consumption of diesel oil by the generator is 4639 liters for an operation time of 3090 hours. Despite the lowest NPC, this set up may not be a good choice as the contribution by the renewable resource is only 21 %.

In the third row a setup of PV-Wind turbine-Gen-Battery-Converter generates power at a renewable proportion of 37 %. This is an increase by 76 % over the earlier setup but with only some 6 % increase of NPC. This can be a better choice than the earlier one. Still further down in the list, there is another setup of a 4 kW PV-1 WES 5 Tulipo type Wind turbine-5 kW Generator- 10 S6CS25P type batteries-4 kW Converter combination. This setup generates 51 % of the power from the renewable resources and can be a good choice for implementation. The NPC for this setup is \$ 76,833, the COE 0.375 \$/kWh and the operating cost is 3,273 \$/Yr. The annual oil consumption of the generator is 3,435 liters in an operation time of 2,727 hours. More details of the investigation can be obtained in the paper attached at the end of the thesis.

4.2 Results for Mekele

The monthly average wind speed for Mekele, determined in the ways described in the published paper [Bekele and Palm, 2009b], together with other related data, such as values of k , diurnal pattern etc., was fed into HOMER. Figure 4-1 shows the wind resource data.

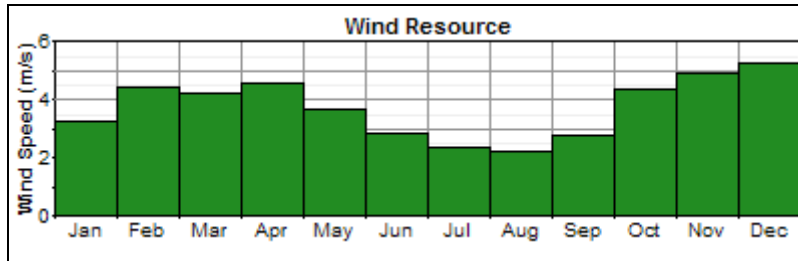


Figure4-1 Mekele monthly average wind resource

In a similar way, the solar energy potential for Mekele, determined as explained in the paper, was fed into HOMER and this is shown in figure 4-2. This figure also shows the clearness index which HOMER generated for the analysis. The clearness index is the fraction of solar radiation transmitted through the atmosphere which strikes the surface of the Earth and hence it is a measure of the clearness of the atmosphere. It is a dimensionless number between 0 and 1, defined as the surface radiation divided by the extraterrestrial radiation. Typical values for the monthly average clearness index range from 0.25 (a very cloudy month) to 0.75 (a very sunny month).

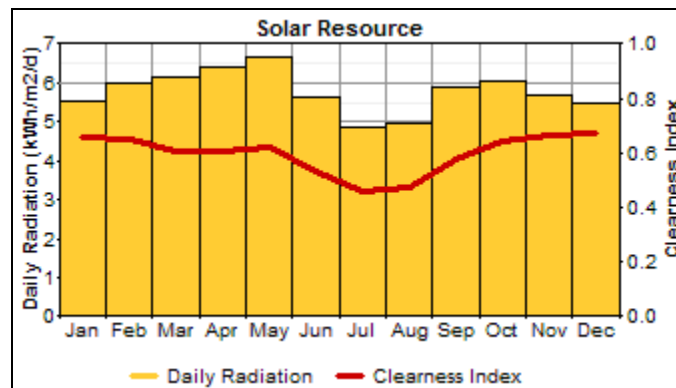


Figure 4-2 Mekele monthly average solar resource

Once the wind and solar resource data are entered into the software, by changing the most important variables HOMER is run repeatedly to obtain optimum results. Consequently, a list of realizable optimal combinations of PV, wind-turbine, generator, converter, and battery hybrid system set-up is obtained. The list is generated in either of two forms; an overall form in which the top-ranked system configurations are listed according to their net present cost (NPC) and in a categorized form; where only the least-cost system configuration is considered for each system type. Table 4-3 shows a list of the possible combinations of system components in an overall form. The table is generated based on a particular set of inputs selected from the input summary table (table 3-3) and the solar and wind resource data for Mekele. The diesel price is 0.5 \$/L and the PV capital multiplier is 0.6 (3600 \$/kW). The price of PV has been checked using different sources on the internet and the price ranged from \$2.5 to \$4.74 per watt. [Digitimes, 2009], [SolarBuzz, 2009]. The solar and wind data inputs are the results of previous investigations into solar and wind potentials [Bekele and Palm, 2009a,b]; the diesel price is the current price of diesel in the country and the price of PV is also believed to be the current price.

Table 4-3 is an extract from the long table that contained all the possible optimal combinations of the realizable systems given by the software. The complete table is available in the appendix (Table 5-2). The extract is based on the contribution to the systems made by renewable resources and those systems with a renewable fraction of more than 50 % have been selected except for the system in the third row which indicated a contribution of 45 % by renewable resources. The reasons for the inclusion of this system will be explained later.

The following favorable results can be noted from the table. The most cost effective system, i.e. the system with the lowest net present cost, is the generator-battery-converter set-up with the generator operating under a cycle charging (CC) strategy (a dispatch strategy whereby the generator operates at full output power to serve the primary load: surplus electrical production goes towards lower-priority objectives). For this set-up, the total net present cost (NPC) is \$201 609, the cost of energy (COE) is 0.322 \$/kWh, there is no contribution from renewable resources, the amount of diesel oil used annually is 18623 liters and the generator operates for 1785 hours per year. In spite of the fact that the net present cost is the smallest, renewable resources make no contribution towards the energy supply.

Table 4-3 Extracts from the overall optimization results table for Mekele

PV (kW)	G20 wind turbine	Generator (kW)	Battery	Converter (kW)	Dispatch Strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (l./yr)	Generator (hrs/yr)
		44	40	20	CC	\$ 58,320	\$ 201,609	0.322	0	18,623	1,785
20		44	40	20	LF	\$ 130,320	\$ 234,980	0.376	0.58	12,049	1,906
5	1	44	40	20	LF	\$ 121,320	\$ 245,448	0.392	0.45	13,467	1,927
20		44	40	40	LF	\$ 144,320	\$ 250,763	0.401	0.57	11,817	1,733
20		44	60	20	LF	\$ 146,980	\$ 251,496	0.402	0.61	10,716	1,736
10	1	44	40	20	LF	\$ 139,320	\$ 252,681	0.404	0.57	11,820	1,801
20		44	60	20	CC	\$ 146,980	\$ 259,829	0.415	0.59	11,900	1,911
30		44	40	20	LF	\$ 166,320	\$ 261,958	0.419	0.71	10,731	1,747
20	1	44	40	20	LF	\$ 175,320	\$ 276,727	0.443	0.71	10,032	1,628
30		44	40	40	LF	\$ 180,320	\$ 278,911	0.446	0.7	10,372	1,572
30		44	60	40	LF	\$ 196,980	\$ 279,583	0.447	0.78	7,008	1,058
	2	44	40	20	LF	\$ 148,320	\$ 280,725	0.449	0.51	13,140	1,845
20	1	44	40	40	LF	\$ 189,320	\$ 284,266	0.455	0.73	8,535	1,285
20		44	40	80	LF	\$ 172,320	\$ 287,397	0.46	0.57	11,801	1,727
30		88	80	40	LF	\$ 224,640	\$ 302,976	0.484	0.84	5,196	437
20	1	88	60	40	LF	\$ 216,980	\$ 307,740	0.492	0.79	6,730	565
10	1	44	80	40	LF	\$ 186,640	\$ 308,095	0.493	0.59	9,893	1,353

The next most cost effective system, row 2 in Table 4-3, is the PV-Gen-battery-Converter set-up, with the generator operating under a Load Following (LF) strategy (a dispatch strategy whereby the generator operates to produce just enough power to meet the primary load; lower-priority objectives, such as charging the battery bank or serving the deferrable load, is left to the renewable power sources). For this set-up the total net present cost (NPC) is \$234,980, the cost of energy (COE) is 0.376 \$/kWh, the contribution made by renewable resources is 58 %, annual diesel oil usage is 12,049 liters and the generator operates for 1,906 hours in the year.

In this set-up the part that renewable resources (solar) contribute to the supply system is quite significant, 58 %, without a wind turbine operating in the system. This could be a good choice for implementation. Figure 4-3 shows the monthly average electrical production of this system. Table

4-6 gives some of the principal information about the system. The cost breakdown supported by a pie-chart for this set-up is also given in figure 4-6.

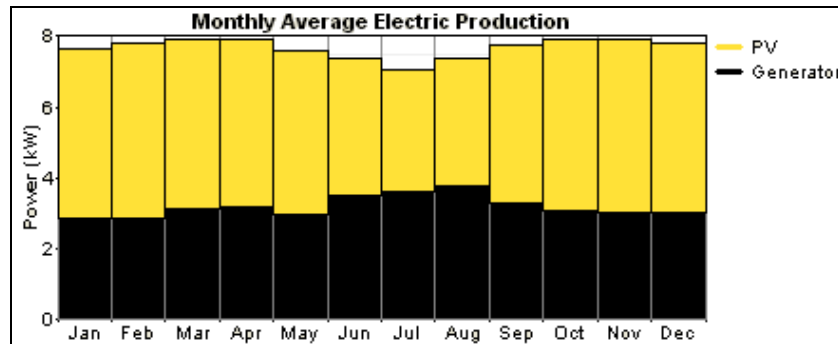


Figure 4-3 Contribution of the power units with a 58 % proportion of renewables for Mekele, second row in Table 4-3

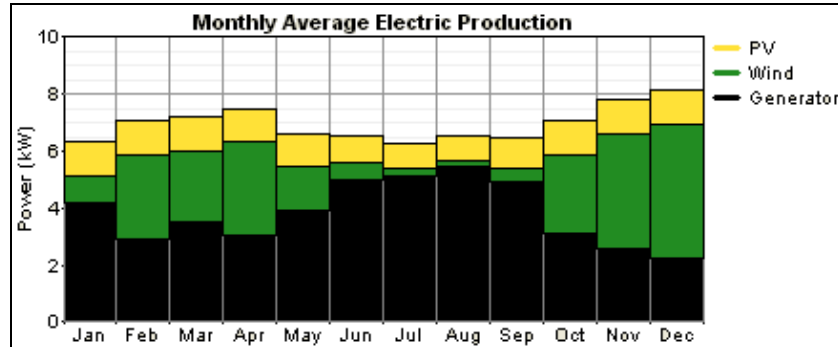


Figure 4-4 Contribution of the power units with a 45 % proportion of renewables for Mekele, the 3rd row in Table 4-3.

The most cost effective system which comprises both the renewable resources, a PV-Wind-Gen-Battery-Converter set-up, is the system in the 3rd row. For this set-up the contribution made by renewable resources is 45 %, which is less than the earlier set-up by 13 %. Nonetheless, the NPC has increased to \$245,448 and the COE to 0.392 \$/kWh. This could also be a good choice if there is a motive for utilizing the available wind energy, which would , however, be at the cost of a 4.5 % increase in the total NPC. Once again the monthly average electric production for this set-up is shown in figure 4-4 and table 4-7 gives the most important information about this set-up. Furthermore, the cost breakdown, supported by a pie-chart, is given in figure 4-7

Further down in the list, there is another set-up with an 84 % contribution by renewable resources, which is a good percentage indeed. However, for this set-up the NPC increases to \$302,976 and the COE to 0.484 \$/kWh, which is a 28.9 % increase over the NPC of the set-up with a 58 % renewable resource contribution.

This could be another attractive system if the 84 % contribution made by renewable resources is to be given due merit and also if consideration is given to other related issues, such as the future price trend of the components which make up the system and also the unprecedented rise in current diesel prices. Enquiries into such issues would enable a decision to be made on whether to consider the set-up or simply omit it. As mentioned earlier, the price variation for diesel oil, considered in this study, is between \$0.4 and \$1.1. To give an example of what the situation would look like if the current diesel price reached the highest level considered, the first few lines of the optimization results are given in Table 4-4. As can be seen from the table the minimum total NPC in the list is \$321,792, which is about 50% more than the minimum total NPC given in Table 4-3.

Table 4-4 The first few lines of the optimization results for Mekele for a diesel price of \$1.10

PV (kW)	G20	Gen (kW)	Battery	Converter(kW)	Dispatch Strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (L)	Gen (hrs)
20		44	40	20	LF	\$ 130,320	\$ 321,792	0.515	0.58	12,049	1,906
30		44	80	40	LF	\$ 213,640	\$ 324,298	0.519	0.83	4,778	700
20		44	60	20	LF	\$ 146,980	\$ 328,704	0.526	0.61	10,716	1,736
30		44	60	40	LF	\$ 196,980	\$ 330,074	0.528	0.78	7,008	1,058
20	1	44	60	40	LF	\$ 205,980	\$ 335,069	0.536	0.79	5,969	866
20		44	40	40	LF	\$ 144,320	\$ 335,902	0.537	0.57	11,817	1,733
20		44	60	40	LF	\$ 160,980	\$ 336,688	0.538	0.61	10,072	1,434
10		44	40	20	LF	\$ 94,320	\$ 337,535	0.54	0.33	15,777	2,269

The monthly average power production for the 84% renewable fraction set-up in table 4-3 is given in figure 4-5. Information about the system is

also given in table 4-8. The cost breakdown supported by a pie-chart is also given in figure 4-8.

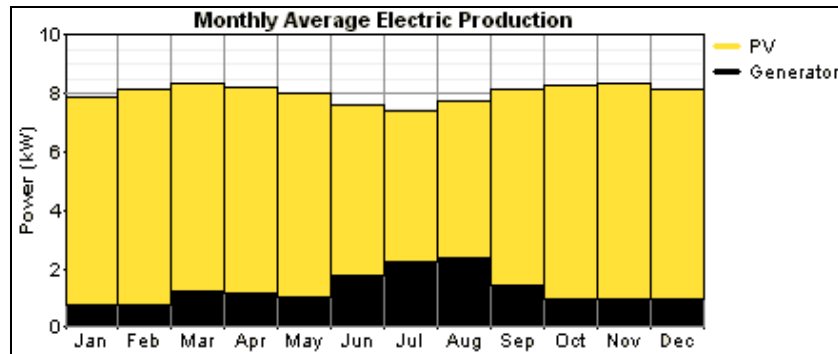


Figure 4-5 Contribution of the power units with an 84 % proportion of renewables for Mekele, 3rd row from the bottom of Table 4-3.

The result, in a categorized form, is given in table 4-5, where only the least cost effective system is considered for each system type. The set-up in the first row of this table and also that with a 45 % contribution made by renewable resources in the same table (4th row) are the same as those listed in table 4-3 and have been discussed earlier. In this table two system set-ups are of greatest interest, each supplying 100 % from renewable resources. One comprises PV, Battery and Converter and the other PV, Wind turbine, Battery, and Converter. The NPC for the first is \$492,244 and that for the second is \$ 529,666. The COE is 0.787 \$/kWh and 0.847 \$/kWh respectively.

The net present costs for these two set-ups are more than double that of the setup with a 58 % renewable contribution. However, these set-ups could also be considered as options if issues discussed earlier are given consideration; the issues of future price trends of the components which constitute the system and also the rapidly rising diesel price.

Table 4-5 optimization results in a Categorized form; ranked according to the NPC of each system type

PV (kW)	G20	Gen1 (kW)	Battery	Converter(kW)	Dispatch strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (L)	Gen (hrs)
		44	40	20	CC	\$ 58,320	\$ 201,609	0.322	0	18623	1785
5		44	40	20	LF	\$ 76,320	\$ 219,157	0.35	0.17	17872	2375
	1	44	40	20	CC	\$ 103,320	\$ 237,336	0.38	0.28	15520	1574
5	1	44	40	20	LF	\$ 121,320	\$ 245,448	0.392	0.45	13467	1927
10		44		20	LF	\$ 61,000	\$ 345,060	0.552	0.17	40366	5913
10	1	44		20	LF	\$ 106,000	\$ 348,667	0.558	0.31	32971	4797
	1	44			LF	\$ 56,000	\$ 364,171	0.582	0.14	42717	6436
		44			LF	\$ 11,000	\$ 412,070	0.659	0	57291	8760
50			200	60	CC	\$ 388,600	\$ 492,244	0.787	1		
50	1		200	40	CC	\$ 419,600	\$ 529,666	0.847	1		

Table 4-6 System report for the 58 % renewable resource contribution for Mekele

System architecture		Sensitivity case		Annual electric production (kWh/yr)			Annual electric energy consumption (kWh/yr)			Emissions (kg/yr)							
PV Array	20 kW	Solar Data	5.77 kWh/m ² /d	PV array	39168	58%	AC primary load	50,772	97%	CO2	31728						
Wind turbine		Wind Data	3.75 m/s				Deferrable load	1,306	3%	CO	78.3						
Gen.	44kW	Diesel Price	0.5 \$/L	Generator	28066	42%	Total	52,077	100%	Unburned HC	8.68						
Battery	40 Surrette 6CS25P	PV Capital Cost Multiplier	0.6	Excess electricity	5812		<table><tr><th colspan="2">Cost summary</th></tr><tr><td>Total NPC</td><td>\$ 234 980</td></tr><tr><td>Cost of energy</td><td>0.376 \$/kWh</td></tr></table>			Cost summary		Total NPC	\$ 234 980	Cost of energy	0.376 \$/kWh	Particulate matter	5.9
Cost summary																	
Total NPC	\$ 234 980																
Cost of energy	0.376 \$/kWh																
Inverter	20 kW	PV Replacement Cost Multiplier	0.6	Unmet load:	0 ≐		SO2	63.7									
Rectifier	20 kW			Capacity shortage	0		NOx	699									

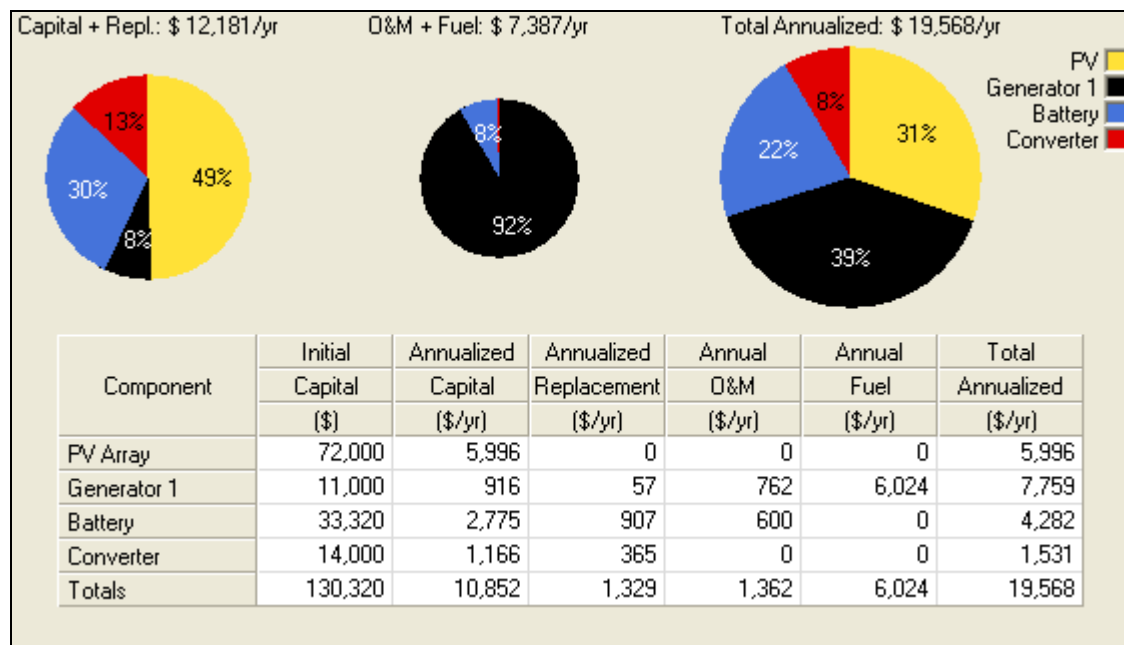


Figure 4-6 Cost summary for the 58 % renewable resource contribution for Mekele

Table 4-7 System report for the 45 % renewable resource contribution for Mekele

System architecture		Sensitivity case		Annual electric production (kWh/yr)			Annual electric energy consumption (kWh/yr)			Emissions (kg/yr)	
PV Array	5 kW	Solar Data	5.77 kWh/m²/d	PV array	9,792	16%	AC primary load	50,772	97%	CO2	35,874
Wind turbine	1Generic 20kW	Wind Data	3.75 m/s	Wind turbine	17,509	29%	Deferrable load	1,306	3%	CO	88.5
Gen.	44kW	Diesel Price	0.5 \$/L	Generator	33,804	55%	Total	52,077	100%	Unburned HC	9.81
Battery	40 Surrette 6CS25P	PV Capital Cost Multiplier	0.6	Excess electricity	2535		Cost summary Total NPC\$ 246,608 Cost of energy0.394 \$/kWh			Particulate matter	6.68
Inverter	20 kW	PV Replacement Cost Multiplier	0.6	Unmet load:	0					SO2	72
Rectifier	20 kW			Capacity shortage	0					NOx	790

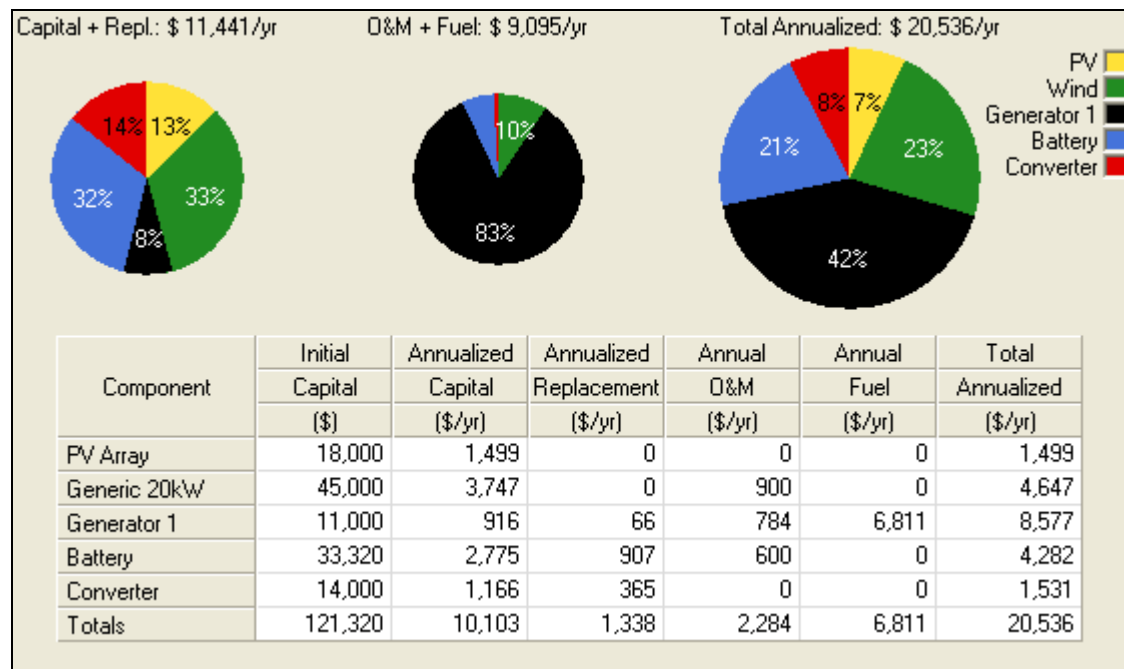


Figure 4-7 Cost summary for the 45 % renewable resource contribution for Mekele

Table 4-8 System report for the 84 % renewable resource contribution for Mekele

System architecture		Sensitivity case		Annual electric production (kWh/yr)			Annual electric energy consumption (kWh/yr)			Emissions (kg/yr)	
PV Array	30 kW	Solar Data	5.77 kWh/m ² /d	PV array	58,751	84%	AC primary load	50,772	97%	CO2	13,682
Wind turbine		Wind Data	3.75 m/s	Wind turbine			Deferrable load	1,304	3%	CO	33.8
Gen.	88 kW	Diesel Price	0.5 \$/L	Generator	11,554	16%	Total	52,076	100%	Unburned HC	3.74
Battery	80 Surrette 6CS25P	PV Capital Cost Multiplier	0.6	Excess electricity	2,895		Cost summary Total NPC\$ 302 976 Cost of energy0.484 \$/kWh			Particulate matter	2.55
Inverter	40 kW	PV Replacement Cost Multiplier	0.6	Unmet load:	0					SO2	27.5
Rectifier	40 kW			Capacity shortage	0				NOx	301	

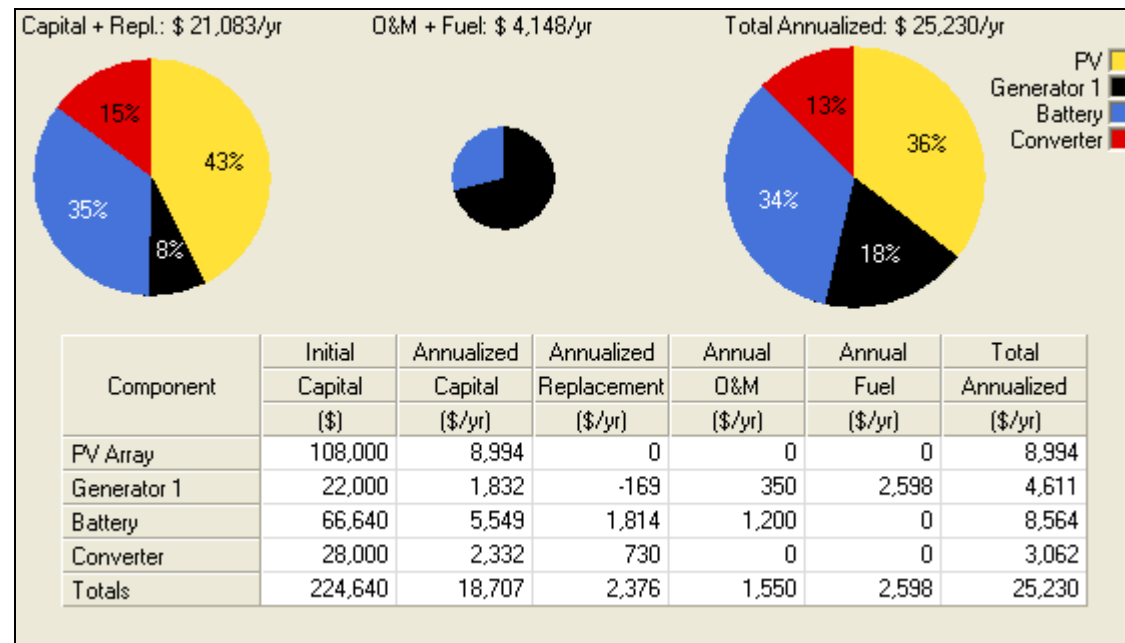


Figure 4-8 Cost summary for the 84 % renewable resource contribution for Mekele

Sensitivity analysis, where multiple values for a particular input variable are entered into the software for performance analysis, was also carried out and figure 4-9 shows the variation of the PV capital multiplier against diesel price for a fixed annual wind speed of 3.75 m/s and an annual solar radiation of 5.77 kWh/m²/day. In the figure the net present cost of the most cost-effective set-up for a particular set of diesel and PV prices is shown.

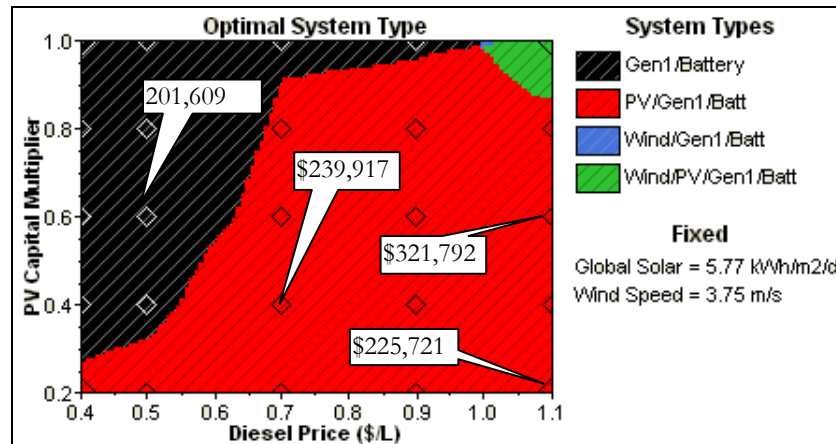


Figure 4-9 Sensitivity of PV cost to diesel price for Mekele with some important NPCs labeled

As can be seen in figure 4-9, the wind energy potential is low and this has been explained earlier in the study [Bekele and Palm, 2009b]. The potentials at Mekele were evaluated with respect to the wind power classification of the US Department of Energy (DOE), and have been found to be of class 1 type. Class 1 potential is generally considered to be unsuitable for wind energy development. However, average annual wind speeds of 3–4 m/s may be adequate for non grid-connected electrical and mechanical applications such as battery charging and water pumping. The current PV price is assumed to be \$3600/kW and the price varies between this and the least price, \$1200/kW, which assumes a future fall in price.

4.3 Results at Nazret

As in the case of Mekele, the monthly average wind energy and monthly average solar energy [Bekele and Palm, 2009a,b] of Nazret were fed into the software. Figures 4-10 and 4-11 show the data which was input. The clearness index for the solar energy, which the software produced, is also indicated in figure 4-11.

After entering data for the wind and solar resources, the software was run and the resulting list of optimal combinations of realizable supply systems was obtained in both an overall form and a categorized form. The listing is arranged in the manner described for Mekele and table 4-9 shows the list which was extracted from the complete overall table which is placed in the appendix.

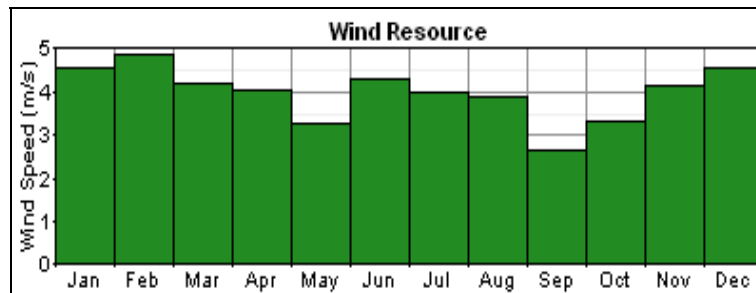


Figure 4-10 Nazret monthly average wind resource

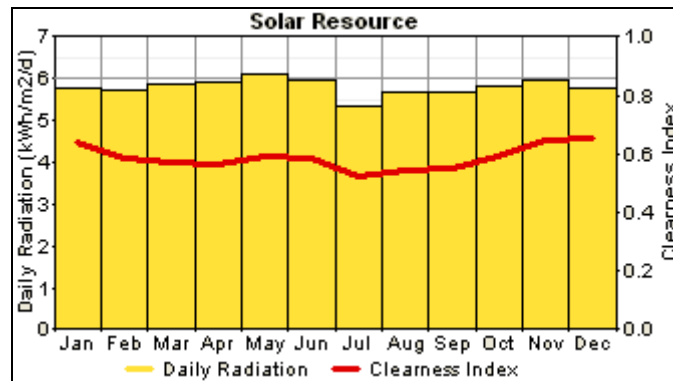


Figure 4-11 Nazret monthly average solar resource

Table 4-9 Extracts from the overall optimization results table for Nazret

PV (kW)	G20 wind turbine	Generator (kW)	Battery	Converter (kW)	Dispatch strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (l.)	Generator (hrs)
		44	40	20	CC	\$ 58,320	\$ 201,609	0.322	0	18,623	1,785
20		44	40	20	LF	\$ 130,320	\$ 235,586	0.377	0.58	12,131	1,922
5	1	44	40	20	LF	\$ 121,320	\$ 238,266	0.381	0.51	12,381	1,832
10	1	44	40	20	LF	\$ 139,320	\$ 245,095	0.392	0.62	10,695	1,683
10	1	44	40	40	LF	\$ 153,320	\$ 257,109	0.411	0.63	9,894	1,428
	2	44	40	20	LF	\$ 148,320	\$ 268,212	0.429	0.61	11,275	1,658
20	1	44	40	20	LF	\$ 175,320	\$ 270,473	0.433	0.75	9,105	1,513
10	1	44	60	40	LF	\$ 169,980	\$ 275,732	0.441	0.65	8,883	1,249
5	2	44	40	20	LF	\$ 166,320	\$ 276,115	0.442	0.7	9,745	1,512
30		44	40	40	LF	\$ 180,320	\$ 278,749	0.446	0.7	10,379	1,572
30		44	60	40	LF	\$ 196,980	\$ 280,101	0.448	0.77	7,084	1,068
20	1	44	60	40	LF	\$ 205,980	\$ 283,955	0.454	0.84	4,766	710
10	2	44	40	20	LF	\$ 184,320	\$ 287,106	0.459	0.76	8,672	1,412
30		44	80	40	LF	\$ 213,640	\$ 290,993	0.465	0.83	4,940	726
10	2	44	40	40	LF	\$ 198,320	\$ 293,861	0.47	0.79	7,083	1,044
20	1	44	60	60	LF	\$ 219,980	\$ 302,259	0.483	0.84	4,756	706
20	1	44	80	40	LF	\$ 222,640	\$ 302,958	0.484	0.87	3,793	557

Once again, the extract is based on the contribution made by renewable resources. Those set-ups with a renewable fraction greater than 50% are considered. It can be noted from the table that the system with the least NPC is again the non-renewable set-up. The NPC for this set-up is the same, \$201 609 at a COE of 0.322 \$/kWh. Despite the low cost no contribution is made by renewable resources.

The following cost effective setup is that in the second row with a renewable resource proportion of 58 % with the generator operating under Load following (LF) strategy. The NPC for this setup is \$ 235,586 and the COE is 0.377\$/kWh. The diesel oil used annually is 12,131 liters and the corresponding generator operation time is 1,922 hours. There is no wind turbine in this system. However, the penetration into the renewable resource is quite significant and this set up can be a good choice to implement. Figure 4-12 shows the monthly average electrical

production of this system. Important information about the setup is given in table 4-11. The cost breakdown supported by a pie-chart for this setup is also given in figure 4-15.

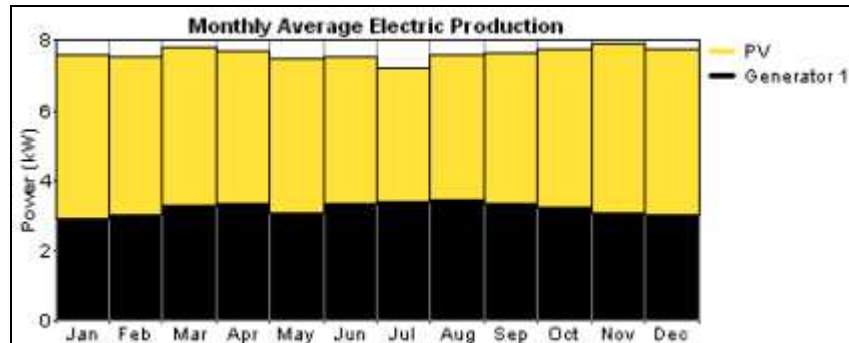


Figure 4-12 Contribution of the power units with a 58 % proportion of renewables for Nazret

One place ahead in the list, there is another PV-Wind turbine-Gen-Battery-Converter set-up, with a renewable proportion of 62 %. The NPC for this set-up is \$245,095 and the COE is 0.392 \$/kWh. This is also a good choice for implementation. The monthly average electricity production for this set-up is shown in figure 4-13, and table 4-12 gives important information about this set-up. Furthermore, the cost breakdown, supported by a pie-chart, is given in figure 4-16.

Another interesting set-up in the list is that in the last row, with an 87 % contribution made by renewable resources. This can be said to be an excellent proportion of renewable resources, but at the cost of a 23.6 % increase in the total NPC (\$ 302,958) over the set-up with a 62 % renewable contribution (\$ 245,095). However, this is once again a matter of priority, such as giving due merit to the utilization of renewable resources in view of the future benefits.

The monthly average electrical production for this set-up is given in figure 4-14. Information about the system is given in table 4-13. The cost breakdown for this set-up, supported by a pie-chart, is also given in figure 4-17.

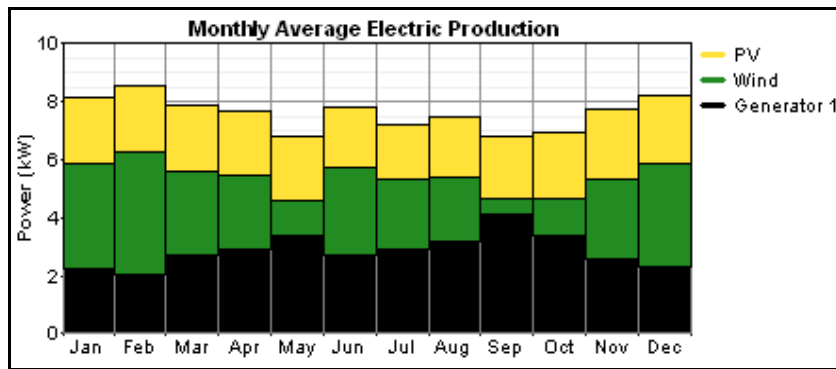


Figure 4-13 Contribution of the power units with a 62 % proportion of renewables for Nazret

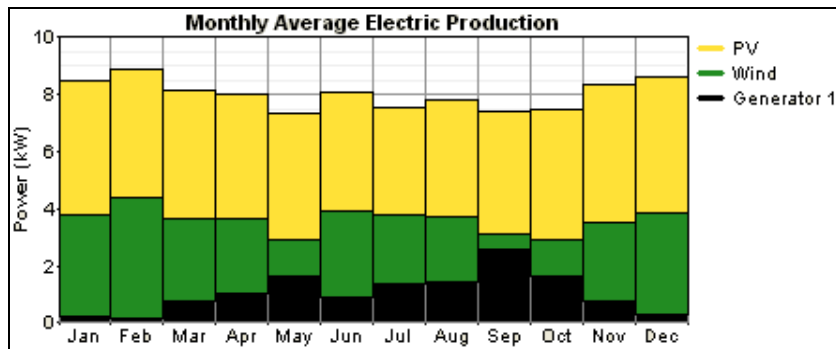


Figure 4-14 Contribution of the power units with an 87 % proportion of renewables for Nazret

The result in a categorized form, where only the least cost system is considered from each system type, is given in table 4-10. The first row of the table contains the same set-up as the earlier table, which was discussed previously. The two set-ups of greatest interest in this table are again those with a 100 % renewable component. One is a PV, Wind-turbine, Battery, and Converter set-up and the other comprises PV, Battery, and Converter. The NPC for the first is \$ 473,123 and that for the second is \$ 492,244. The corresponding COEs are 0.757 \$/kWh and 0.787 \$/kWh respectively.

Table 4-10 Optimization results in a Categorized form at Nazret; ranking is according to the NPC of each system type

PV (kW)	G20	Gen1 (kW)	Battery	Converter(kW)	Dispatch strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (L)	Gen1 (hrs)
		44	40	20	CC	\$ 58,320	\$ 201,609	0.322	0	18,623	1,785
5		44	40	20	LF	\$ 76,320	\$ 219,264	0.351	0.17	17,891	2,374
	1	44	40	20	LF	\$ 103,320	\$ 234,729	0.375	0.37	14,561	2,030
5	1	44	40	20	LF	\$ 121,320	\$ 238,266	0.381	0.51	12,381	1,832
10	1	44		20	LF	\$ 106,000	\$ 334,901	0.536	0.35	31,002	4,507
10		44		20	LF	\$ 61,000	\$ 346,411	0.554	0.16	40,553	5,946
	1	44			LF	\$ 56,000	\$ 346,501	0.554	0.18	40,208	6,054
		44			LF	\$ 11,000	\$ 412,070	0.659	0	57,291	8,760
70	1		100	40	CC	\$ 408,300	\$ 473,123	0.757	1		
50			200	60	CC	\$ 388,600	\$ 492,244	0.787	1		

Table 4-11 System report for the 58 % renewable resource contribution for Nazret

System architecture		Sensitivity case		Annual electric production (kWh/yr)			Annual electric energy consumption (kWh/yr)			Emissions (kg/yr)							
PV Array	30 kW	Solar Data	5.77 kWh/m ² /d	PV array	38,645	58%	AC primary load	50,772	97%	CO2	31,945						
Wind turbine		Wind Data	3.75 m/s	Wind turbine			Deferrable load	1,306	3%	CO	78.9						
Gen.	44 kW	Diesel Price	0.5 \$/L	Generator	28,227	42%	Total	52,077	100%	Unburned HC	8.73						
Battery	40 Surrette 6CS25P	PV Capital Cost Multiplier	0.6	Excess electricity	5,459		<table><tr><th colspan="2">Cost summary</th></tr><tr><td>Total NPC</td><td>\$ 235,586</td></tr><tr><td>Cost of energy</td><td>0. 377 \$/kWh</td></tr></table>			Cost summary		Total NPC	\$ 235,586	Cost of energy	0. 377 \$/kWh	Particulate matter	5.94
Cost summary																	
Total NPC	\$ 235,586																
Cost of energy	0. 377 \$/kWh																
Inverter	20 kW	PV Replacement Cost Multiplier	0.6	Unmet load:	0		SO2	64.2									
Rectifier	20 kW			Capacity shortage	0		NOx	704									

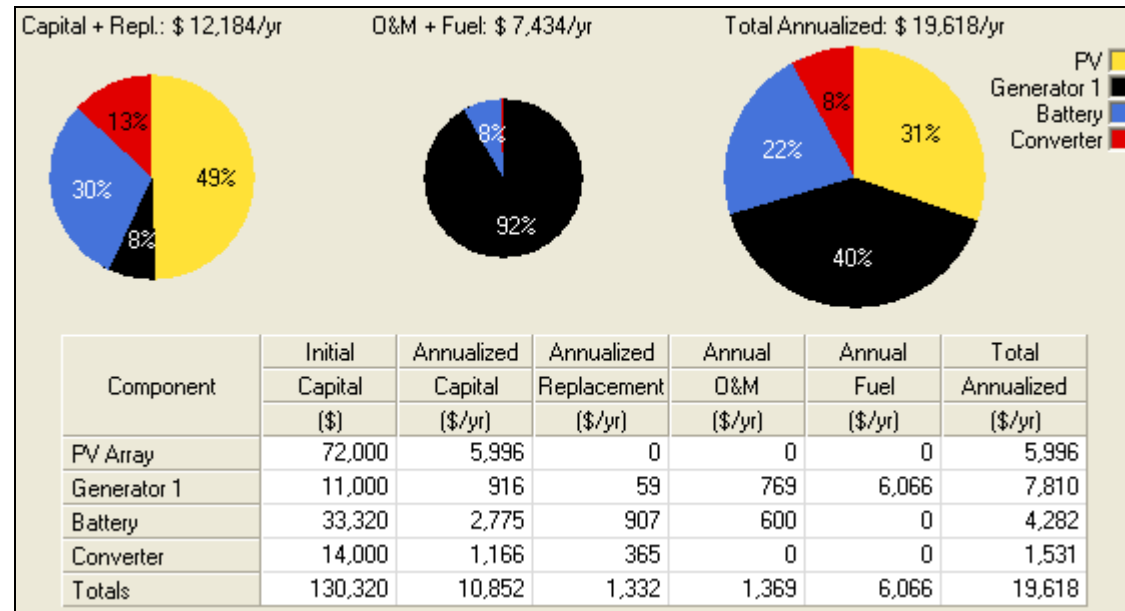


Figure 4-15 Cost summary for the 58 % renewable resource contribution for Nazret

Table 4-12 System report for the 62 % renewable resource contribution for Nazret

System architecture		Sensitivity case		Annual electric production (kWh/yr)			Annual electric energy consumption (kWh/yr)			Emissions (kg/yr)							
PV Array	10 kW	Solar Data	5.78 kWh/m ² /d	PV array	19,322	29%	AC primary load	50,772	97%	CO2	28,361						
Wind turbine	1Generic 20kW	Wind Data	3.99 m/s	Wind turbine	22,018	33%	Deferrable load	1,305	3%	CO	70						
Gen.	44 kW	Diesel Price	0.5 \$/L	Generator	25,212	38%	Total	52,076	100%	Unburned HC	7.75						
Battery	40 Surrette 6CS25P	PV Capital Cost Multiplier	0.6	Excess electricity	5,994		<table><tr><th colspan="2">Cost summary</th></tr><tr><td>Total NPC</td><td>\$ 245,612</td></tr><tr><td>Cost of energy</td><td>0.393 \$/kWh</td></tr></table>			Cost summary		Total NPC	\$ 245,612	Cost of energy	0.393 \$/kWh	Particulate matter	5.28
Cost summary																	
Total NPC	\$ 245,612																
Cost of energy	0.393 \$/kWh																
Inverter	20 kW	PV Replacement Cost Multiplier	0.6	Unmet load:	0		SO2	57									
Rectifier	20 kW			Capacity shortage	0				NOx	625							

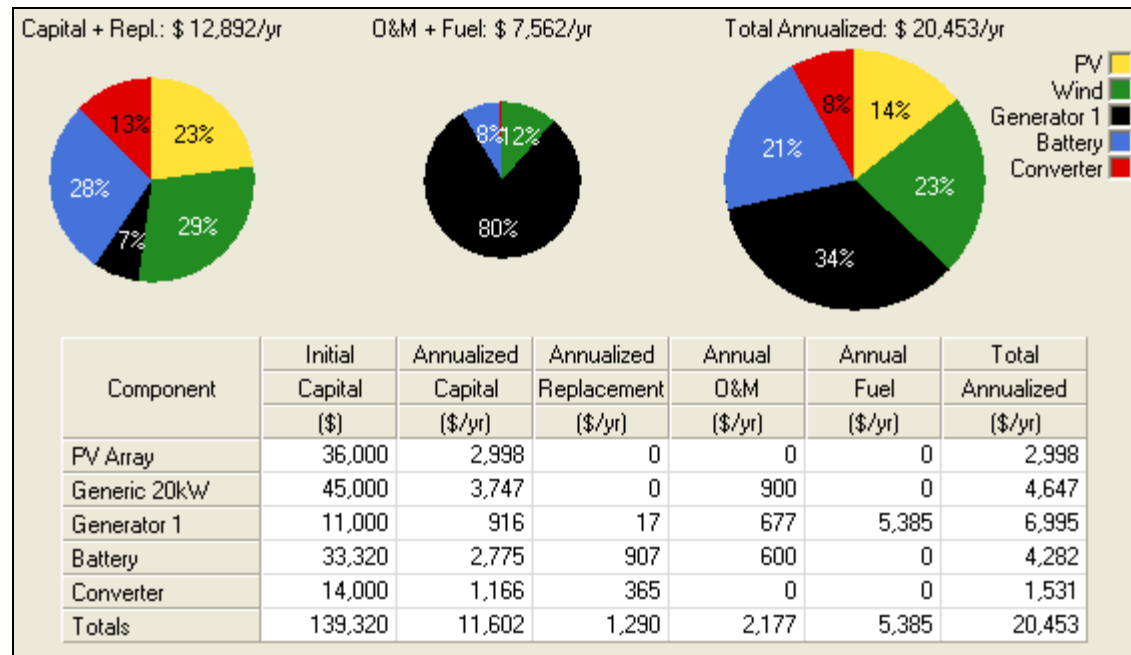


Figure 4-16 Cost summary for the 62 % renewable resource contribution for Nazret

Table 4-13 System report for the 87 % renewable resource contribution for Nazret

System architecture		Sensitivity case		Annual electric production (kWh/yr)			Annual electric energy consumption (kWh/yr)			Emissions (kg/yr)							
PV Array	20 kW	Solar Data	5.78 kWh/m ² /d	PV array	38,645	55%	AC primary load	50,772	97%	CO2	10,217						
Wind turbine	1Generic 20kW	Wind Data	3.99 m/s	Wind turbine	22,018	31%	Deferrable load	1,305	3%	CO	25.2						
Gen.	44 kW	Diesel Price	0.5 \$/L	Generator	9,521	14%	Total	52,076	100%	Unburned HC	2.79						
Battery	80 Surrette 6CS25P	PV Capital Cost Multiplier	0.6	Excess electricity	5,994		<table><tr><th colspan="2">Cost summary</th></tr><tr><td>Total NPC</td><td>\$ 303,538</td></tr><tr><td>Cost of energy</td><td>0.485 \$/kWh</td></tr></table>			Cost summary		Total NPC	\$ 303,538	Cost of energy	0.485 \$/kWh	Particulate matter	1.901
Cost summary																	
Total NPC	\$ 303,538																
Cost of energy	0.485 \$/kWh																
Inverter	40 kW	PV Replacement Cost Multiplier	0.6	Unmet load:	0		SO2	20.5									
Rectifier	40 kW			Capacity shortage	0		NOx	225									

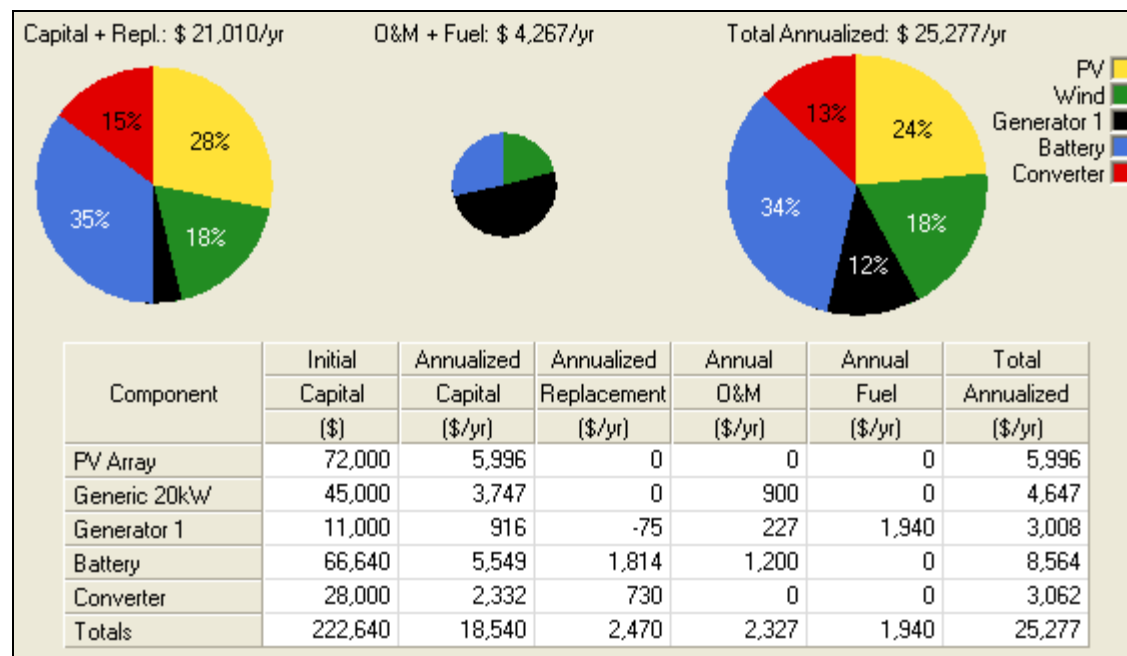


Figure 4-17 Cost summary for the 87 % renewable resource contribution for Nazret

Sensitivity analysis was also carried out and figure 4-18 shows the variation of PV capital cost multiplier against diesel price for a fixed average wind speed of 3.99 m/s and solar radiation of 5.8 kWh/m²/day. In the figure, the net present cost of the most cost-effective set-up for a particular set of diesel and PV prices is also included.

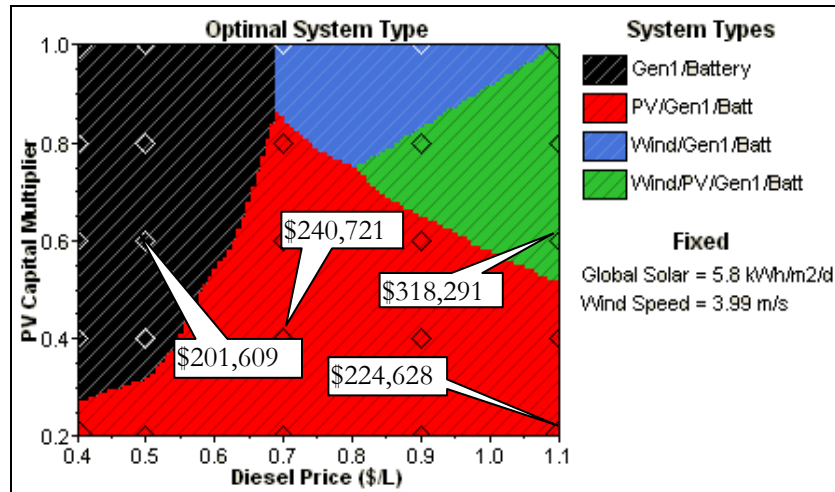


Figure 4-18 Sensitivity of PV cost to diesel price for Nazret with some important NPCs labeled

4.4 Results at Debrezeit

As carried out for the previous locations, Mekele and Nazret, the monthly average wind energy [Bekele and Palm, 2009b] and monthly average solar energy of Debrezeit [Bekele and Palm, 2009a] were fed into the software. Figures 4-19 and 4-20 show the wind and solar energy potential of Debrezeit, which was input to the software. .

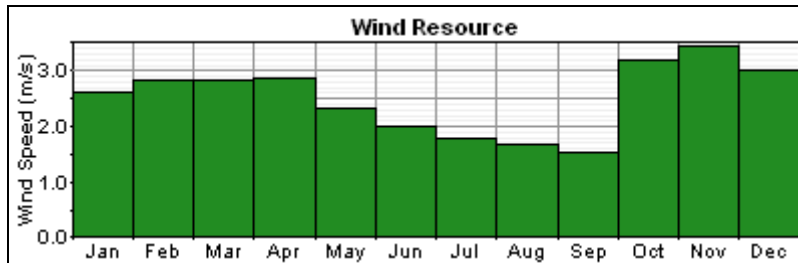


Figure 4-19 Debrezeit monthly average wind resource

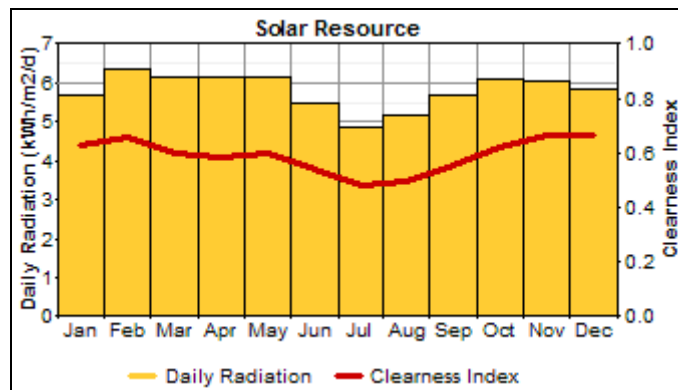


Figure 4-20 Debrezeit monthly average solar resource

The resulting list of optimal combinations of realizable setups, obtained by running the software, is given in both overall and categorized forms. Table 4-12 shows the list extracted from the complete overall table, which is found in the appendix.

The extract from the table is again based on the contribution made by renewable resources in the realizable set-ups. The set-up in the first row

is the same as for the other sites. The other set-ups are selected for having a proportion of renewable resources greater than 50 %.

Table 4-14 Extracts from the overall optimization results table for Debrezeit

PV(kW)	G20	Gen (kW)	Battery	Converter (kW)	Dispatch strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (L)	Generator (hrs)
		44	40	20	CC	\$ 58,320	\$ 201,609	0.322	0	18623	1785
20		44	40	20	LF	\$ 130,320	\$ 235,177	0.376	0.58	12078	1909
15	1	44	40	20	LF	\$ 157,320	\$ 276,081	0.441	0.53	12550	1947
20		44	80	20	LF	\$ 163,640	\$ 276,560	0.442	0.62	10617	1729
30		44	40	20	CC	\$ 166,320	\$ 278,443	0.445	0.66	13037	2127
30		44	60	40	LF	\$ 196,980	\$ 279,851	0.448	0.77	7048	1062
20	1	44	40	20	LF	\$ 175,320	\$ 285,862	0.457	0.62	11339	1811
30		44	80	40	LF	\$ 213,640	\$ 290,597	0.465	0.83	4883	716
20	1	44	60	20	LF	\$ 191,980	\$ 305,266	0.488	0.64	10417	1701
30		44	100	40	LF	\$ 230,300	\$ 310,604	0.497	0.85	4053	590

In the table the first row contains the same set-up as in the earlier tables for the other locations and it is a set-up with no contribution (0 %) made by renewable resources. The next row contains a PV-Gen-battery-Converter set-up. For just a 16.7 % increase in total NPC over the first set-up (\$201,609 to \$235,177), the percentage contribution made by renewables increased from 0 to 58 %. This is therefore an attractive set-up for implementation. Of course, there is no wind turbine involved in the system; the wind energy potential at this location is quite low, as can be seen from figure 4-19 and also from previous investigation [Bekele and Palm, 2009b].

Figure 4-21 shows the monthly average electrical production. Important information about the setup is also given in table 4-15. Figure 4-23 shows the cost breakdown supported by a pie-chart.

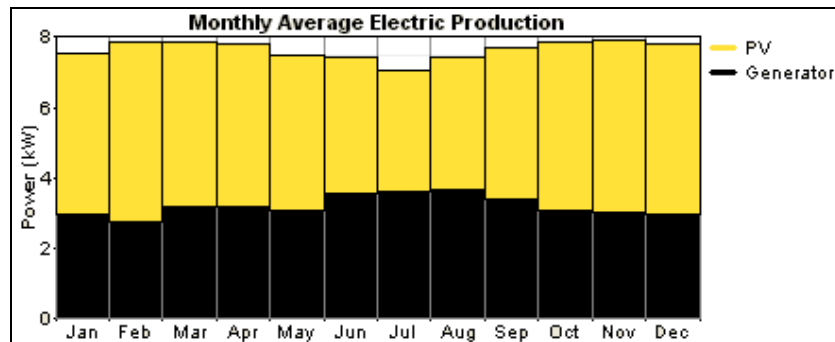


Figure 4-21 Contribution of the power units with a 58 % proportion of renewables for Debrezeit

The following set-up involves all the available power generating units; PV, wind turbine, generator, battery, and converter. For this set-up the total NPC is \$ 276,081 but with a renewable proportion of 53 %. This is an increase in NPC of over 17 % compared to the earlier setup but a 5% decrease in the contribution made by renewable resources. Unless there is a motive for using wind energy as well, this setup shouldn't be considered as a better choice.

The maximum contribution by renewables, 85 %, is achieved by the set-up given at the end of the table. For this set-up the NPC is \$310,604, which is a 32 % increase in the total NPC over the setup with a renewable contribution of 58 %. Of course, the percentage increase in the proportion of renewable resources is quite significant, 46 %. With the same arguments, set out in the previous cases, this setup can also be seen as an alternative for implementation. It should be noted that this set-up does not include a wind turbine.

The monthly average electrical production for this set-up is given in figure 4-22. Information about the system is given in table 4-16. The cost breakdown for this set-up, supported by a pie-chart, is also given in figure 4-24.

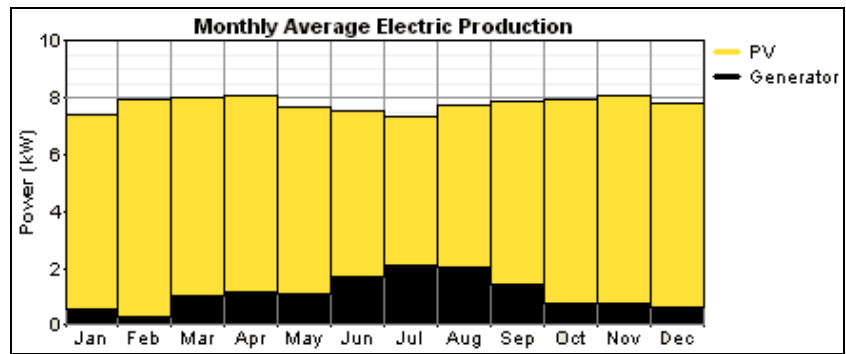


Figure 4-22 Contribution of the power units with an 85 % proportion of renewables for Debrezeit

Table 4-15 System report for the 58 % renewable resource contribution for Debrezeit

System architecture		Sensitivity case		Annual electric production (kWh/yr)			Annual electric energy consumption (kWh/yr)			Emissions (kg/yr)							
PV Array	20 kW	Solar Data	5. 81 kWh/m²/d	PV array	38,823	58%	AC primary load	50,772	97%	CO2	31,806						
Wind turbine		Wind Data	2.51 m/s	Wind turbine			Deferrable load	1,306	3%	CO	78.5						
Gen.	44 kW	Diesel Price	0.5 \$/L	Generator	28,152	42%	Total	52,077	100%	Unburned HC	8.7						
Battery	40 Surrette 6CS25P	PV Capital Cost Multiplier	0.6	Excess electricity	5,591		<table><tr><th colspan="2">Cost summary</th></tr><tr><td>Total NPC</td><td>\$ 235,177</td></tr><tr><td>Cost of energy</td><td>0. 376 \$/kWh</td></tr></table>			Cost summary		Total NPC	\$ 235,177	Cost of energy	0. 376 \$/kWh	Particulate matter	5.92
Cost summary																	
Total NPC	\$ 235,177																
Cost of energy	0. 376 \$/kWh																
Inverter	20 kW	PV Replacement Cost Multiplier	0.6	Unmet load:	0		SO2	63.9									
Rectifier	20 kW			Capacity shortage	0		NOx	701									

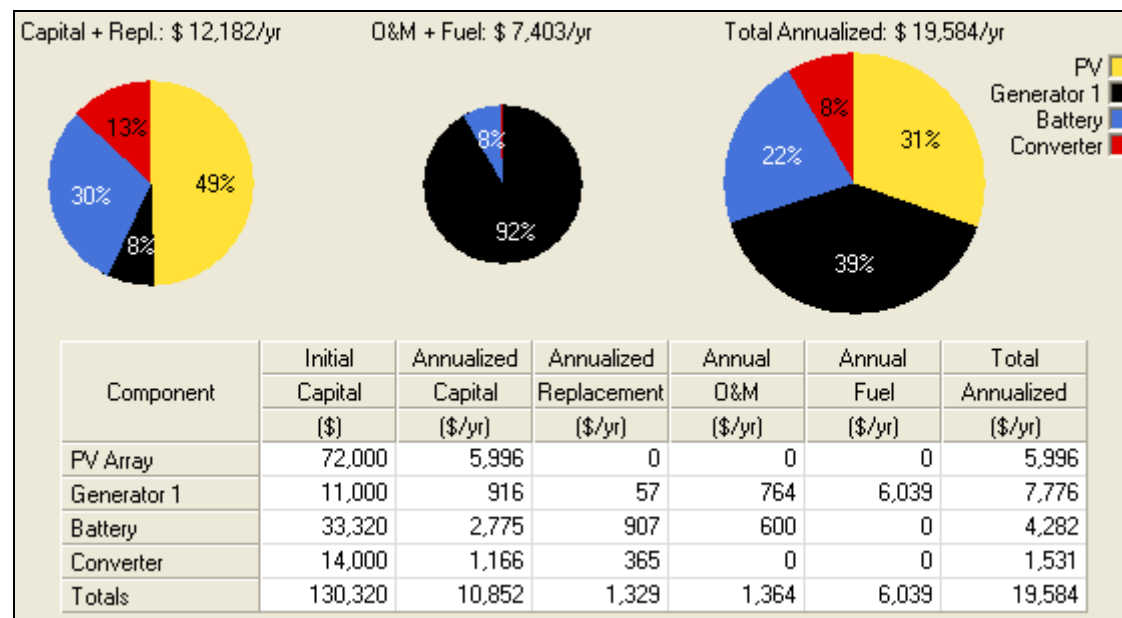


Figure 4-23 Cost summary for the 58 % renewable resource contribution for Debrezeit

Table 4-16 System report for the 85 % renewable resource contribution for Debrezeit

System architecture		Sensitivity case		Annual electric production (kWh/yr)			Annual electric energy consumption (kWh/yr)			Emissions (kg/yr)							
PV Array	30 kW	Solar Data	5. 81 kWh/m²/d	PV array	58,234	85%	AC primary load	50,772	97%	CO2	10,673						
Wind turbine		Wind Data	2.51 m/s	Wind turbine			Deferrable load	1,304	3%	CO	26.3						
Gen.	44 kW	Diesel Price	0.5 \$/L	Generator	9,981	15%	Total	52,076	100%	Unburned HC	2.92						
Battery	100 Surrette 6CS25P	PV Capital Cost Multiplier	0.6	Excess electricity	354		<table><tr><th colspan="2">Cost summary</th></tr><tr><td>Total NPC</td><td>\$ 310,604</td></tr><tr><td>Cost of energy</td><td>0. 497 \$/kWh</td></tr></table>			Cost summary		Total NPC	\$ 310,604	Cost of energy	0. 497 \$/kWh	Particulate matter	1.99
Cost summary																	
Total NPC	\$ 310,604																
Cost of energy	0. 497 \$/kWh																
Inverter	40 kW	PV Replacement Cost Multiplier	0.6	Unmet load:	0		SO2	21.4									
Rectifier	40 kW			Capacity shortage	0		NOx	235									

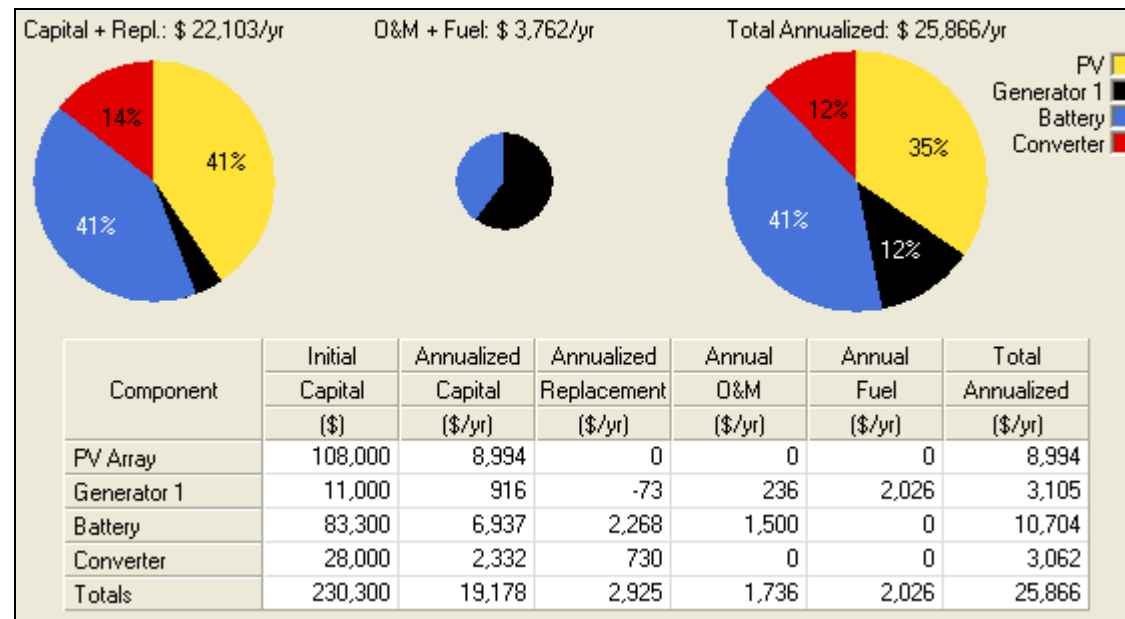


Figure 4-24 Cost summary for the 85 % renewable resource contribution for Debrezeit

The results obtained in a categorized form are given in table 4-17. The set-up in the first row is again the same as those in the first row of the other tables. The two set-ups with a 100 % renewable contribution are: the set-up with PV, Battery, and Converter and the set-up with PV, Wind-turbine, Battery, and Converter. The total NPC for the first is \$ 492,244 and that for the second is \$ 548,052. The corresponding COEs are 0.787 \$/kWh and 0.876 \$/kWh respectively.

Table 4-17 Optimization results in a Categorized form; ranking is according to the NPC of each system type

PV (kW)	G20	Gen1 (kW)	Battery	Converter (kW)	Dispatch strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (L)	Gen1 (hrs)
		44	40	20	CC	\$ 58,320	\$ 201,609	0.322	0	18,623	1,785
5		44	40	20	LF	\$ 76,320	\$ 219,243	0.351	0.17	17,886	2,375
	1	44	40	20	CC	\$ 103,320	\$ 250,466	0.401	0.07	17,572	1,694
5	1	44	40	20	LF	\$ 121,320	\$ 267,525	0.428	0.24	16,752	2,269
15		44		20	LF	\$ 79,000	\$ 343,000	0.548	0.24	37,563	5,445
10	1	44		20	LF	\$ 106,000	\$ 382,675	0.612	0.2	37,797	5,512
		44			LF	\$ 11,000	\$ 412,070	0.659	0	57,291	8,760
	1	44			LF	\$ 56,000	\$ 434,886	0.695	0.03	52,671	8,012
50			200	60	CC	\$ 388,600	\$ 492,244	0.787	1		
50	1		200	60	CC	\$ 433,600	\$ 548,052	0.876	1		

With a similar argument to that stated in the previous cases, these set-ups could also be considered for implementation. Otherwise, the NPC for each does indeed seem significantly high and compared to the set-up with a 58 % proportion of renewables, discussed earlier, it is again more than double.

In the sensitivity analysis, figure 4-25 shows the PV capital cost multiplier against diesel price. Here again, the net present cost of the most cost effective set-up for a particular set of diesel and PV price is shown.

In this figure it can be seen that the wind plays no role in the supply of energy to the community. This is understandable as the annual average wind speed at the location is just 2.51 m/s.

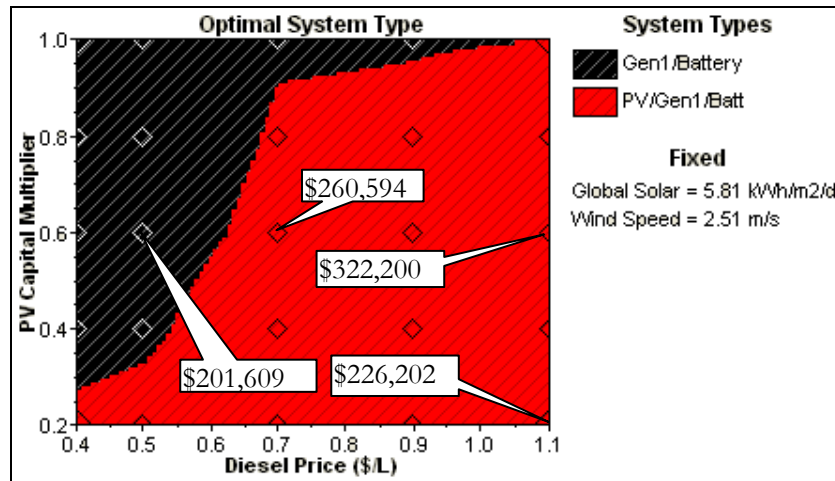


Figure 4-25 Sensitivity of PV cost to diesel price for Debrezeit with some important NPCs labeled

5 Conclusion

In simple terms what has been accomplished in this work is firstly the determination of solar and wind energy potentials at four typical locations in Ethiopia. Then, based on these potentials, a feasibility study for a standalone electric power supply system for a model community of 200 families in a village as been conducted.

In determining the wind energy potential of the sites, the study is based on relatively recent (2000 – 2003) synoptic wind data obtained from the NMSA. The data was recorded only five times a day from dawn to dusk and no data was recorded during the night. Hence, a way needed to be found to compensate for the missing night-time data and this is one of the core components of this study. The method is well explained in the published paper [Bekele and Palm, 2009b] and the authors believe that this method is of great importance to researchers working on similar issues in most developing countries, where properly recorded data is not available. A piece of software (HOMER) was used as an aid for the study. The results obtained have been confirmed by recovering the measured daytime data to an accuracy of better than 2%.

From the results, the wind energy potential of one of the sites, Debrezeit, is considerably lower than the other three locations. However, it can be concluded that, generally-speaking, although the potential may not be sufficient for a large, independent wind energy farm, the analysis has shown that wind energy may in some cases be a viable option if integrated into other energy conversion systems such as PV, diesel generator and battery. The results of this study can be considered to be applicable to most regions in the country with similar climatic conditions.

Regarding solar energy potential, as in the case of the wind, there is no accurately measured solar radiation database. Only sunshine hour data was available. Therefore, mathematical models able to incorporate the available sunshine hour data and provide the required solar radiation data were used for determining the potential at each location. The findings were also cross-checked against satellite data obtained from other sources [NASA, 2008] and [Meteonorm, Ver. 5.1]. The results

demonstrated the availability of extensive utilizable solar energy at each location.

The feasibility study for the hybrid system is based on the findings of the wind and solar energy potentials at the particular locations. With the potentials determined, three different approaches have been followed in the hybrid system design. In the first approach, system components that are commonly available have been included in the design without much concern about the efficiency. This is done for both the load and the supply side. In the second approach [Bekele and Palm, 2009d] with a thorough market-survey the best technologies available are compared and those with the highest efficiency are selected. A third approach followed is to see if cost is minimized by considering a self contained system, i.e., every household having its own supply system [Bekele and Palm, 2009d]. The results obtained in the second approach have shown that the net present cost is less than 50 % of that for the first approach. The third approach is found to be of a higher cost.

In the results, numerous alternative feasible hybrid set-ups, with different levels of contribution by the renewable resources, were obtained. Despite the numerous alternatives, the choice is restricted by the varying net present cost of each set-up. Compared to the current global electricity tariff [Wikipedia, 2008] and the tariff in the country (<5 cents) where the main source of electric energy is hydro-power; the costs of the feasible set-ups obtained in this study are high, in the range of 30 to 40 cents per kilowatt hour and this is slightly worrisome. However, considering the shortage of power in the country (only 15 % coverage), its role in the protection of vegetation and forestry and therefore the prevention of soil degradation, the improvement to the quality of life of the many people residing in the countryside, the future situation regarding fossil fuel sources, and its contribution to the reduction of pollutant emissions into the environment, this cost should not be seen as a significant impairment.

It should also be noted that free solar and wind energy will also be utilized, load will be satisfied in an optimal way; help is given to the mobilization of investments towards clean energy; and, most of all, the poor will benefit from the electric light provided.

Nomenclature

A	Area	(m ²)
A_c	Array area	(m ²)
COE	Cost of energy	\$/kWh
c	Speed of light	m/s)
c'	A scale parameter (Eq. 1-6)	(m/s)
E_g	Band gap energy	(eV)
E_a	Armature voltage	(V)
E_{ph}	Photon energy	
f_j	Frequency of occurrence in the j th class (Eq. 1-7)	
G_{sc}	The solar constant =1367	(W/m ²)
G_T	Incident solar radiation on the array	(W/m ²)
g	Gravitational constant	(m/s ²)
h	Planck constant	
$\overline{H}$	Monthly average daily radiation on a horizontal surface	(MJ/m ²)
$\overline{H}_0$	Monthly average daily extraterrestrial radiation on a horizontal surface	(MJ/m ²)
$\overline{H}_e$	Mean value of the estimated radiation	(MJ/m ²)
$\overline{H}_{ob}$	Mean value of the obtained radiation	(MJ/m ²)
I	Load current	(A)
I_a	Armature current	(A)
I_D	Diode current	(A)
$I_{sc}(G)$	Maximum power current	(A)
I_{sh}	Shunt resistance current	(A)
I_L	Current produced by the cell	(A)
I_{mp}	Maximum power point	(A)
I_o	Reverse saturation current of diode	(A)
jX_s	Synchronous reactance	(Ω)

K	Boltzmann's constant	
k	A constant known as shape factor(Eq. 1-6)	
M	Number of data points (Eq. 2-7)	
m	Diode quality factor	
N	Number of photons	
NPC	<i>Net present cost</i>	(\$)
$\bar{n}$	Monthly average daily number of hours of bright sunshine	
n	Number of wind speed readings (Eq. 1-2)	
n_s	Synchronous speed	(RPM)
P	Power	(W)
P_o	Standard sea level atmospheric pressure	(Pa)
p_l	Number of pole pairs	
P_{mp}	Maximum power point	(W)
p_r	Pressure	(Pa)
P_o	Standard sea level atmospheric pressure	(Pa)
q	Charge on an electron	(C)
R	Specific gas constant	(Jkg ⁻¹ K ⁻¹)
R_a	Armature winding resistance	(Ω)
T	Temperature	(^o K)
u	Wind velocity	(m/s)
$\bar{u}$	Mean wind speed	(m/s)
u_j	The j th wind speed	(m/s)
V	Output voltage	(V)
V_a	Terminal voltage	(V)
V_j	Median velocity in class j	(m/s)
V_{mp}	Maximum power voltage	(V)
V_{oc}	Open circuit voltage	(V)
$\tilde{z}$	Region's elevation	(m)
$\tilde{z}_0$	Roughness length (Eq. 1-9)	(m)

z_r	Reference height (Eq. 1-9)	(m)
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Greek alphabets

δ	Declination angle	
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ϕ	Latitude angle	
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λ	The wavelength of light	(m/s)
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η_{mp}	Maximum power point efficiency (Eq. 3-5)	
-------------	--	--

η_e	Efficiency of power conditioning equipment (Eq. 3-5)	
----------	--	--

ρ	Density of the air	kg/m ³
--------	--------------------	-------------------

ρ_j	The j th readings of the air density	kg/m ³
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σ_u	Standard deviation of wind speed	m/s
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ω	The sunset hour angle	
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Appendix A:

Overall Optimization Results Tables

Table 5-1 Overall optimization results table for Addis Ababa

	PV(kW)	G20	Gen (kW)	Battery	Converter (kW)	Dispatch strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (L)	Generator (hrs)
			44	40	20	CC	\$ 58,320	\$ 201,609	0.322	0	18,623	1,785
5			44	40	20	LF	\$ 76,320	\$ 220,728	0.353	0.16	18,115	2,391
			44	40	20	LF	\$ 58,320	\$ 222,616	0.356	0	21,056	2,725
10			44	40	20	LF	\$ 94,320	\$ 226,668	0.362	0.3	16,217	2,293
			44	60	20	CC	\$ 74,980	\$ 227,227	0.363	0	18,605	1,788
5			44	60	20	CC	\$ 92,980	\$ 230,456	0.369	0.15	16,320	1,638
15			44	40	20	LF	\$ 112,320	\$ 231,855	0.371	0.43	14,260	2,137
	1		44	40	20	LF	\$ 103,320	\$ 233,435	0.373	0.39	14,355	2,021
5			44	40	40	LF	\$ 90,320	\$ 238,842	0.382	0.16	18,083	2,379
5	1		44	40	20	LF	\$ 121,320	\$ 239,756	0.383	0.51	12,599	1,858
20			44	40	20	LF	\$ 130,320	\$ 240,104	0.384	0.54	12,815	1,981
10			44	40	40	LF	\$ 108,320	\$ 240,814	0.385	0.3	15,726	2,104
			44	40	40	LF	\$ 72,320	\$ 241,008	0.386	0	21,057	2,725
15			44	40	40	LF	\$ 126,320	\$ 245,888	0.393	0.43	13,780	1,924
5			44	60	20	LF	\$ 92,980	\$ 246,224	0.394	0.16	18,084	2,387
			44	40	40	CC	\$ 72,320	\$ 247,995	0.397	0	22,461	2,508
			44	60	20	LF	\$ 74,980	\$ 248,124	0.397	0	21,027	2,721
10	1		44	40	20	LF	\$ 139,320	\$ 248,587	0.398	0.61	11,214	1,736

	1	44	40	40	LF	\$ 117,320	\$ 248,751	0.398	0.4	13,987	1,897
		88	40	20	CC	\$ 69,320	\$ 249,872	0.4	0	23,503	1,712
10		44	60	20	LF	\$ 110,980	\$ 252,225	0.403	0.3	16,194	2,291
5		88	40	20	CC	\$ 87,320	\$ 252,616	0.404	0.15	21,112	1,648
5	1	44	40	40	LF	\$ 135,320	\$ 252,697	0.404	0.52	11,931	1,654
		44	80	20	CC	\$ 91,640	\$ 253,015	0.405	0	18,619	1,787
20		44	40	40	LF	\$ 144,320	\$ 255,595	0.409	0.53	12,537	1,805
15		44	60	20	LF	\$ 128,980	\$ 255,715	0.409	0.44	13,989	2,105
5		44	80	20	CC	\$ 109,640	\$ 256,104	0.41	0.15	16,315	1,634
5		44	40	60	LF	\$ 104,320	\$ 257,228	0.411	0.16	18,083	2,379
	1	44	60	20	LF	\$ 119,980	\$ 257,499	0.412	0.4	14,109	1,997
5		88	40	20	LF	\$ 87,320	\$ 257,829	0.412	0.15	21,601	1,810
20		44	60	20	LF	\$ 146,980	\$ 258,016	0.413	0.56	11,679	1,841
	1	44	60	20	CC	\$ 119,980	\$ 258,420	0.413	0.35	14,793	1,529
10		44	40	60	LF	\$ 122,320	\$ 259,155	0.414	0.3	15,721	2,102
		44	40	60	LF	\$ 86,320	\$ 259,394	0.415	0	21,057	2,725
10	1	44	40	40	LF	\$ 153,320	\$ 259,616	0.415	0.62	10,271	1,470
5		44	40	20	CC	\$ 76,320	\$ 259,965	0.416	0.14	23,348	3,550
15	1	44	40	20	LF	\$ 157,320	\$ 260,371	0.416	0.68	10,288	1,643
10		88	40	20	LF	\$ 105,320	\$ 260,442	0.416	0.29	19,446	1,636
10		44	40	20	CC	\$ 94,320	\$ 261,755	0.419	0.26	20,956	3,289
	1	44	40	20	CC	\$ 103,320	\$ 262,600	0.42	0.35	18,327	2,810
5	1	44	60	20	LF	\$ 137,980	\$ 262,614	0.42	0.52	12,176	1,813
15		44	40	20	CC	\$ 112,320	\$ 262,902	0.42	0.38	18,524	2,951
15		44	40	60	LF	\$ 140,320	\$ 264,160	0.422	0.43	13,767	1,919
5		44	60	40	LF	\$ 106,980	\$ 264,322	0.423	0.16	18,051	2,374
		44	40	60	CC	\$ 86,320	\$ 266,036	0.425	0	22,417	2,496
10		88	40	20	CC	\$ 105,320	\$ 266,165	0.426	0.27	20,366	1,650
10		44	60	40	LF	\$ 124,980	\$ 266,210	0.426	0.3	15,684	2,095
30		44	40	20	LF	\$ 166,320	\$ 266,410	0.426	0.67	11,390	1,818
		44	60	40	LF	\$ 88,980	\$ 266,521	0.426	0	21,029	2,721
20		44	40	20	CC	\$ 130,320	\$ 266,677	0.426	0.48	16,486	2,650
5	1	44	40	20	CC	\$ 121,320	\$ 266,760	0.427	0.46	16,274	2,582
	1	44	40	60	LF	\$ 131,320	\$ 267,114	0.427	0.4	13,985	1,896
10	1	44	40	20	CC	\$ 139,320	\$ 267,141	0.427	0.56	13,752	2,215
	2	44	40	20	LF	\$ 148,320	\$ 267,956	0.428	0.63	11,228	1,662
15		44	60	40	LF	\$ 142,980	\$ 268,208	0.429	0.44	13,340	1,817
15	1	44	40	40	LF	\$ 171,320	\$ 268,568	0.429	0.7	8,917	1,309
20		44	60	20	CC	\$ 146,980	\$ 269,487	0.431	0.53	13,308	2,085
15		44	60	20	CC	\$ 128,980	\$ 270,116	0.432	0.41	16,007	2,441
5	1	44	40	60	LF	\$ 149,320	\$ 271,037	0.433	0.52	11,925	1,652
		88	60	20	CC	\$ 85,980	\$ 271,307	0.434	0	22,958	1,641
20		44	60	40	LF	\$ 160,980	\$ 271,430	0.434	0.56	11,160	1,572

10	1	44	60	20	LF	\$ 155,980	\$ 271,629	0.434	0.62	10,821	1,692
5		44	80	20	LF	\$ 109,640	\$ 271,775	0.435	0.16	18,060	2,385
	1	44	60	40	LF	\$ 133,980	\$ 272,233	0.435	0.4	13,683	1,840
15	1	44	40	20	CC	\$ 157,320	\$ 273,502	0.437	0.64	12,093	1,971
15		88	40	40	LF	\$ 137,320	\$ 273,506	0.437	0.42	16,117	1,351
		44	80	20	LF	\$ 91,640	\$ 273,732	0.438	0	21,010	2,721
5	1	44	60	40	LF	\$ 151,980	\$ 273,798	0.438	0.53	11,280	1,544
20		44	40	60	LF	\$ 158,320	\$ 273,844	0.438	0.53	12,522	1,799
10		44	60	20	CC	\$ 110,980	\$ 274,101	0.438	0.27	19,217	2,849
	1	88	40	20	LF	\$ 114,320	\$ 275,047	0.44	0.36	18,718	1,572
20	1	44	40	20	LF	\$ 175,320	\$ 275,203	0.44	0.72	9,819	1,591
10		88	40	40	LF	\$ 119,320	\$ 275,300	0.44	0.29	18,968	1,587
5		44	40	80	LF	\$ 118,320	\$ 275,614	0.441	0.16	18,083	2,379
20		88	40	40	LF	\$ 155,320	\$ 277,117	0.443	0.53	14,046	1,179
	1	88	40	20	CC	\$ 114,320	\$ 277,293	0.443	0.34	19,195	1,517
10		44	40	80	LF	\$ 136,320	\$ 277,541	0.444	0.3	15,721	2,102
5		88	60	20	CC	\$ 103,980	\$ 277,605	0.444	0.15	21,033	1,631
10	1	44	60	40	LF	\$ 169,980	\$ 277,664	0.444	0.64	9,176	1,279
10		44	80	20	LF	\$ 127,640	\$ 277,718	0.444	0.3	16,163	2,287
		44	40	80	LF	\$ 100,320	\$ 277,780	0.444	0	21,057	2,725
15		88	40	20	LF	\$ 123,320	\$ 277,952	0.444	0.38	19,376	1,631
10	1	44	40	60	LF	\$ 167,320	\$ 277,981	0.445	0.62	10,269	1,469
5	2	44	40	20	LF	\$ 166,320	\$ 278,329	0.445	0.7	10,075	1,553
5	1	44	60	20	CC	\$ 137,980	\$ 278,729	0.446	0.48	14,411	2,203
		44	100	20	CC	\$ 108,300	\$ 278,746	0.446	0	18,628	1,782
5	1	88	40	40	LF	\$ 146,320	\$ 279,636	0.447	0.51	14,149	1,187
5		88	40	40	LF	\$ 101,320	\$ 279,989	0.448	0.15	22,137	1,850
	2	44	40	40	LF	\$ 162,320	\$ 280,122	0.448	0.64	10,408	1,449
5		88	60	20	LF	\$ 103,980	\$ 280,643	0.449	0.16	21,196	1,777
20	1	44	40	40	LF	\$ 189,320	\$ 281,170	0.45	0.75	8,105	1,216
15		44	80	20	LF	\$ 145,640	\$ 281,214	0.45	0.44	13,959	2,101
		88	40	20	LF	\$ 69,320	\$ 281,406	0.45	0	27,444	2,281
5		44	60	40	CC	\$ 106,980	\$ 281,470	0.45	0.13	20,931	2,353
30		44	40	40	LF	\$ 180,320	\$ 281,673	0.45	0.66	11,043	1,640
	1	88	40	40	LF	\$ 128,320	\$ 281,721	0.451	0.38	17,041	1,427
10	1	44	60	20	CC	\$ 155,980	\$ 281,784	0.451	0.59	12,245	1,922
5		44	100	20	CC	\$ 126,300	\$ 281,787	0.451	0.15	16,312	1,633
10	1	88	40	40	LF	\$ 164,320	\$ 282,000	0.451	0.61	11,901	999
	1	44	80	20	LF	\$ 136,640	\$ 282,431	0.452	0.4	13,995	1,984
15		44	40	80	LF	\$ 154,320	\$ 282,546	0.452	0.43	13,767	1,919
20		44	80	20	LF	\$ 163,640	\$ 282,557	0.452	0.57	11,508	1,821
15		88	40	20	CC	\$ 123,320	\$ 282,616	0.452	0.37	20,125	1,643
15	1	44	60	40	LF	\$ 187,980	\$ 282,681	0.452	0.73	7,252	1,027

5		44	60	60	LF	\$ 120,980	\$ 282,708	0.452	0.16	18,051	2,374
5		44	40	40	CC	\$ 90,320	\$ 282,881	0.452	0.13	24,086	3,564
	1	44	80	20	CC	\$ 136,640	\$ 283,686	0.454	0.35	14,739	1,508
15	1	44	60	20	LF	\$ 173,980	\$ 283,799	0.454	0.69	9,956	1,602
		44	60	40	CC	\$ 88,980	\$ 283,880	0.454	0	23,972	2,674
30		44	60	20	LF	\$ 182,980	\$ 284,412	0.455	0.69	10,265	1,682
		44	40	80	CC	\$ 100,320	\$ 284,422	0.455	0	22,417	2,496
10		44	60	60	LF	\$ 138,980	\$ 284,551	0.455	0.3	15,679	2,093
30		44	40	20	CC	\$ 166,320	\$ 284,618	0.455	0.62	13,922	2,253
		44	60	60	LF	\$ 102,980	\$ 284,907	0.456	0	21,029	2,721
10		44	40	40	CC	\$ 108,320	\$ 284,939	0.456	0.25	21,743	3,265
	1	44	40	80	LF	\$ 145,320	\$ 285,500	0.457	0.4	13,985	1,896
10		88	60	20	LF	\$ 121,980	\$ 285,661	0.457	0.29	19,376	1,631
20	1	44	40	20	CC	\$ 175,320	\$ 285,860	0.457	0.7	11,297	1,847
15		44	60	60	LF	\$ 156,980	\$ 286,526	0.458	0.44	13,332	1,814
5	1	88	40	20	LF	\$ 132,320	\$ 286,784	0.459	0.45	17,812	1,499
15	1	44	40	60	LF	\$ 185,320	\$ 286,911	0.459	0.7	8,911	1,307
5	1	44	80	20	LF	\$ 154,640	\$ 287,191	0.459	0.53	12,011	1,793
30		44	60	40	LF	\$ 196,980	\$ 287,613	0.46	0.72	8,204	1,207
	1	44	40	40	CC	\$ 117,320	\$ 287,632	0.46	0.33	19,367	2,871
5	2	44	40	40	LF	\$ 180,320	\$ 287,708	0.46	0.72	8,851	1,261
15	1	88	40	40	LF	\$ 182,320	\$ 288,248	0.461	0.69	10,210	858
15	1	44	60	20	CC	\$ 173,980	\$ 288,870	0.462	0.67	10,674	1,710
20		44	80	20	CC	\$ 163,640	\$ 288,913	0.462	0.55	12,432	1,937
5	1	44	40	80	LF	\$ 163,320	\$ 289,424	0.463	0.52	11,925	1,652
	2	44	60	20	LF	\$ 164,980	\$ 289,485	0.463	0.65	10,608	1,597
20		44	60	60	LF	\$ 174,980	\$ 289,665	0.463	0.56	11,141	1,565
5		44	80	40	LF	\$ 123,640	\$ 289,811	0.463	0.16	18,020	2,369
20	1	44	60	40	LF	\$ 205,980	\$ 289,942	0.464	0.81	5,662	817
10	2	44	40	20	LF	\$ 184,320	\$ 290,411	0.464	0.75	9,177	1,460
	1	44	60	60	LF	\$ 147,980	\$ 290,573	0.465	0.4	13,677	1,838
5	1	88	40	20	CC	\$ 132,320	\$ 290,945	0.465	0.43	18,492	1,506
	2	44	40	20	CC	\$ 148,320	\$ 291,043	0.465	0.58	14,401	2,244
5	1	44	40	40	CC	\$ 135,320	\$ 291,252	0.466	0.44	17,253	2,616
10		88	60	20	CC	\$ 121,980	\$ 291,371	0.466	0.27	20,325	1,632
10		44	80	40	LF	\$ 141,640	\$ 291,693	0.466	0.3	15,652	2,090
30		44	60	20	CC	\$ 182,980	\$ 291,747	0.467	0.67	11,323	1,822
15		88	40	60	LF	\$ 151,320	\$ 291,893	0.467	0.42	16,118	1,351
		44	80	40	LF	\$ 105,640	\$ 291,983	0.467	0	20,994	2,715
15		44	80	20	CC	\$ 145,640	\$ 292,009	0.467	0.41	15,512	2,316
5	1	44	60	60	LF	\$ 165,980	\$ 292,119	0.467	0.53	11,272	1,541
20		44	40	80	LF	\$ 172,320	\$ 292,230	0.467	0.53	12,522	1,799
20		88	60	40	LF	\$ 171,980	\$ 293,210	0.469	0.55	12,664	1,063

15		44	80	40	LF	\$ 159,640	\$ 293,673	0.47	0.44	13,307	1,811
10		88	40	60	LF	\$ 133,320	\$ 293,686	0.47	0.29	18,968	1,587
5		44	40	100	LF	\$ 132,320	\$ 294,000	0.47	0.16	18,083	2,379
10		44	80	20	CC	\$ 127,640	\$ 294,077	0.47	0.27	18,515	2,620
15		88	60	40	LF	\$ 153,980	\$ 295,110	0.472	0.43	15,528	1,301
20		88	40	60	LF	\$ 169,320	\$ 295,505	0.473	0.53	14,046	1,179
	1	44	60	40	CC	\$ 133,980	\$ 295,533	0.473	0.34	17,197	2,158
		88	60	20	LF	\$ 85,980	\$ 295,644	0.473	0	25,829	2,151
20		44	80	40	LF	\$ 177,640	\$ 295,651	0.473	0.56	10,952	1,530
20		88	40	20	LF	\$ 141,320	\$ 295,867	0.473	0.45	19,364	1,630
10		44	40	100	LF	\$ 150,320	\$ 295,927	0.473	0.3	15,721	2,102
10	1	44	60	60	LF	\$ 183,980	\$ 296,028	0.473	0.64	9,173	1,278
		44	40	100	LF	\$ 114,320	\$ 296,166	0.474	0	21,057	2,725
10	1	44	80	20	LF	\$ 172,640	\$ 296,256	0.474	0.62	10,664	1,673
10		44	60	40	CC	\$ 124,980	\$ 296,275	0.474	0.25	20,088	2,631
10	1	44	40	80	LF	\$ 181,320	\$ 296,367	0.474	0.62	10,269	1,469
		88	80	20	CC	\$ 102,640	\$ 296,603	0.474	0	22,913	1,631
5	2	44	40	20	CC	\$ 166,320	\$ 296,731	0.474	0.66	12,599	2,026
	2	44	60	40	LF	\$ 178,980	\$ 296,853	0.475	0.67	9,102	1,248
10	2	44	40	40	LF	\$ 198,320	\$ 297,172	0.475	0.78	7,574	1,108
5		44	100	20	LF	\$ 126,300	\$ 297,333	0.475	0.16	18,037	2,383
	1	44	80	40	LF	\$ 150,640	\$ 297,352	0.475	0.4	13,602	1,825
10	1	44	40	40	CC	\$ 153,320	\$ 297,726	0.476	0.53	15,415	2,394
10		88	60	40	LF	\$ 135,980	\$ 297,896	0.476	0.3	18,520	1,550
5	1	88	40	60	LF	\$ 160,320	\$ 298,188	0.477	0.51	14,173	1,189
5	1	44	80	20	CC	\$ 154,640	\$ 298,262	0.477	0.49	13,588	2,024
5		88	40	60	LF	\$ 115,320	\$ 298,375	0.477	0.15	22,137	1,850
20	1	44	60	20	LF	\$ 191,980	\$ 298,383	0.477	0.73	9,443	1,544
	2	44	40	60	LF	\$ 176,320	\$ 298,508	0.477	0.64	10,408	1,449
5	1	44	80	40	LF	\$ 168,640	\$ 298,599	0.477	0.53	11,148	1,523
5	1	44	60	40	CC	\$ 151,980	\$ 298,713	0.478	0.46	14,919	1,992
10	1	88	60	40	LF	\$ 180,980	\$ 298,816	0.478	0.63	10,622	892
5	1	88	60	40	LF	\$ 162,980	\$ 298,896	0.478	0.52	13,221	1,110
	1	88	60	20	LF	\$ 130,980	\$ 299,038	0.478	0.36	18,471	1,551
		44	100	20	LF	\$ 108,300	\$ 299,284	0.479	0	20,986	2,719
30		88	40	40	LF	\$ 191,320	\$ 299,391	0.479	0.66	12,072	1,014
20	1	44	40	60	LF	\$ 203,320	\$ 299,591	0.479	0.75	8,111	1,216
20		88	40	20	CC	\$ 141,320	\$ 299,837	0.479	0.44	20,009	1,637
30		44	40	60	LF	\$ 194,320	\$ 299,967	0.48	0.66	11,033	1,636
20	1	88	40	40	LF	\$ 200,320	\$ 300,009	0.48	0.74	9,312	783
	1	88	40	60	LF	\$ 142,320	\$ 300,107	0.48	0.38	17,041	1,427
10	1	88	40	60	LF	\$ 178,320	\$ 300,303	0.48	0.61	11,889	998
5	2	44	60	20	LF	\$ 182,980	\$ 300,481	0.481	0.71	9,542	1,490

15	1	88	60	40	LF	\$ 198,980	\$ 300,620	0.481	0.73	8,295	696
5		44	60	60	CC	\$ 120,980	\$ 300,645	0.481	0.13	21,036	2,376
30		44	80	40	LF	\$ 213,640	\$ 300,698	0.481	0.77	6,379	914
10	1	44	80	40	LF	\$ 186,640	\$ 300,866	0.481	0.65	8,812	1,222
		44	80	40	CC	\$ 105,640	\$ 300,887	0.481	0	23,005	2,243
		88	40	40	LF	\$ 83,320	\$ 300,892	0.481	0	27,598	2,294
15		44	40	100	LF	\$ 168,320	\$ 300,932	0.481	0.43	13,767	1,919
15	1	44	60	60	LF	\$ 201,980	\$ 300,959	0.481	0.73	7,239	1,022
5		44	60	80	LF	\$ 134,980	\$ 301,094	0.481	0.16	18,051	2,374
5		44	40	60	CC	\$ 104,320	\$ 301,644	0.482	0.13	24,133	3,577

Table 5-2 Overall optimization results table for Mekele

	PV(kW)	G20	Gen (kW)	Battery	Converter (kW)	Dispatch strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (L)	Generator (hrs)
			44	40	20	CC	\$ 58,320	\$ 201,609	0.322	0	18,623	1,785
5			44	40	20	LF	\$ 76,320	\$ 219,157	0.35	0.17	17,872	2,375
			44	40	20	LF	\$ 58,320	\$ 222,616	0.356	0	21,056	2,725
10			44	40	20	LF	\$ 94,320	\$ 223,862	0.358	0.33	15,777	2,269
			44	60	20	CC	\$ 74,980	\$ 227,227	0.363	0	18,605	1,788
20			44	40	20	LF	\$ 130,320	\$ 234,980	0.376	0.58	12,049	1,906
10			44	40	40	LF	\$ 108,320	\$ 237,301	0.379	0.33	15,203	2,050
5			44	40	40	LF	\$ 90,320	\$ 237,323	0.38	0.17	17,846	2,365
	1		44	40	20	CC	\$ 103,320	\$ 237,336	0.38	0.28	15,520	1,574
			44	40	40	LF	\$ 72,320	\$ 241,008	0.386	0	21,057	2,725
	1		44	40	20	LF	\$ 103,320	\$ 242,962	0.389	0.3	15,799	2,146
5			44	60	20	LF	\$ 92,980	\$ 244,680	0.391	0.17	17,844	2,372
5	1		44	40	20	LF	\$ 121,320	\$ 245,448	0.392	0.45	13,467	1,927
			44	40	40	CC	\$ 72,320	\$ 247,995	0.397	0	22,461	2,508
			44	60	20	LF	\$ 74,980	\$ 248,124	0.397	0	21,027	2,721
10			44	60	20	LF	\$ 110,980	\$ 249,382	0.399	0.33	15,749	2,266
			88	40	20	CC	\$ 69,320	\$ 249,872	0.4	0	23,503	1,712
20			44	40	40	LF	\$ 144,320	\$ 250,763	0.401	0.57	11,817	1,733
5			88	40	20	CC	\$ 87,320	\$ 251,474	0.402	0.17	20,934	1,643
20			44	60	20	LF	\$ 146,980	\$ 251,496	0.402	0.61	10,716	1,736
10	1		44	40	20	LF	\$ 139,320	\$ 252,681	0.404	0.57	11,820	1,801
			44	80	20	CC	\$ 91,640	\$ 253,015	0.405	0	18,619	1,787
5			88	40	20	LF	\$ 87,320	\$ 253,746	0.406	0.17	21,026	1,765
5			44	80	20	CC	\$ 109,640	\$ 255,382	0.408	0.17	16,185	1,642
10			44	40	60	LF	\$ 122,320	\$ 255,619	0.409	0.33	15,195	2,047
5			44	40	60	LF	\$ 104,320	\$ 255,709	0.409	0.17	17,846	2,365
5			44	40	20	CC	\$ 76,320	\$ 258,890	0.414	0.15	23,166	3,552
			44	40	60	LF	\$ 86,320	\$ 259,394	0.415	0	21,057	2,725
	1		44	40	40	LF	\$ 117,320	\$ 259,584	0.415	0.3	15,591	2,071
10			44	40	20	CC	\$ 94,320	\$ 259,643	0.415	0.29	20,628	3,269

20		44	60	20	CC	\$ 146,980	\$ 259,829	0.415	0.59	11,900	1,911
10		88	40	20	LF	\$ 105,320	\$ 259,953	0.416	0.31	19,377	1,631
20		44	40	20	CC	\$ 130,320	\$ 260,439	0.416	0.52	15,573	2,536
5	1	44	40	40	LF	\$ 135,320	\$ 260,735	0.417	0.45	13,109	1,791
30		44	40	20	LF	\$ 166,320	\$ 261,958	0.419	0.71	10,731	1,747
	1	44	60	20	CC	\$ 119,980	\$ 262,150	0.419	0.28	15,418	1,525
10		44	60	40	LF	\$ 124,980	\$ 262,646	0.42	0.33	15,155	2,039
5		44	60	40	LF	\$ 106,980	\$ 262,801	0.42	0.17	17,813	2,360
20		44	60	40	LF	\$ 160,980	\$ 264,117	0.422	0.61	10,072	1,434
10	1	44	40	40	LF	\$ 153,320	\$ 264,201	0.422	0.57	10,966	1,543
10		88	40	20	CC	\$ 105,320	\$ 265,056	0.424	0.3	20,193	1,645
		44	40	60	CC	\$ 86,320	\$ 266,036	0.425	0	22,417	2,496
		44	60	40	LF	\$ 88,980	\$ 266,521	0.426	0	21,029	2,721
	1	44	60	20	LF	\$ 119,980	\$ 267,569	0.428	0.3	15,635	2,129
10		44	60	20	CC	\$ 110,980	\$ 268,547	0.429	0.3	18,412	2,740
20		44	40	60	LF	\$ 158,320	\$ 269,011	0.43	0.57	11,801	1,727
5	1	44	60	20	LF	\$ 137,980	\$ 269,552	0.431	0.45	13,225	1,905
20		88	40	40	LF	\$ 155,320	\$ 269,739	0.431	0.58	12,983	1,091
5		44	80	20	LF	\$ 109,640	\$ 270,232	0.432	0.17	17,820	2,370
10		88	40	40	LF	\$ 119,320	\$ 270,320	0.432	0.32	18,248	1,529
		88	60	20	CC	\$ 85,980	\$ 271,307	0.434	0	22,958	1,641
5		44	60	20	CC	\$ 92,980	\$ 271,640	0.434	0.15	21,567	3,068
	1	44	40	40	CC	\$ 117,320	\$ 273,424	0.437	0.26	17,920	2,050
5	1	44	40	20	CC	\$ 121,320	\$ 273,729	0.438	0.4	17,278	2,723
		44	80	20	LF	\$ 91,640	\$ 273,732	0.438	0	21,010	2,721
10	1	44	40	20	CC	\$ 139,320	\$ 273,820	0.438	0.52	14,695	2,364
10		44	40	80	LF	\$ 136,320	\$ 274,005	0.438	0.33	15,195	2,047
5		44	40	80	LF	\$ 118,320	\$ 274,095	0.438	0.17	17,846	2,365
10		44	80	20	LF	\$ 127,640	\$ 274,876	0.44	0.33	15,718	2,262
20		44	80	20	LF	\$ 163,640	\$ 275,730	0.441	0.62	10,501	1,710
5		88	40	40	LF	\$ 101,320	\$ 276,112	0.442	0.17	21,591	1,807
10	1	44	60	20	LF	\$ 155,980	\$ 276,358	0.442	0.57	11,515	1,772
20	1	44	40	20	LF	\$ 175,320	\$ 276,727	0.443	0.71	10,032	1,628
5		88	60	20	CC	\$ 103,980	\$ 276,843	0.443	0.17	20,906	1,631
5		88	60	20	LF	\$ 103,980	\$ 277,474	0.444	0.17	20,750	1,742
		44	40	80	LF	\$ 100,320	\$ 277,780	0.444	0	21,057	2,725
	1	44	40	60	LF	\$ 131,320	\$ 277,970	0.444	0.3	15,591	2,071
		44	100	20	CC	\$ 108,300	\$ 278,746	0.446	0	18,628	1,782
30		44	40	40	LF	\$ 180,320	\$ 278,911	0.446	0.7	10,372	1,572

5	1	44	40	60	LF	\$ 149,320	\$ 279,075	0.446	0.45	13,103	1,789
30		44	40	20	CC	\$ 166,320	\$ 279,475	0.447	0.66	13,175	2,157
30		44	60	40	LF	\$ 196,980	\$ 279,583	0.447	0.78	7,008	1,058
5		44	60	40	CC	\$ 106,980	\$ 279,740	0.447	0.14	20,661	2,337
20		44	80	20	CC	\$ 163,640	\$ 279,935	0.448	0.6	11,116	1,783
	2	44	40	20	LF	\$ 148,320	\$ 280,725	0.449	0.51	13,140	1,845
5		44	40	40	CC	\$ 90,320	\$ 280,802	0.449	0.14	23,786	3,524
10		44	60	60	LF	\$ 138,980	\$ 280,918	0.449	0.33	15,142	2,034
5		44	60	60	LF	\$ 120,980	\$ 281,187	0.45	0.17	17,813	2,360
5		44	100	20	CC	\$ 126,300	\$ 281,195	0.45	0.17	16,198	1,646
		88	40	20	LF	\$ 69,320	\$ 281,406	0.45	0	27,444	2,281
30		44	60	20	LF	\$ 182,980	\$ 281,792	0.451	0.73	9,878	1,640
10		44	40	40	CC	\$ 108,320	\$ 282,005	0.451	0.28	21,236	3,202
20		44	60	60	LF	\$ 174,980	\$ 282,395	0.452	0.61	10,059	1,429
10	1	44	40	60	LF	\$ 167,320	\$ 282,543	0.452	0.57	10,961	1,541
	1	88	40	20	CC	\$ 114,320	\$ 282,620	0.452	0.27	20,030	1,545
	2	44	40	20	CC	\$ 148,320	\$ 282,694	0.452	0.46	13,912	1,434
5	1	44	60	40	LF	\$ 151,980	\$ 283,176	0.453	0.45	12,657	1,712
20		88	60	40	LF	\$ 171,980	\$ 283,672	0.454	0.61	11,294	947
	1	44	60	40	LF	\$ 133,980	\$ 283,793	0.454	0.3	15,385	2,033
		44	60	40	CC	\$ 88,980	\$ 283,880	0.454	0	23,972	2,674
10	1	44	60	40	LF	\$ 169,980	\$ 284,100	0.454	0.59	10,142	1,391
20	1	44	40	40	LF	\$ 189,320	\$ 284,266	0.455	0.73	8,535	1,285
		44	40	80	CC	\$ 100,320	\$ 284,422	0.455	0	22,417	2,496
		44	60	60	LF	\$ 102,980	\$ 284,907	0.456	0	21,029	2,721
30		44	60	20	CC	\$ 182,980	\$ 285,538	0.457	0.71	10,426	1,705
10		88	60	20	LF	\$ 121,980	\$ 285,575	0.457	0.31	19,364	1,630
	1	88	40	20	LF	\$ 114,320	\$ 285,673	0.457	0.28	20,218	1,692
5	1	44	60	20	CC	\$ 137,980	\$ 286,786	0.459	0.42	15,612	2,328
20		44	40	80	LF	\$ 172,320	\$ 287,397	0.46	0.57	11,801	1,727
10	1	88	40	40	LF	\$ 164,320	\$ 287,895	0.46	0.57	12,750	1,069
	1	44	80	20	CC	\$ 136,640	\$ 287,941	0.46	0.28	15,433	1,523
10		44	80	40	LF	\$ 141,640	\$ 288,135	0.461	0.33	15,124	2,034
20		44	80	40	LF	\$ 177,640	\$ 288,158	0.461	0.61	9,835	1,391
5	2	44	40	20	LF	\$ 166,320	\$ 288,224	0.461	0.61	11,549	1,710
5		44	80	40	LF	\$ 123,640	\$ 288,291	0.461	0.17	17,783	2,355
20		88	40	60	LF	\$ 169,320	\$ 288,419	0.461	0.57	13,027	1,094
10		88	40	60	LF	\$ 133,320	\$ 288,709	0.462	0.32	18,249	1,529
10	1	44	60	20	CC	\$ 155,980	\$ 288,891	0.462	0.54	13,255	2,072

10		44	80	20	CC	\$ 127,640	\$ 289,157	0.462	0.31	17,776	2,547
5	1	88	40	40	LF	\$ 146,320	\$ 289,232	0.463	0.44	15,530	1,302
20	1	44	40	20	CC	\$ 175,320	\$ 289,560	0.463	0.68	11,805	1,940
30		44	80	40	LF	\$ 213,640	\$ 289,873	0.464	0.83	4,778	700
5	1	88	40	20	LF	\$ 132,320	\$ 289,987	0.464	0.4	18,276	1,536
10		88	60	20	CC	\$ 121,980	\$ 290,554	0.465	0.3	20,194	1,630
	1	44	40	60	CC	\$ 131,320	\$ 291,529	0.466	0.26	17,884	2,041
		44	80	40	LF	\$ 105,640	\$ 291,983	0.467	0	20,994	2,715
20	1	44	60	40	LF	\$ 205,980	\$ 292,064	0.467	0.79	5,969	866
10		44	40	100	LF	\$ 150,320	\$ 292,391	0.468	0.33	15,195	2,047
5		44	40	100	LF	\$ 132,320	\$ 292,481	0.468	0.17	17,846	2,365
10		88	60	40	LF	\$ 135,980	\$ 292,976	0.468	0.33	17,810	1,492
	1	44	80	20	LF	\$ 136,640	\$ 292,979	0.468	0.3	15,589	2,126
30		88	40	40	LF	\$ 191,320	\$ 293,093	0.469	0.7	11,165	939
30		88	60	40	LF	\$ 207,980	\$ 293,155	0.469	0.78	7,481	628
5	1	88	40	20	CC	\$ 132,320	\$ 293,635	0.47	0.39	18,895	1,530
10		44	60	40	CC	\$ 124,980	\$ 293,661	0.47	0.28	19,666	2,619
	2	44	40	40	LF	\$ 162,320	\$ 294,407	0.471	0.52	12,538	1,690
5		88	40	60	LF	\$ 115,320	\$ 294,498	0.471	0.17	21,591	1,807
5	1	44	80	20	LF	\$ 154,640	\$ 294,623	0.471	0.46	13,130	1,895
		88	60	20	LF	\$ 85,980	\$ 295,644	0.473	0	25,829	2,151
5		44	100	20	LF	\$ 126,300	\$ 295,754	0.473	0.17	17,793	2,367
20		88	40	20	LF	\$ 141,320	\$ 295,867	0.473	0.48	19,364	1,630
		44	40	100	LF	\$ 114,320	\$ 296,166	0.474	0	21,057	2,725
	1	44	40	80	LF	\$ 145,320	\$ 296,356	0.474	0.3	15,591	2,071
		88	80	20	CC	\$ 102,640	\$ 296,603	0.474	0	22,913	1,631
30		44	40	60	LF	\$ 194,320	\$ 297,166	0.475	0.7	10,355	1,566
5	1	44	40	80	LF	\$ 163,320	\$ 297,461	0.476	0.45	13,103	1,789
30		44	60	60	LF	\$ 210,980	\$ 297,914	0.476	0.78	7,002	1,054
	1	88	40	40	LF	\$ 128,320	\$ 298,092	0.477	0.29	19,396	1,619
20		44	60	40	CC	\$ 160,980	\$ 298,455	0.477	0.51	15,044	2,108
10	2	44	40	20	LF	\$ 184,320	\$ 298,617	0.478	0.69	10,392	1,617
5		88	60	40	LF	\$ 117,980	\$ 298,670	0.478	0.17	21,153	1,770
20	1	44	60	20	LF	\$ 191,980	\$ 298,698	0.478	0.72	9,484	1,556
5		44	60	60	CC	\$ 120,980	\$ 298,789	0.478	0.14	20,750	2,356
5	2	44	40	40	LF	\$ 180,320	\$ 299,154	0.478	0.62	10,563	1,466
		44	100	20	LF	\$ 108,300	\$ 299,284	0.479	0	20,986	2,719
10		44	60	80	LF	\$ 152,980	\$ 299,304	0.479	0.33	15,142	2,034
5		44	60	80	LF	\$ 134,980	\$ 299,573	0.479	0.17	17,813	2,360

20		88	40	20	CC	\$ 141,320	\$ 299,712	0.479	0.46	19,986	1,638
5		44	40	60	CC	\$ 104,320	\$ 299,905	0.48	0.14	23,881	3,545
5	1	44	80	20	CC	\$ 154,640	\$ 300,311	0.48	0.39	14,497	1,521
10		44	100	20	LF	\$ 144,300	\$ 300,382	0.48	0.33	15,688	2,258
5	1	44	40	40	CC	\$ 135,320	\$ 300,710	0.481	0.38	18,610	2,814
20		44	60	80	LF	\$ 188,980	\$ 300,781	0.481	0.61	10,059	1,429
		44	80	40	CC	\$ 105,640	\$ 300,887	0.481	0	23,005	2,243
		88	40	40	LF	\$ 83,320	\$ 300,892	0.481	0	27,598	2,294
10	1	44	40	80	LF	\$ 181,320	\$ 300,929	0.481	0.57	10,961	1,541
20		44	100	20	LF	\$ 180,300	\$ 300,946	0.481	0.62	10,429	1,701
	1	44	60	40	CC	\$ 133,980	\$ 300,984	0.481	0.25	18,214	2,063
10	1	44	80	20	LF	\$ 172,640	\$ 301,044	0.481	0.58	11,369	1,752
10		44	80	40	CC	\$ 141,640	\$ 301,118	0.482	0.28	17,524	1,829
5	1	44	60	60	LF	\$ 165,980	\$ 301,517	0.482	0.45	12,652	1,710
20		88	60	60	LF	\$ 185,980	\$ 301,962	0.483	0.61	11,280	946
	1	44	60	60	LF	\$ 147,980	\$ 302,156	0.483	0.3	15,382	2,032
5		88	80	20	LF	\$ 120,640	\$ 302,236	0.483	0.18	20,618	1,731
10		44	40	60	CC	\$ 122,320	\$ 302,304	0.483	0.28	21,472	3,247
10	1	44	60	60	LF	\$ 183,980	\$ 302,421	0.484	0.59	10,134	1,388
5		88	80	20	CC	\$ 120,640	\$ 302,600	0.484	0.17	20,916	1,630
20	1	44	40	60	LF	\$ 203,320	\$ 302,645	0.484	0.73	8,534	1,284
20	1	88	40	40	LF	\$ 200,320	\$ 302,730	0.484	0.72	9,705	815
		44	40	100	CC	\$ 114,320	\$ 302,808	0.484	0	22,417	2,496
30		88	80	40	LF	\$ 224,640	\$ 302,976	0.484	0.84	5,196	437
20	1	44	60	20	CC	\$ 191,980	\$ 303,069	0.485	0.71	10,113	1,649
		44	60	60	CC	\$ 102,980	\$ 303,161	0.485	0	24,092	2,700
	2	44	60	20	LF	\$ 164,980	\$ 303,238	0.485	0.52	12,662	1,799
5		44	80	40	CC	\$ 123,640	\$ 303,262	0.485	0.14	20,558	2,107
		44	60	80	LF	\$ 116,980	\$ 303,293	0.485	0	21,029	2,721
20		44	100	20	CC	\$ 180,300	\$ 303,984	0.486	0.61	10,872	1,755
10	1	44	40	40	CC	\$ 153,320	\$ 305,484	0.488	0.49	16,482	2,553
10	1	88	40	20	LF	\$ 150,320	\$ 305,585	0.489	0.48	17,927	1,509
20		44	40	100	LF	\$ 186,320	\$ 305,783	0.489	0.57	11,801	1,727
10	2	44	40	40	LF	\$ 198,320	\$ 305,785	0.489	0.71	8,860	1,265
30		44	80	40	CC	\$ 213,640	\$ 306,114	0.49	0.75	7,285	909
10	1	88	40	60	LF	\$ 178,320	\$ 306,281	0.49	0.57	12,750	1,069
20		44	80	60	LF	\$ 191,640	\$ 306,307	0.49	0.61	9,806	1,380
10		44	80	60	LF	\$ 155,640	\$ 306,407	0.49	0.33	15,111	2,029
5		44	80	60	LF	\$ 137,640	\$ 306,677	0.49	0.17	17,783	2,355

	1	88	60	20	CC	\$ 130,980	\$ 306,758	0.491	0.27	19,829	1,513
20		88	40	80	LF	\$ 183,320	\$ 306,805	0.491	0.57	13,027	1,094
10	1	88	60	40	LF	\$ 180,980	\$ 306,821	0.491	0.58	11,775	987
30		44	80	20	LF	\$ 199,640	\$ 306,849	0.491	0.73	9,778	1,633
20		88	80	40	LF	\$ 188,640	\$ 306,983	0.491	0.62	10,948	919
10		88	40	80	LF	\$ 147,320	\$ 307,095	0.491	0.32	18,249	1,529
5	1	88	40	60	LF	\$ 160,320	\$ 307,623	0.492	0.44	15,531	1,302
30		44	80	20	CC	\$ 199,640	\$ 307,684	0.492	0.72	9,907	1,642
20	1	88	60	40	LF	\$ 216,980	\$ 307,740	0.492	0.79	6,730	565
	1	88	60	20	LF	\$ 130,980	\$ 307,765	0.492	0.28	19,712	1,651
10	1	44	80	40	LF	\$ 186,640	\$ 308,095	0.493	0.59	9,893	1,353
	2	44	60	20	CC	\$ 164,980	\$ 308,130	0.493	0.46	13,887	1,413
30		44	80	60	LF	\$ 227,640	\$ 308,194	0.493	0.83	4,772	695
5	1	44	80	40	LF	\$168,640	\$308,324	0.493	0.46	12,578	1,700

Table 5-3 Overall optimization results table for Nazret

PV(kW)	G20	Gen (kW)	Battery	Converter (kW)	Dispatch strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (l.)	Generator (hrs)
		44	40	20	CC	\$ 58,320	\$ 201,609	0.322	0	18,623	1,785
5		44	40	20	LF	\$ 76,320	\$ 219,264	0.351	0.17	17,891	2,374
		44	40	20	LF	\$ 58,320	\$ 222,616	0.356	0	21,056	2,725
10		44	40	20	LF	\$ 94,320	\$ 224,051	0.358	0.33	15,811	2,267
		44	60	20	CC	\$ 74,980	\$ 227,227	0.363	0	18,605	1,788
	1	44	40	20	LF	\$ 103,320	\$ 234,729	0.375	0.37	14,561	2,030
20		44	40	20	LF	\$ 130,320	\$ 235,586	0.377	0.58	12,131	1,922
5		44	40	40	LF	\$ 90,320	\$ 237,407	0.38	0.17	17,863	2,363
10		44	40	40	LF	\$ 108,320	\$ 237,676	0.38	0.33	15,258	2,056
5	1	44	40	20	LF	\$ 121,320	\$ 238,266	0.381	0.51	12,381	1,832
		44	40	40	LF	\$ 72,320	\$ 241,008	0.386	0	21,057	2,725
5		44	60	20	LF	\$ 92,980	\$ 244,787	0.391	0.17	17,863	2,371
10	1	44	40	20	LF	\$ 139,320	\$ 245,095	0.392	0.62	10,695	1,683
		44	40	40	CC	\$ 72,320	\$ 247,995	0.397	0	22,461	2,508
		44	60	20	LF	\$ 74,980	\$ 248,124	0.397	0	21,027	2,721
10		44	60	20	LF	\$ 110,980	\$ 249,603	0.399	0.33	15,787	2,265
		88	40	20	CC	\$ 69,320	\$ 249,872	0.4	0	23,503	1,712
20		44	40	40	LF	\$ 144,320	\$ 251,127	0.402	0.57	11,867	1,742
5		88	40	20	CC	\$ 87,320	\$ 251,647	0.402	0.16	20,958	1,645
	1	44	40	40	LF	\$ 117,320	\$ 251,660	0.402	0.37	14,405	1,955
20		44	60	20	LF	\$ 146,980	\$ 251,976	0.403	0.61	10,777	1,752
5	1	44	40	40	LF	\$ 135,320	\$ 252,769	0.404	0.51	11,935	1,660
		44	80	20	CC	\$ 91,640	\$ 253,015	0.405	0	18,619	1,787
5		88	40	20	LF	\$ 87,320	\$ 254,184	0.406	0.17	21,088	1,770
5		44	80	20	CC	\$ 109,640	\$ 255,420	0.408	0.17	16,192	1,642
5		44	40	60	LF	\$ 104,320	\$ 255,793	0.409	0.17	17,863	2,363
10		44	40	60	LF	\$ 122,320	\$ 256,017	0.409	0.33	15,253	2,054
10	1	44	40	40	LF	\$ 153,320	\$ 257,109	0.411	0.63	9,894	1,428
5		44	60	20	CC	\$ 92,980	\$ 257,129	0.411	0.16	19,712	2,556
10		44	40	20	CC	\$ 94,320	\$ 258,896	0.414	0.29	20,531	3,246

5		44	40	20	CC	\$ 76,320	\$ 258,999	0.414	0.15	23,187	3,550
	1	44	60	20	LF	\$ 119,980	\$ 259,087	0.414	0.38	14,354	2,014
		44	40	60	LF	\$ 86,320	\$ 259,394	0.415	0	21,057	2,725
20		44	40	20	CC	\$ 130,320	\$ 259,601	0.415	0.52	15,474	2,500
10		88	40	20	LF	\$ 105,320	\$ 259,866	0.416	0.31	19,364	1,630
20		44	60	20	CC	\$ 146,980	\$ 260,502	0.417	0.58	12,006	1,916
5	1	44	60	20	LF	\$ 137,980	\$ 261,054	0.417	0.52	11,946	1,788
30		44	40	20	LF	\$ 166,320	\$ 262,002	0.419	0.7	10,735	1,750
5		44	60	40	LF	\$ 106,980	\$ 262,885	0.42	0.17	17,830	2,358
10		44	60	40	LF	\$ 124,980	\$ 263,046	0.421	0.33	15,214	2,046
10	1	44	40	20	CC	\$ 139,320	\$ 263,290	0.421	0.58	13,204	2,133
10		88	40	20	CC	\$ 105,320	\$ 265,157	0.424	0.3	20,208	1,646
20		44	60	40	LF	\$ 160,980	\$ 265,318	0.424	0.6	10,242	1,466
5	1	44	40	20	CC	\$ 121,320	\$ 265,434	0.424	0.46	16,078	2,559
		44	40	60	CC	\$ 86,320	\$ 266,036	0.425	0	22,417	2,496
		44	60	40	LF	\$ 88,980	\$ 266,521	0.426	0	21,029	2,721
10	1	44	60	20	LF	\$ 155,980	\$ 267,337	0.427	0.64	10,183	1,627
	1	44	40	20	CC	\$ 103,320	\$ 267,381	0.428	0.33	18,990	2,932
	2	44	40	20	LF	\$ 148,320	\$ 268,212	0.429	0.61	11,275	1,658
20		44	40	60	LF	\$ 158,320	\$ 269,398	0.431	0.57	11,854	1,737
10		44	60	20	CC	\$ 110,980	\$ 269,728	0.431	0.3	18,570	2,775
	1	44	40	60	LF	\$ 131,320	\$ 270,046	0.432	0.37	14,405	1,955
20		88	40	40	LF	\$ 155,320	\$ 270,098	0.432	0.57	13,034	1,096
5		44	80	20	LF	\$ 109,640	\$ 270,338	0.432	0.17	17,839	2,369
20	1	44	40	20	LF	\$ 175,320	\$ 270,473	0.433	0.75	9,105	1,513
10		88	40	40	LF	\$ 119,320	\$ 270,663	0.433	0.32	18,298	1,533
5	1	44	40	60	LF	\$ 149,320	\$ 271,132	0.434	0.51	11,933	1,659
		88	60	20	CC	\$ 85,980	\$ 271,307	0.434	0	22,958	1,641
	1	88	40	20	LF	\$ 114,320	\$ 272,219	0.435	0.35	18,311	1,538
		44	80	20	LF	\$ 91,640	\$ 273,732	0.438	0	21,010	2,721
5	1	44	60	20	CC	\$ 137,980	\$ 273,906	0.438	0.49	13,735	2,092
5		44	40	80	LF	\$ 118,320	\$ 274,179	0.438	0.17	17,863	2,363
10		44	40	80	LF	\$ 136,320	\$ 274,403	0.439	0.33	15,253	2,054
5	1	44	60	40	LF	\$ 151,980	\$ 274,654	0.439	0.52	11,397	1,571
10		44	80	20	LF	\$ 127,640	\$ 275,097	0.44	0.33	15,756	2,261
	1	44	60	40	LF	\$ 133,980	\$ 275,397	0.44	0.38	14,132	1,907
10	1	44	40	60	LF	\$ 167,320	\$ 275,452	0.44	0.63	9,888	1,426
10	1	44	60	40	LF	\$ 169,980	\$ 275,732	0.441	0.65	8,883	1,249
10	1	44	60	20	CC	\$ 155,980	\$ 275,999	0.441	0.61	11,407	1,814

5	2	44	40	20	LF	\$ 166,320	\$ 276,115	0.442	0.7	9,745	1,512
20		44	80	20	LF	\$ 163,640	\$ 276,339	0.442	0.62	10,581	1,728
5		88	60	20	CC	\$ 103,980	\$ 276,865	0.443	0.16	20,909	1,631
5		88	40	40	LF	\$ 101,320	\$ 276,941	0.443	0.17	21,706	1,817
10	1	88	40	40	LF	\$ 164,320	\$ 277,648	0.444	0.62	11,272	948
		44	40	80	LF	\$ 100,320	\$ 277,780	0.444	0	21,057	2,725
5		88	60	20	LF	\$ 103,980	\$ 277,932	0.444	0.17	20,815	1,747
20	1	44	40	40	LF	\$ 189,320	\$ 277,980	0.445	0.77	7,491	1,148
	1	88	40	20	CC	\$ 114,320	\$ 278,211	0.445	0.33	19,320	1,532
	1	44	60	20	CC	\$ 119,980	\$ 278,218	0.445	0.34	17,035	2,459
		44	100	20	CC	\$ 108,300	\$ 278,746	0.446	0	18,628	1,782
30		44	40	40	LF	\$ 180,320	\$ 278,749	0.446	0.7	10,379	1,572
5	1	88	40	40	LF	\$ 146,320	\$ 278,801	0.446	0.5	14,027	1,178
20	1	44	40	20	CC	\$ 175,320	\$ 278,945	0.446	0.72	10,308	1,707
30		44	40	20	CC	\$ 166,320	\$ 279,494	0.447	0.66	13,176	2,159
5		44	60	40	CC	\$ 106,980	\$ 279,869	0.448	0.14	20,681	2,338
30		44	60	40	LF	\$ 196,980	\$ 280,101	0.448	0.77	7,084	1,068
5		44	40	40	CC	\$ 90,320	\$ 280,366	0.448	0.14	23,737	3,504
20		44	80	20	CC	\$ 163,640	\$ 280,907	0.449	0.6	11,253	1,804
5		44	100	20	CC	\$ 126,300	\$ 281,211	0.45	0.17	16,202	1,645
5		44	60	60	LF	\$ 120,980	\$ 281,271	0.45	0.17	17,830	2,358
10		44	60	60	LF	\$ 138,980	\$ 281,319	0.45	0.33	15,200	2,041
	2	44	40	40	LF	\$ 162,320	\$ 281,340	0.45	0.62	10,590	1,471
		88	40	20	LF	\$ 69,320	\$ 281,406	0.45	0	27,444	2,281
30		44	60	20	LF	\$ 182,980	\$ 281,536	0.45	0.72	9,837	1,639
20		44	60	60	LF	\$ 174,980	\$ 283,575	0.453	0.6	10,226	1,460
5	1	88	40	20	LF	\$ 132,320	\$ 283,699	0.454	0.45	17,368	1,462
10		44	40	40	CC	\$ 108,320	\$ 283,820	0.454	0.27	21,469	3,247
		44	60	40	CC	\$ 88,980	\$ 283,880	0.454	0	23,972	2,674
20	1	44	60	40	LF	\$ 205,980	\$ 283,955	0.454	0.84	4,766	710
		44	40	80	CC	\$ 100,320	\$ 284,422	0.455	0	22,417	2,496
	1	88	40	40	LF	\$ 128,320	\$ 284,554	0.455	0.36	17,447	1,462
	1	44	80	20	LF	\$ 136,640	\$ 284,556	0.455	0.38	14,319	2,010
30		44	60	20	CC	\$ 182,980	\$ 284,819	0.455	0.71	10,316	1,697
		44	60	60	LF	\$ 102,980	\$ 284,907	0.456	0	21,029	2,721
20		88	60	40	LF	\$ 171,980	\$ 285,035	0.456	0.6	11,487	965
10		88	60	20	LF	\$ 121,980	\$ 285,575	0.457	0.31	19,364	1,630
5	2	44	40	40	LF	\$ 180,320	\$ 286,182	0.458	0.71	8,630	1,226
5	1	44	80	20	LF	\$ 154,640	\$ 286,216	0.458	0.52	11,864	1,780

10	2	44	40	20	LF	\$ 184,320	\$ 287,106	0.459	0.76	8,672	1,412
20		44	40	80	LF	\$ 172,320	\$ 287,784	0.46	0.57	11,854	1,737
5		44	80	40	LF	\$ 123,640	\$ 288,375	0.461	0.17	17,799	2,353
	1	44	40	80	LF	\$ 145,320	\$ 288,432	0.461	0.37	14,405	1,955
10		44	80	40	LF	\$ 141,640	\$ 288,535	0.461	0.33	15,183	2,041
20		88	40	60	LF	\$ 169,320	\$ 288,752	0.462	0.57	13,073	1,099
5	1	88	40	20	CC	\$ 132,320	\$ 288,753	0.462	0.43	18,180	1,478
10		88	40	60	LF	\$ 133,320	\$ 288,888	0.462	0.32	18,275	1,531
	2	44	60	20	LF	\$ 164,980	\$ 289,011	0.462	0.63	10,540	1,585
5	1	44	40	80	LF	\$ 163,320	\$ 289,518	0.463	0.51	11,933	1,659
20		44	80	40	LF	\$ 177,640	\$ 289,696	0.463	0.61	10,052	1,433
10		44	80	20	CC	\$ 127,640	\$ 289,966	0.464	0.3	17,889	2,567
10		88	60	20	CC	\$ 121,980	\$ 290,631	0.465	0.29	20,204	1,631
30		44	80	40	LF	\$ 213,640	\$ 290,993	0.465	0.83	4,940	726
10	1	44	80	20	LF	\$ 172,640	\$ 291,670	0.466	0.64	9,981	1,604
		44	80	40	LF	\$ 105,640	\$ 291,983	0.467	0	20,994	2,715
5		44	40	100	LF	\$ 132,320	\$ 292,565	0.468	0.17	17,863	2,363
10		44	40	100	LF	\$ 150,320	\$ 292,789	0.468	0.33	15,253	2,054
5	1	44	40	40	CC	\$ 135,320	\$ 292,807	0.468	0.43	17,465	2,658
5	1	44	60	60	LF	\$ 165,980	\$ 292,954	0.468	0.52	11,386	1,567
10		44	60	40	CC	\$ 124,980	\$ 293,000	0.469	0.27	19,593	2,586
30		88	40	40	LF	\$ 191,320	\$ 293,084	0.469	0.7	11,163	939
20	1	88	40	40	LF	\$ 200,320	\$ 293,664	0.47	0.76	8,399	707
30		88	60	40	LF	\$ 207,980	\$ 293,692	0.47	0.78	7,557	635
	1	44	60	60	LF	\$ 147,980	\$ 293,760	0.47	0.38	14,129	1,906
20	1	44	60	20	LF	\$ 191,980	\$ 293,802	0.47	0.76	8,753	1,466
10	1	44	40	80	LF	\$ 181,320	\$ 293,838	0.47	0.63	9,888	1,426
10	2	44	40	40	LF	\$ 198,320	\$ 293,861	0.47	0.79	7,083	1,044
10	1	44	60	60	LF	\$ 183,980	\$ 294,054	0.47	0.65	8,875	1,246
	2	44	40	20	CC	\$ 148,320	\$ 294,059	0.47	0.56	14,820	2,317
10		88	60	40	LF	\$ 135,980	\$ 294,311	0.471	0.32	18,002	1,508
	1	44	40	40	CC	\$ 117,320	\$ 294,332	0.471	0.31	20,297	3,042
20	1	44	60	20	CC	\$ 191,980	\$ 294,575	0.471	0.75	8,876	1,473
5		88	40	60	LF	\$ 115,320	\$ 295,327	0.472	0.17	21,706	1,817
10	1	88	60	40	LF	\$ 180,980	\$ 295,372	0.472	0.65	10,126	851
	1	88	60	20	LF	\$ 130,980	\$ 295,512	0.473	0.35	17,961	1,510
		88	60	20	LF	\$ 85,980	\$ 295,644	0.473	0	25,829	2,151
20		88	40	20	LF	\$ 141,320	\$ 295,867	0.473	0.47	19,364	1,630
5		44	100	20	LF	\$ 126,300	\$ 295,867	0.473	0.17	17,812	2,366

10	1	44	80	20	CC	\$ 172,640	\$ 296,008	0.473	0.62	10,622	1,673
		44	40	100	LF	\$ 114,320	\$ 296,166	0.474	0	21,057	2,725
20	1	44	40	60	LF	\$ 203,320	\$ 296,326	0.474	0.77	7,487	1,146
10	1	88	40	60	LF	\$ 178,320	\$ 296,381	0.474	0.62	11,322	952
5	2	44	40	20	CC	\$ 166,320	\$ 296,512	0.474	0.65	12,557	2,031
		88	80	20	CC	\$ 102,640	\$ 296,603	0.474	0	22,913	1,631
5	1	44	80	20	CC	\$ 154,640	\$ 296,717	0.474	0.49	13,364	1,995
10	1	44	40	40	CC	\$ 153,320	\$ 296,765	0.475	0.54	15,205	2,366
30		44	40	60	LF	\$ 194,320	\$ 296,985	0.475	0.7	10,361	1,565
10	2	44	40	20	CC	\$ 184,320	\$ 297,174	0.475	0.73	10,108	1,654
5	1	88	40	60	LF	\$ 160,320	\$ 297,272	0.475	0.5	14,040	1,179
5	2	44	60	20	LF	\$ 182,980	\$ 297,523	0.476	0.72	9,097	1,440
20	1	88	60	40	LF	\$ 216,980	\$ 297,599	0.476	0.84	5,272	443
	2	44	60	40	LF	\$ 178,980	\$ 297,936	0.476	0.65	9,266	1,266
30		44	60	60	LF	\$ 210,980	\$ 298,355	0.477	0.77	7,068	1,062
5		88	60	40	LF	\$ 117,980	\$ 298,896	0.478	0.17	21,188	1,771
5		44	60	60	CC	\$ 120,980	\$ 298,973	0.478	0.14	20,777	2,359
		44	100	20	LF	\$ 108,300	\$ 299,284	0.479	0	20,986	2,719
20		44	60	40	CC	\$ 160,980	\$ 299,334	0.479	0.5	15,174	2,122
20		88	40	20	CC	\$ 141,320	\$ 299,603	0.479	0.46	19,970	1,637
5		44	60	80	LF	\$ 134,980	\$ 299,657	0.479	0.17	17,830	2,358
10		44	60	80	LF	\$ 152,980	\$ 299,705	0.479	0.33	15,200	2,041
10	1	44	80	40	LF	\$ 186,640	\$ 299,714	0.479	0.66	8,630	1,212
	2	44	40	60	LF	\$ 176,320	\$ 299,729	0.479	0.62	10,590	1,471
5	1	88	60	40	LF	\$ 162,980	\$ 299,803	0.479	0.52	13,351	1,121
5	1	44	80	40	LF	\$ 168,640	\$ 299,872	0.48	0.52	11,325	1,560
	1	44	80	20	CC	\$ 136,640	\$ 300,265	0.48	0.34	16,585	2,317
5		44	40	60	CC	\$ 104,320	\$ 300,409	0.48	0.14	23,953	3,555
	1	44	60	40	CC	\$ 133,980	\$ 300,595	0.481	0.32	17,873	2,305
10		44	100	20	LF	\$ 144,300	\$ 300,603	0.481	0.33	15,726	2,257
	1	44	80	40	LF	\$ 150,640	\$ 300,817	0.481	0.38	14,092	1,900
		44	80	40	CC	\$ 105,640	\$ 300,887	0.481	0	23,005	2,243
		88	40	40	LF	\$ 83,320	\$ 300,892	0.481	0	27,598	2,294
	1	88	60	20	CC	\$ 130,980	\$ 301,215	0.482	0.33	18,994	1,466
5	2	44	60	40	LF	\$ 196,980	\$ 301,308	0.482	0.75	7,097	983
10		44	80	40	CC	\$ 141,640	\$ 301,469	0.482	0.27	17,581	1,830
10	1	88	40	20	LF	\$ 150,320	\$ 301,617	0.482	0.52	17,357	1,461
20		44	100	20	LF	\$ 180,300	\$ 301,636	0.482	0.62	10,522	1,720
	2	44	60	20	CC	\$ 164,980	\$ 301,694	0.482	0.6	12,333	1,861

20		44	60	80	LF	\$ 188,980	\$ 301,961	0.483	0.6	10,226	1,460
5	1	44	60	40	CC	\$ 151,980	\$ 302,164	0.483	0.44	15,395	2,078
20	1	44	60	60	LF	\$ 219,980	\$ 302,259	0.483	0.84	4,756	706
5		88	80	20	CC	\$ 120,640	\$ 302,627	0.484	0.16	20,921	1,630
5		88	80	20	LF	\$ 120,640	\$ 302,703	0.484	0.17	20,682	1,737
		44	40	100	CC	\$ 114,320	\$ 302,808	0.484	0	22,417	2,496
	1	88	40	60	LF	\$ 142,320	\$ 302,869	0.484	0.36	17,437	1,461
20	1	44	80	40	LF	\$ 222,640	\$ 302,958	0.484	0.87	3,793	557
		44	60	60	CC	\$ 102,980	\$ 303,161	0.485	0	24,092	2,700
20		88	60	60	LF	\$ 185,980	\$ 303,289	0.485	0.6	11,469	963
		44	60	80	LF	\$ 116,980	\$ 303,293	0.485	0	21,029	2,721
5		44	80	40	CC	\$ 123,640	\$ 303,399	0.485	0.14	20,580	2,107
10	1	44	60	40	CC	\$ 169,980	\$ 303,474	0.485	0.56	12,936	1,801
10		44	40	60	CC	\$ 122,320	\$ 303,613	0.486	0.27	21,636	3,281

Table 5-4 Overall optimization results table for Debrezeit

	PV(kW)	G20	Gen (kW)	Battery	Converter (kW)	Dispatch strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (L)	Generator (hrs)
			44	40	20	CC	\$ 58,320	\$ 201,609	0.322	0	18,623	1,785
5			44	40	20	LF	\$ 76,320	\$ 219,243	0.351	0.17	17,886	2,375
			44	40	20	LF	\$ 58,320	\$ 222,616	0.356	0	21,056	2,725
10			44	40	20	LF	\$ 94,320	\$ 223,971	0.358	0.33	15,797	2,267
			44	60	20	CC	\$ 74,980	\$ 227,227	0.363	0	18,605	1,788
15			44	40	20	LF	\$ 112,320	\$ 227,831	0.364	0.47	13,642	2,092
5			44	60	20	CC	\$ 92,980	\$ 229,636	0.367	0.17	16,179	1,642
20			44	40	20	LF	\$ 130,320	\$ 235,177	0.376	0.58	12,078	1,909
10			44	40	40	LF	\$ 108,320	\$ 237,335	0.38	0.33	15,214	2,045
5			44	40	40	LF	\$ 90,320	\$ 237,431	0.38	0.17	17,863	2,366
			44	40	40	LF	\$ 72,320	\$ 241,008	0.386	0	21,057	2,725
15			44	40	40	LF	\$ 126,320	\$ 241,580	0.386	0.47	13,136	1,861
5			44	60	20	LF	\$ 92,980	\$ 244,767	0.391	0.17	17,859	2,372
			44	40	40	CC	\$ 72,320	\$ 247,995	0.397	0	22,461	2,508
			44	60	20	LF	\$ 74,980	\$ 248,124	0.397	0	21,027	2,721
10			44	60	20	LF	\$ 110,980	\$ 249,493	0.399	0.33	15,770	2,264
			88	40	20	CC	\$ 69,320	\$ 249,872	0.4	0	23,503	1,712
	1		44	40	20	CC	\$ 103,320	\$ 250,466	0.401	0.07	17,572	1,694
20			44	40	40	LF	\$ 144,320	\$ 250,976	0.401	0.57	11,847	1,737
15			44	60	20	LF	\$ 128,980	\$ 250,994	0.401	0.48	13,274	2,043
5			88	40	20	CC	\$ 87,320	\$ 251,669	0.402	0.17	20,962	1,645
20			44	60	20	LF	\$ 146,980	\$ 252,176	0.403	0.61	10,810	1,752
			44	80	20	CC	\$ 91,640	\$ 253,015	0.405	0	18,619	1,787
5			88	40	20	LF	\$ 87,320	\$ 254,268	0.407	0.17	21,102	1,770
5			44	80	20	CC	\$ 109,640	\$ 255,412	0.408	0.17	16,190	1,642
10			44	40	60	LF	\$ 122,320	\$ 255,676	0.409	0.33	15,209	2,043
5			44	40	60	LF	\$ 104,320	\$ 255,817	0.409	0.17	17,863	2,366
15			44	40	20	CC	\$ 112,320	\$ 257,108	0.411	0.42	17,664	2,855
10			44	40	20	CC	\$ 94,320	\$ 258,186	0.413	0.29	20,443	3,220
5			44	40	20	CC	\$ 76,320	\$ 259,218	0.415	0.15	23,211	3,560
			44	40	60	LF	\$ 86,320	\$ 259,394	0.415	0	21,057	2,725

20		44	40	20	CC	\$ 130,320	\$ 259,542	0.415	0.52	15,446	2,516
15		44	40	60	LF	\$ 140,320	\$ 259,829	0.415	0.47	13,120	1,855
10		88	40	20	LF	\$ 105,320	\$ 259,952	0.416	0.31	19,376	1,631
20		44	60	20	CC	\$ 146,980	\$ 260,783	0.417	0.58	12,038	1,929
30		44	40	20	LF	\$ 166,320	\$ 262,123	0.419	0.7	10,756	1,749
10		44	60	40	LF	\$ 124,980	\$ 262,705	0.42	0.33	15,169	2,035
5		44	60	40	LF	\$ 106,980	\$ 262,911	0.42	0.17	17,831	2,361
15		44	60	40	LF	\$ 142,980	\$ 263,305	0.421	0.48	12,617	1,737
15		44	60	20	CC	\$ 128,980	\$ 264,041	0.422	0.45	15,111	2,338
20		44	60	40	LF	\$ 160,980	\$ 264,826	0.423	0.6	10,173	1,452
10		88	40	20	CC	\$ 105,320	\$ 265,220	0.424	0.3	20,221	1,645
		44	40	60	CC	\$ 86,320	\$ 266,036	0.425	0	22,417	2,496
15		88	40	40	LF	\$ 137,320	\$ 266,059	0.425	0.47	15,043	1,263
		44	60	40	LF	\$ 88,980	\$ 266,521	0.426	0	21,029	2,721
	1	44	40	20	LF	\$ 103,320	\$ 267,017	0.427	0.07	19,399	2,506
5	1	44	40	20	LF	\$ 121,320	\$ 267,525	0.428	0.24	16,752	2,269
10		44	60	20	CC	\$ 110,980	\$ 268,879	0.43	0.3	18,453	2,753
20		44	40	60	LF	\$ 158,320	\$ 269,225	0.431	0.57	11,832	1,731
5		44	80	20	LF	\$ 109,640	\$ 270,319	0.432	0.17	17,835	2,370
20		88	40	40	LF	\$ 155,320	\$ 270,561	0.433	0.57	13,101	1,101
10		88	40	40	LF	\$ 119,320	\$ 270,632	0.433	0.32	18,291	1,534
		88	60	20	CC	\$ 85,980	\$ 271,307	0.434	0	22,958	1,641
10	1	44	40	20	LF	\$ 139,320	\$ 271,786	0.435	0.39	14,615	2,136
		44	80	20	LF	\$ 91,640	\$ 273,732	0.438	0	21,010	2,721
10		44	40	80	LF	\$ 136,320	\$ 274,062	0.438	0.33	15,209	2,043
5		44	40	80	LF	\$ 118,320	\$ 274,203	0.438	0.17	17,863	2,366
10		44	80	20	LF	\$ 127,640	\$ 275,016	0.44	0.33	15,742	2,261
	1	44	60	20	CC	\$ 119,980	\$ 275,795	0.441	0.07	17,512	1,691
15	1	44	40	20	LF	\$ 157,320	\$ 276,081	0.441	0.53	12,550	1,947
5		88	40	40	LF	\$ 101,320	\$ 276,467	0.442	0.17	21,641	1,811
15		44	80	20	LF	\$ 145,640	\$ 276,522	0.442	0.48	13,247	2,040
20		44	80	20	LF	\$ 163,640	\$ 276,560	0.442	0.62	10,617	1,729
5		88	60	20	CC	\$ 103,980	\$ 276,887	0.443	0.16	20,913	1,631
5		88	60	20	LF	\$ 103,980	\$ 277,639	0.444	0.17	20,773	1,744
		44	40	80	LF	\$ 100,320	\$ 277,780	0.444	0	21,057	2,725
15		88	40	20	LF	\$ 123,320	\$ 277,867	0.444	0.4	19,364	1,630
15		44	40	80	LF	\$ 154,320	\$ 278,215	0.445	0.47	13,120	1,855
30		44	40	20	CC	\$ 166,320	\$ 278,443	0.445	0.66	13,037	2,127
30		44	40	40	LF	\$ 180,320	\$ 278,617	0.446	0.7	10,361	1,570

		44	100	20	CC	\$ 108,300	\$ 278,746	0.446	0	18,628	1,782
5		44	60	40	CC	\$ 106,980	\$ 279,772	0.447	0.14	20,666	2,337
30		44	60	40	LF	\$ 196,980	\$ 279,851	0.448	0.77	7,048	1,062
5		44	40	40	CC	\$ 90,320	\$ 280,154	0.448	0.14	23,705	3,501
20		44	80	20	CC	\$ 163,640	\$ 280,807	0.449	0.6	11,240	1,801
10		44	60	60	LF	\$ 138,980	\$ 280,977	0.449	0.33	15,156	2,030
5		44	100	20	CC	\$ 126,300	\$ 281,158	0.45	0.17	16,195	1,643
5		44	60	60	LF	\$ 120,980	\$ 281,297	0.45	0.17	17,831	2,361
		88	40	20	LF	\$ 69,320	\$ 281,406	0.45	0	27,444	2,281
15		44	60	60	LF	\$ 156,980	\$ 281,576	0.45	0.48	12,604	1,732
30		44	60	20	LF	\$ 182,980	\$ 281,804	0.451	0.72	9,879	1,641
15		88	40	20	CC	\$ 123,320	\$ 281,915	0.451	0.39	20,020	1,638
20		44	60	60	LF	\$ 174,980	\$ 283,134	0.453	0.6	10,164	1,448
		44	60	40	CC	\$ 88,980	\$ 283,880	0.454	0	23,972	2,674
10		44	40	40	CC	\$ 108,320	\$ 284,035	0.454	0.27	21,497	3,254
		44	40	80	CC	\$ 100,320	\$ 284,422	0.455	0	22,417	2,496
15		88	40	60	LF	\$ 151,320	\$ 284,515	0.455	0.46	15,055	1,263
15		44	80	20	CC	\$ 145,640	\$ 284,546	0.455	0.46	14,412	2,190
5	1	44	40	40	LF	\$ 135,320	\$ 284,599	0.455	0.24	16,602	2,209
20		88	60	40	LF	\$ 171,980	\$ 284,708	0.455	0.6	11,442	960
		44	60	60	LF	\$ 102,980	\$ 284,907	0.456	0	21,029	2,721
	1	44	40	40	LF	\$ 117,320	\$ 285,334	0.456	0.07	19,392	2,502
10	1	44	40	40	LF	\$ 153,320	\$ 285,473	0.456	0.39	14,082	1,919
10		88	60	20	LF	\$ 121,980	\$ 285,575	0.457	0.31	19,364	1,630
30		44	60	20	CC	\$ 182,980	\$ 285,577	0.457	0.71	10,434	1,704
20	1	44	40	20	LF	\$ 175,320	\$ 285,862	0.457	0.62	11,339	1,811
20		44	40	80	LF	\$ 172,320	\$ 287,611	0.46	0.57	11,832	1,731
10		44	80	40	LF	\$ 141,640	\$ 288,194	0.461	0.33	15,138	2,030
5		44	80	40	LF	\$ 123,640	\$ 288,400	0.461	0.17	17,800	2,356
15		88	60	40	LF	\$ 153,980	\$ 288,432	0.461	0.47	14,565	1,222
15		44	80	40	LF	\$ 159,640	\$ 288,787	0.462	0.48	12,585	1,732
20		44	80	40	LF	\$ 177,640	\$ 288,941	0.462	0.61	9,948	1,410
10		44	80	20	CC	\$ 127,640	\$ 288,984	0.462	0.3	17,758	2,537
10		88	40	60	LF	\$ 133,320	\$ 289,018	0.462	0.32	18,291	1,534
20		88	40	60	LF	\$ 169,320	\$ 289,388	0.463	0.57	13,165	1,106
15	1	44	40	40	LF	\$ 171,320	\$ 290,432	0.464	0.52	12,123	1,736
10		88	60	20	CC	\$ 121,980	\$ 290,586	0.465	0.3	20,199	1,630
30		44	80	40	LF	\$ 213,640	\$ 290,597	0.465	0.83	4,883	716
		44	80	40	LF	\$ 105,640	\$ 291,983	0.467	0	20,994	2,715

10		44	40	100	LF	\$ 150,320	\$ 292,448	0.468	0.33	15,209	2,043
	1	44	60	20	LF	\$ 119,980	\$ 292,560	0.468	0.07	19,374	2,504
5		44	40	100	LF	\$ 132,320	\$ 292,589	0.468	0.17	17,863	2,366
30		88	40	40	LF	\$ 191,320	\$ 292,750	0.468	0.7	11,115	935
5	1	44	60	20	LF	\$ 137,980	\$ 292,920	0.468	0.24	16,706	2,264
10		44	60	40	CC	\$ 124,980	\$ 293,593	0.469	0.28	19,670	2,605
10		88	60	40	LF	\$ 135,980	\$ 293,771	0.47	0.33	17,923	1,502
30		88	60	40	LF	\$ 207,980	\$ 293,918	0.47	0.78	7,589	638
5		88	40	60	LF	\$ 115,320	\$ 294,853	0.471	0.17	21,641	1,811
		88	60	20	LF	\$ 85,980	\$ 295,644	0.473	0	25,829	2,151
20		88	40	20	LF	\$ 141,320	\$ 295,867	0.473	0.47	19,364	1,630
5		44	100	20	LF	\$ 126,300	\$ 295,870	0.473	0.17	17,811	2,368
	1	44	40	40	CC	\$ 117,320	\$ 295,934	0.473	0.06	21,265	2,406
		44	40	100	LF	\$ 114,320	\$ 296,166	0.474	0	21,057	2,725
15		44	40	100	LF	\$ 168,320	\$ 296,601	0.474	0.47	13,120	1,855
		88	80	20	CC	\$ 102,640	\$ 296,603	0.474	0	22,913	1,631
10	1	44	60	20	LF	\$ 155,980	\$ 296,822	0.475	0.4	14,517	2,124
30		44	40	60	LF	\$ 194,320	\$ 296,871	0.475	0.7	10,345	1,564
	1	88	40	20	CC	\$ 114,320	\$ 297,724	0.476	0.07	22,281	1,668
15		44	40	40	CC	\$ 126,320	\$ 297,913	0.476	0.38	20,492	3,214
15	1	44	60	20	LF	\$ 173,980	\$ 298,004	0.477	0.54	11,998	1,879
30		44	60	60	LF	\$ 210,980	\$ 298,168	0.477	0.77	7,041	1,058
5		88	60	40	LF	\$ 117,980	\$ 298,527	0.477	0.17	21,134	1,768
5		44	60	60	CC	\$ 120,980	\$ 299,002	0.478	0.14	20,779	2,361
15		44	60	40	CC	\$ 142,980	\$ 299,060	0.478	0.39	17,759	2,447
		44	100	20	LF	\$ 108,300	\$ 299,284	0.479	0	20,986	2,719
5		44	40	60	CC	\$ 104,320	\$ 299,326	0.479	0.14	23,808	3,525
10		44	60	80	LF	\$ 152,980	\$ 299,363	0.479	0.33	15,156	2,030
5		44	60	80	LF	\$ 134,980	\$ 299,683	0.479	0.17	17,831	2,361
20		44	60	40	CC	\$ 160,980	\$ 299,703	0.479	0.5	15,222	2,134
20		88	40	20	CC	\$ 141,320	\$ 299,704	0.479	0.46	19,985	1,638
15		44	60	80	LF	\$ 170,980	\$ 299,962	0.48	0.48	12,604	1,732
20	1	44	40	40	LF	\$ 189,320	\$ 300,244	0.48	0.62	10,910	1,608
10		44	100	20	LF	\$ 144,300	\$ 300,493	0.481	0.33	15,709	2,256
		44	80	40	CC	\$ 105,640	\$ 300,887	0.481	0	23,005	2,243
		88	40	40	LF	\$ 83,320	\$ 300,892	0.481	0	27,598	2,294
10		44	80	40	CC	\$ 141,640	\$ 301,241	0.482	0.28	17,547	1,826
20		44	60	80	LF	\$ 188,980	\$ 301,520	0.482	0.6	10,164	1,448
	1	44	80	20	CC	\$ 136,640	\$ 301,551	0.482	0.07	17,520	1,691

20		44	100	20	LF	\$ 180,300	\$ 301,802	0.483	0.62	10,548	1,721
10	1	44	40	20	CC	\$ 139,320	\$ 301,842	0.483	0.35	18,697	2,966
5	1	88	40	20	LF	\$ 132,320	\$ 301,949	0.483	0.24	19,966	1,677
15		44	100	20	LF	\$ 162,300	\$ 302,044	0.483	0.48	13,220	2,037
15	1	44	40	20	CC	\$ 157,320	\$ 302,053	0.483	0.47	16,113	2,621
5		88	80	20	LF	\$ 120,640	\$ 302,486	0.484	0.17	20,653	1,734
5		88	80	20	CC	\$ 120,640	\$ 302,645	0.484	0.16	20,924	1,630
		44	40	100	CC	\$ 114,320	\$ 302,808	0.484	0	22,417	2,496
15		88	40	80	LF	\$ 165,320	\$ 302,901	0.484	0.46	15,055	1,263
5		44	80	40	CC	\$ 123,640	\$ 302,941	0.484	0.14	20,517	2,096
5	1	44	40	60	LF	\$ 149,320	\$ 302,962	0.484	0.24	16,599	2,208
20		88	60	60	LF	\$ 185,980	\$ 302,994	0.485	0.6	11,428	959
		44	60	60	CC	\$ 102,980	\$ 303,161	0.485	0	24,092	2,700
10		44	40	60	CC	\$ 122,320	\$ 303,214	0.485	0.27	21,590	3,271
		44	60	80	LF	\$ 116,980	\$ 303,293	0.485	0	21,029	2,721
15		88	60	20	LF	\$ 139,980	\$ 303,575	0.485	0.4	19,364	1,630
30		88	80	40	LF	\$ 224,640	\$ 303,617	0.486	0.83	5,289	444
	1	44	40	60	LF	\$ 131,320	\$ 303,720	0.486	0.07	19,392	2,502
10	1	44	40	60	LF	\$ 167,320	\$ 303,836	0.486	0.39	14,079	1,918
5	1	44	40	20	CC	\$ 121,320	\$ 304,758	0.487	0.21	21,732	3,364
5	1	88	40	20	CC	\$ 132,320	\$ 305,129	0.488	0.22	20,552	1,653
20	1	44	60	20	LF	\$ 191,980	\$ 305,266	0.488	0.64	10,417	1,701
20	1	44	40	20	CC	\$ 175,320	\$ 305,559	0.489	0.57	14,064	2,295
20		44	100	20	CC	\$ 180,300	\$ 305,801	0.489	0.6	11,139	1,785
20		44	40	100	LF	\$ 186,320	\$ 305,997	0.489	0.57	11,832	1,731
10		44	80	60	LF	\$ 155,640	\$ 306,466	0.49	0.33	15,125	2,025
15		88	60	60	LF	\$ 167,980	\$ 306,495	0.49	0.47	14,519	1,218
30		44	80	20	LF	\$ 199,640	\$ 306,745	0.491	0.73	9,763	1,631
5		44	80	60	LF	\$ 137,640	\$ 306,786	0.491	0.17	17,800	2,356
15		44	80	60	LF	\$ 173,640	\$ 306,990	0.491	0.48	12,564	1,724
20		44	80	60	LF	\$ 191,640	\$ 307,068	0.491	0.61	9,916	1,398
10		88	40	80	LF	\$ 147,320	\$ 307,405	0.492	0.32	18,291	1,534
20		88	80	40	LF	\$ 188,640	\$ 307,484	0.492	0.61	11,020	925
15	1	44	60	20	CC	\$ 173,980	\$ 307,709	0.492	0.51	13,360	2,100
15		88	60	20	CC	\$ 139,980	\$ 307,772	0.492	0.39	20,063	1,630
20		88	40	80	LF	\$ 183,320	\$ 307,774	0.492	0.57	13,165	1,106
30		44	80	20	CC	\$ 199,640	\$ 307,813	0.492	0.72	9,928	1,642
30		44	80	40	CC	\$ 213,640	\$ 307,877	0.492	0.74	7,541	949
5	1	44	80	20	CC	\$ 154,640	\$ 308,455	0.493	0.23	15,753	1,622

15	1	44	40	60	LF	\$ 185,320	\$ 308,704	0.494	0.52	12,110	1,731
30		44	80	60	LF	\$ 227,640	\$ 308,944	0.494	0.83	4,882	711
15		44	100	20	CC	\$ 162,300	\$ 309,167	0.494	0.46	14,264	2,161
20		44	80	40	CC	\$ 177,640	\$ 309,204	0.494	0.52	13,127	1,613
20		44	40	40	CC	\$ 144,320	\$ 309,265	0.495	0.46	19,364	3,075
5	1	44	60	40	LF	\$ 151,980	\$ 309,892	0.496	0.24	16,544	2,199
10	1	44	60	40	LF	\$ 169,980	\$ 310,035	0.496	0.39	13,933	1,883
15		44	80	40	CC	\$ 159,640	\$ 310,124	0.496	0.4	15,923	1,917
		44	80	60	LF	\$ 119,640	\$ 310,369	0.497	0	20,994	2,715
30		44	100	40	LF	\$ 230,300	\$ 310,604	0.497	0.85	4,053	590
	1	44	60	40	LF	\$ 133,980	\$ 310,745	0.497	0.07	19,350	2,495
15	1	44	60	40	LF	\$ 187,980	\$ 310,898	0.497	0.53	11,416	1,593
30		88	40	60	LF	\$ 205,320	\$ 310,972	0.497	0.7	11,091	933

Table 5-5 Overall optimization results table for the resettlers in the vicinity of Mekele

PV (kW)	WES5	Gen1 (kW)	SoCS25P	Converter (kW)	Dispatch strategy	Initial capital	Total NPC	COE (\$/kWh)	Renewable fraction	Diesel (L)	Gen1 (hrs)
2		5	10	4	CC	\$ 21,330	\$ 66,940	0.327	0.21	4639	3,090
2		5	10	6	CC	\$ 22,730	\$ 68,783	0.336	0.21	4640	3,090
2	1	5	10	4	CC	\$ 30,330	\$ 71,463	0.349	0.37	3676	2,431
		5	10	4	CC	\$ 14,130	\$ 72,350	0.353	0.00	6,047	4,157
2		5	10	10	CC	\$ 25,530	\$ 72,460	0.353	0.21	4,640	3,090
2		5	15	4	CC	\$ 25,495	\$ 73,138	0.357	0.21	4,620	2,983
		5	10	6	CC	\$ 15,530	\$ 74,183	0.362	0.00	6,046	4,155
	1	5	10	4	CC	\$ 23,130	\$ 74,863	0.365	0.17	4,870	3,299
2		5	15	6	CC	\$ 26,895	\$ 74,993	0.366	0.21	4,622	2,984
2	1	5	10	6	CC	\$ 31,730	\$ 75,754	0.369	0.36	3,939	2,797
	1	5	10	6	CC	\$ 24,530	\$ 76,733	0.374	0.17	4,873	3,304
4	1	5	10	4	CC	\$ 37,530	\$ 76,833	0.375	0.51	3,435	2,727
2		5	10	15	CC	\$ 29,030	\$ 77,057	0.376	0.21	4,640	3,090
2	1	5	15	4	CC	\$ 34,495	\$ 77,450	0.378	0.38	3,630	2,352
		5	10	10	CC	\$ 18,330	\$ 77,860	0.380	0.00	6,046	4,155
		5	15	4	CC	\$ 18,295	\$ 78,184	0.381	0.00	6,000	3,876
4		5	20	4	CC	\$ 36,860	\$ 78,318	0.382	0.40	3,648	2,538
2		5	15	10	CC	\$ 29,695	\$ 78,670	0.384	0.21	4,622	2,984
2	1	5	10	10	CC	\$ 34,530	\$ 79,268	0.387	0.36	3,923	2,755
4	1	5	10	6	CC	\$ 38,930	\$ 79,424	0.387	0.50	3,533	2,611
2		5	20	4	CC	\$ 29,660	\$ 79,554	0.388	0.21	4,619	2,982
		5	15	6	CC	\$ 19,695	\$ 80,037	0.390	0.00	6,002	3,875
	1	5	10	10	CC	\$ 27,330	\$ 80,410	0.392	0.17	4,873	3,304
2	1	5	15	6	CC	\$ 35,895	\$ 80,730	0.394	0.37	3,790	2,493
	2	5	10	4	CC	\$ 32,130	\$ 80,868	0.394	0.33	4,075	2,756
	1	5	15	4	CC	\$ 27,295	\$ 80,893	0.395	0.17	4,837	3,128
2		5	20	6	CC	\$ 31,060	\$ 81,418	0.397	0.21	4,622	2,983
4	1	5	15	4	CC	\$ 41,695	\$ 81,482	0.397	0.53	3,233	2,605
4		5	20	6	CC	\$ 38,260	\$ 81,849	0.399	0.39	3,849	2,547
		5	10	15	CC	\$ 21,830	\$ 82,457	0.402	0.00	6,046	4,155
	2	5	10	6	CC	\$ 33,530	\$ 82,729	0.403	0.33	4,077	2,761
	1	5	15	6	CC	\$ 28,695	\$ 82,741	0.404	0.17	4,838	3,127
2		5	15	15	CC	\$ 33,195	\$ 83,266	0.406	0.21	4,622	2,984

4	1	5	15	6	CC	\$ 43,095	\$ 83,417	0.407	0.52	3,264	2,362
		5	15	10	CC	\$ 22,495	\$ 83,714	0.408	0.00	6,002	3,875
2	1	5	20	4	CC	\$ 38,660	\$ 83,810	0.409	0.38	3,621	2,359
2	1	5	10	15	CC	\$ 38,030	\$ 83,865	0.409	0.36	3,923	2,755
		5	20	4	CC	\$ 22,460	\$ 84,590	0.413	0.00	5,998	3,872
4		5	25	4	CC	\$ 41,025	\$ 84,703	0.413	0.40	3,643	2,539
	1	5	10	15	CC	\$ 30,830	\$ 85,007	0.415	0.17	4,873	3,304
2		5	20	10	CC	\$ 33,860	\$ 85,095	0.415	0.21	4,622	2,983
2		5	25	4	CC	\$ 33,825	\$ 85,953	0.419	0.21	4,615	2,980
	2	5	10	10	CC	\$ 36,330	\$ 86,406	0.421	0.33	4,077	2,761
	1	5	15	10	CC	\$ 31,495	\$ 86,418	0.421	0.17	4,838	3,127
		5	20	6	CC	\$ 23,860	\$ 86,439	0.422	0.00	5,999	3,871
	2	5	15	4	CC	\$ 36,295	\$ 86,598	0.422	0.33	4,004	2,597
4	1	5	10	10	CC	\$ 41,730	\$ 86,638	0.423	0.48	3,904	2,978
2	1	5	15	10	CC	\$ 38,695	\$ 86,675	0.423	0.36	4,038	2,781
2	1	5	20	6	CC	\$ 40,060	\$ 86,923	0.424	0.37	3,766	2,446
	1	5	20	4	CC	\$ 31,460	\$ 87,287	0.426	0.17	4,833	3,122
4	1	5	20	4	CC	\$ 45,860	\$ 87,313	0.426	0.54	3,158	2,656
2		5	25	6	CC	\$ 35,225	\$ 87,844	0.428	0.21	4,621	2,982
4		5	25	6	CC	\$ 42,425	\$ 87,887	0.429	0.39	3,809	2,474
		5	15	15	CC	\$ 25,995	\$ 88,311	0.431	0.00	6,002	3,875
	2	5	15	6	CC	\$ 37,695	\$ 88,461	0.431	0.33	4,007	2,599
4	1	5	20	6	CC	\$ 47,260	\$ 89,049	0.434	0.52	3,178	2,254
	1	5	20	6	CC	\$ 32,860	\$ 89,138	0.435	0.17	4,835	3,120
2		5	20	15	CC	\$ 37,360	\$ 89,692	0.437	0.21	4,622	2,983
4	1	5	15	10	CC	\$ 45,895	\$ 89,708	0.438	0.50	3,549	2,689
		5	20	10	CC	\$ 26,660	\$ 90,116	0.440	0.00	5,999	3,871
2	1	5	25	4	CC	\$ 42,825	\$ 90,175	0.440	0.38	3,614	2,356
	2	5	10	15	CC	\$ 39,830	\$ 91,003	0.444	0.33	4,077	2,761
		5	25	4	CC	\$ 26,625	\$ 91,004	0.444	0.00	5,996	3,871
	1	5	15	15	CC	\$ 34,995	\$ 91,014	0.444	0.17	4,838	3,127
4		5	20	10	CC	\$ 41,060	\$ 91,111	0.444	0.36	4,466	3,188
4		5	30	4	CC	\$ 45,190	\$ 91,164	0.445	0.40	3,647	2,542
4	1	5	10	15	CC	\$ 45,230	\$ 91,235	0.445	0.48	3,904	2,978
2	1	5	15	15	CC	\$ 42,195	\$ 91,272	0.445	0.36	4,038	2,781
2	1	5	20	10	CC	\$ 42,860	\$ 91,469	0.446	0.36	3,862	2,535
2		5	25	10	CC	\$ 38,025	\$ 91,521	0.446	0.21	4,621	2,982
	2	5	15	10	CC	\$ 40,495	\$ 92,138	0.449	0.33	4,007	2,599
2		5	30	4	CC	\$ 37,990	\$ 92,367	0.451	0.21	4,614	2,979
	2	5	20	4	CC	\$ 40,460	\$ 92,708	0.452	0.33	3,968	2,568
	1	5	20	10	CC	\$ 35,660	\$ 92,815	0.453	0.17	4,835	3,120
		5	25	6	CC	\$ 28,025	\$ 92,867	0.453	0.00	5,999	3,871

2	1	5	25	6	CC	\$ 44,225	\$ 93,056	0.454	0.37	3,734	2,412
2	2	5	20	4	CC	\$ 47,660	\$ 93,156	0.454	0.51	3,189	2,275
4	1	5	25	4	CC	\$ 50,025	\$ 93,327	0.455	0.54	3,108	2,662
	1	5	25	4	CC	\$ 35,625	\$ 93,688	0.457	0.17	4,831	3,120
4		5	30	6	CC	\$ 46,590	\$ 94,110	0.459	0.39	3,787	2,444
10		5	25	10	CC	\$ 66,825	\$ 94,155	0.459	0.80	1,634	1,331
2		5	30	6	CC	\$ 39,390	\$ 94,268	0.460	0.21	4,621	2,982
4	1	5	15	15	CC	\$ 49,395	\$ 94,305	0.460	0.50	3,549	2,689
	2	5	20	6	CC	\$ 41,860	\$ 94,562	0.461	0.33	3,970	2,565
		5	20	15	CC	\$ 30,160	\$ 94,713	0.462	0.00	5,999	3,871
10		5	25	6	CC	\$ 64,025	\$ 94,826	0.463	0.76	2,106	1,971
4	1	5	25	6	CC	\$ 51,425	\$ 94,852	0.463	0.53	3,111	2,168
4	1	5	20	10	CC	\$ 50,060	\$ 95,402	0.465	0.50	3,474	2,534
2	2	5	20	6	CC	\$ 49,060	\$ 95,423	0.465	0.50	3,250	2,152
	1	5	25	6	CC	\$ 37,025	\$ 95,572	0.466	0.17	4,836	3,120
4		5	20	15	CC	\$ 44,560	\$ 95,708	0.467	0.36	4,466	3,188
6		5	30	4	CC	\$ 52,390	\$ 95,912	0.468	0.55	3,315	3,080
2	1	5	20	15	CC	\$ 46,360	\$ 96,065	0.469	0.36	3,862	2,535
12		5	25	10	CC	\$ 74,025	\$ 96,090	0.469	0.88	1,028	854
2		5	25	15	CC	\$ 41,525	\$ 96,117	0.469	0.21	4,621	2,982
		5	25	10	CC	\$ 30,825	\$ 96,545	0.471	0.00	5,999	3,871
2	1	5	30	4	CC	\$ 46,990	\$ 96,570	0.471	0.38	3,610	2,355
	2	5	15	15	CC	\$ 43,995	\$ 96,735	0.472	0.33	4,007	2,599
8	1	5	25	6	CC	\$ 65,825	\$ 96,821	0.472	0.79	1,688	1,451
10		5	25	4	CC	\$ 62,625	\$ 96,921	0.473	0.74	2,524	2,580
10		5	30	10	CC	\$ 70,990	\$ 97,051	0.473	0.84	1,230	969
8	1	5	25	10	CC	\$ 68,625	\$ 97,347	0.475	0.82	1,331	1,048
10		5	30	6	CC	\$ 68,190	\$ 97,348	0.475	0.80	1,673	1,617
6		5	30	6	CC	\$ 53,790	\$ 97,406	0.475	0.54	3,302	2,703
	1	5	20	15	CC	\$ 39,160	\$ 97,412	0.475	0.17	4,835	3,120
		5	30	4	CC	\$ 30,790	\$ 97,442	0.475	0.00	5,997	3,874
2	1	5	25	10	CC	\$ 47,025	\$ 97,460	0.475	0.37	3,815	2,470
4		5	25	10	CC	\$ 45,225	\$ 97,652	0.476	0.35	4,493	2,999
2		5	30	10	CC	\$ 42,190	\$ 97,945	0.478	0.21	4,621	2,982
	2	5	20	10	CC	\$ 44,660	\$ 98,239	0.479	0.33	3,970	2,565
10	1	5	20	10	CC	\$ 71,660	\$ 98,364	0.480	0.85	1,355	1,162
12		5	30	10	CC	\$ 78,190	\$ 98,699	0.481	0.93	591	471
10		5	25	15	CC	\$ 70,325	\$ 98,751	0.482	0.80	1,634	1,331
10	1	5	20	6	CC	\$ 68,860	\$ 98,766	0.482	0.81	1,822	1,589
	2	5	25	4	CC	\$ 44,625	\$ 98,873	0.482	0.33	3,939	2,547
12		5	25	6	CC	\$ 71,225	\$ 98,951	0.483	0.82	1,763	1,720
	1	5	25	10	CC	\$ 39,825	\$ 99,250	0.484	0.17	4,836	3,120

		5	30	6	CC	\$ 32,190	\$ 99,271	0.484	0.00	5,997	3,869
10		5	30	4	CC	\$ 66,790	\$ 99,462	0.485	0.78	2,085	2,295
2	1	5	30	6	CC	\$ 48,390	\$ 99,494	0.485	0.37	3,735	2,410
4	1	5	30	4	CC	\$ 54,190	\$ 99,541	0.485	0.55	3,083	2,656
2	2	5	25	4	CC	\$ 51,825	\$ 99,578	0.486	0.51	3,186	2,297
10	1	5	25	10	CC	\$ 75,825	\$ 99,624	0.486	0.91	765	603
8	1	5	25	4	CC	\$ 64,425	\$ 99,629	0.486	0.76	2,170	2,343
14		5	25	10	CC	\$ 81,225	\$ 99,891	0.487	0.93	638	532
4	1	5	20	15	CC	\$ 53,560	\$ 99,998	0.488	0.50	3,474	2,534
	1	5	30	4	CC	\$ 39,790	\$ 100,101	0.488	0.17	4,829	3,119
10	1	5	25	6	CC	\$ 73,025	\$ 100,252	0.489	0.86	1,254	1,138
6	1	5	30	4	CC	\$ 61,390	\$ 100,667	0.491	0.69	2,378	2,443
4	1	5	25	10	CC	\$ 54,225	\$ 100,686	0.491	0.51	3,357	2,296
12		5	25	15	CC	\$ 77,525	\$ 100,687	0.491	0.88	1,028	854
	2	5	25	6	CC	\$ 46,025	\$ 100,783	0.492	0.33	3,947	2,549
6		5	30	10	CC	\$ 56,590	\$ 100,804	0.492	0.53	3,303	2,274
4	1	5	30	6	CC	\$ 55,590	\$ 100,872	0.492	0.53	3,067	2,117
	3	5	20	4	CC	\$ 49,460	\$ 100,980	0.493	0.46	3,428	2,230
6	1	5	30	6	CC	\$ 62,790	\$ 100,997	0.493	0.68	2,253	1,791
		5	25	15	CC	\$ 34,325	\$ 101,141	0.493	0.00	5,999	3,871
12		5	30	6	CC	\$ 75,390	\$ 101,167	0.493	0.87	1,284	1,338
8	1	5	30	6	CC	\$ 69,990	\$ 101,501	0.495	0.81	1,488	1,286
10		5	30	15	CC	\$ 74,490	\$ 101,647	0.496	0.84	1,230	969
2	2	5	25	6	CC	\$ 53,225	\$ 101,666	0.496	0.50	3,232	2,102
2	2	5	20	10	CC	\$ 51,860	\$ 101,798	0.497	0.48	3,547	2,443
8	1	5	25	15	CC	\$ 72,125	\$ 101,944	0.497	0.82	1,331	1,048
	1	5	30	6	CC	\$ 41,190	\$ 101,983	0.497	0.17	4,834	3,119
14		5	20	10	CC	\$ 77,060	\$ 102,034	0.498	0.85	1,617	1,458
2	1	5	25	15	CC	\$ 50,525	\$ 102,057	0.498	0.37	3,815	2,470
14		5	30	10	LF	\$ 85,390	\$ 102,119	0.498	0.99	153	192
4		5	30	10	CC	\$ 49,390	\$ 102,123	0.498	0.36	4,278	2,765
14		5	10	4	CC	\$ 64,530	\$ 102,147	0.498	0.71	3,691	3,050
12	1	5	20	10	CC	\$ 78,860	\$ 102,175	0.498	0.90	966	838
12		5	25	4	CC	\$ 69,825	\$ 102,190	0.498	0.79	2,305	2,440
4		5	25	15	CC	\$ 48,725	\$ 102,249	0.499	0.35	4,493	2,999
8	1	5	30	10	CC	\$ 72,790	\$ 102,396	0.499	0.84	1,175	866
2		5	30	15	CC	\$ 45,690	\$ 102,542	0.500	0.21	4,621	2,982
8	2	5	20	6	CC	\$ 70,660	\$ 102,566	0.500	0.82	1,589	1,353
10	1	5	20	4	CC	\$ 67,460	\$ 102,664	0.501	0.77	2,428	2,490
	3	5	20	6	CC	\$ 50,860	\$ 102,712	0.501	0.46	3,416	2,220
	2	5	20	15	CC	\$ 48,160	\$ 102,836	0.502	0.33	3,970	2,565
14		5	30	10	CC	\$ 85,390	\$ 102,856	0.502	0.97	241	188

8	2	5	20	10	CC	\$ 73,460	\$ 102,915	0.502	0.86	1,209	993
		5	30	10	CC	\$ 34,990	\$ 102,948	0.502	0.00	5,997	3,869
14		5	30	6	LF	\$ 82,590	\$ 102,954	0.502	0.95	657	947
10	1	5	20	15	CC	\$ 75,160	\$ 102,961	0.502	0.85	1,355	1,162
6	2	5	25	6	CC	\$ 67,625	\$ 102,980	0.502	0.77	1,729	1,383
16		5	25	10	LF	\$ 88,425	\$ 103,137	0.503	0.99	181	228
14		5	25	6	CC	\$ 78,425	\$ 103,236	0.504	0.87	1,433	1,466
12		5	30	15	CC	\$ 81,690	\$ 103,296	0.504	0.93	591	471
2	3	5	20	4	CC	\$ 56,660	\$ 103,333	0.504	0.60	2,857	2,159
2	1	5	30	10	CC	\$ 51,190	\$ 103,665	0.506	0.37	3,791	2,446
16		5	25	6	LF	\$ 85,625	\$ 103,746	0.506	0.95	659	952
	1	5	25	15	CC	\$ 43,325	\$ 103,846	0.506	0.17	4,836	3,120
14		5	10	6	CC	\$ 65,930	\$ 103,857	0.507	0.71	3,634	2,993
8	1	5	30	4	CC	\$ 68,590	\$ 103,867	0.507	0.79	1,922	2,188
12	1	5	20	6	CC	\$ 76,060	\$ 104,128	0.508	0.85	1,609	1,440
2	3	5	20	6	CC	\$ 58,060	\$ 104,162	0.508	0.61	2,758	1,914
10	1	5	25	15	CC	\$ 79,325	\$ 104,221	0.508	0.91	765	603
10	1	5	30	10	CC	\$ 79,990	\$ 104,291	0.509	0.93	563	426
6	1	5	30	10	CC	\$ 65,590	\$ 104,318	0.509	0.68	2,229	1,572
14		5	20	6	CC	\$ 74,260	\$ 104,363	0.509	0.81	2,277	2,146
	2	5	25	10	CC	\$ 48,825	\$ 104,462	0.509	0.33	3,947	2,549
12	1	5	25	10	CC	\$ 83,025	\$ 104,470	0.510	0.95	494	390
14		5	25	15	CC	\$ 84,725	\$ 104,488	0.510	0.93	638	532
12		5	30	4	CC	\$ 73,990	\$ 104,517	0.510	0.83	1,842	2,141
10	1	5	30	6	CC	\$ 77,190	\$ 104,526	0.510	0.89	1,006	952
6	2	5	25	10	CC	\$ 70,425	\$ 104,663	0.511	0.79	1,504	1,101
6	2	5	25	4	CC	\$ 66,225	\$ 104,827	0.511	0.76	2,103	2,224
10	1	5	25	4	CC	\$ 71,625	\$ 104,971	0.512	0.81	1,959	2,221
16		5	25	10	CC	\$ 88,425	\$ 105,023	0.512	0.96	401	335
12	1	5	10	4	CC	\$ 66,330	\$ 105,187	0.513	0.73	3,362	2,995
	2	5	30	4	CC	\$ 48,790	\$ 105,194	0.513	0.33	3,927	2,539
12	1	5	25	6	CC	\$ 80,225	\$ 105,230	0.513	0.90	997	962
12	1	5	15	6	CC	\$ 71,895	\$ 105,245	0.513	0.79	2,454	2,117
4	1	5	25	15	CC	\$ 57,725	\$ 105,283	0.514	0.51	3,357	2,296
6		5	30	15	CC	\$ 60,090	\$ 105,401	0.514	0.53	3,303	2,274
	1	5	30	10	CC	\$ 43,990	\$ 105,660	0.515	0.17	4,834	3,119
8	2	5	25	6	CC	\$ 74,825	\$ 105,956	0.517	0.86	1,240	1,072
2	2	5	30	4	CC	\$ 55,990	\$ 105,974	0.517	0.51	3,181	2,312
4	1	5	30	10	CC	\$ 58,390	\$ 106,196	0.518	0.51	3,261	2,131
8	2	5	20	4	CC	\$ 69,260	\$ 106,229	0.518	0.79	2,174	2,272
14		5	30	6	CC	\$ 82,590	\$ 106,234	0.518	0.91	1,039	1,149